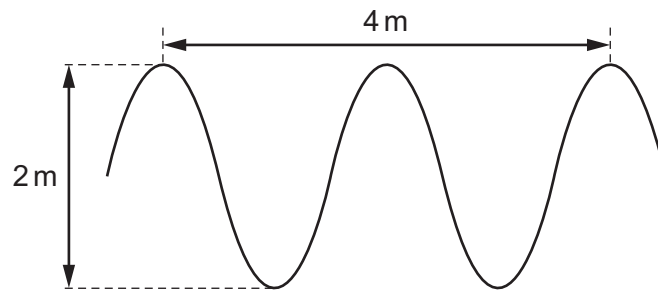


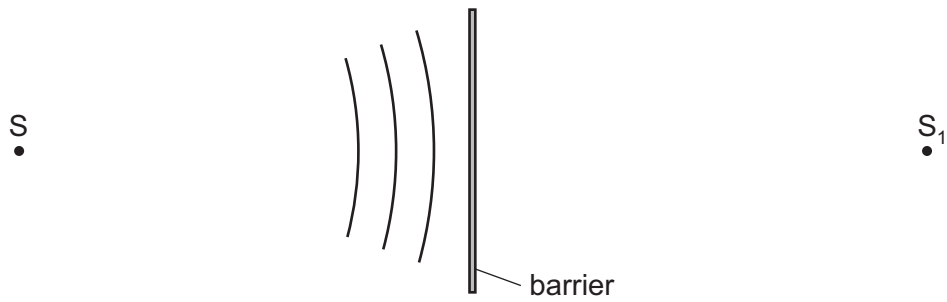
18 The diagram represents a water wave.



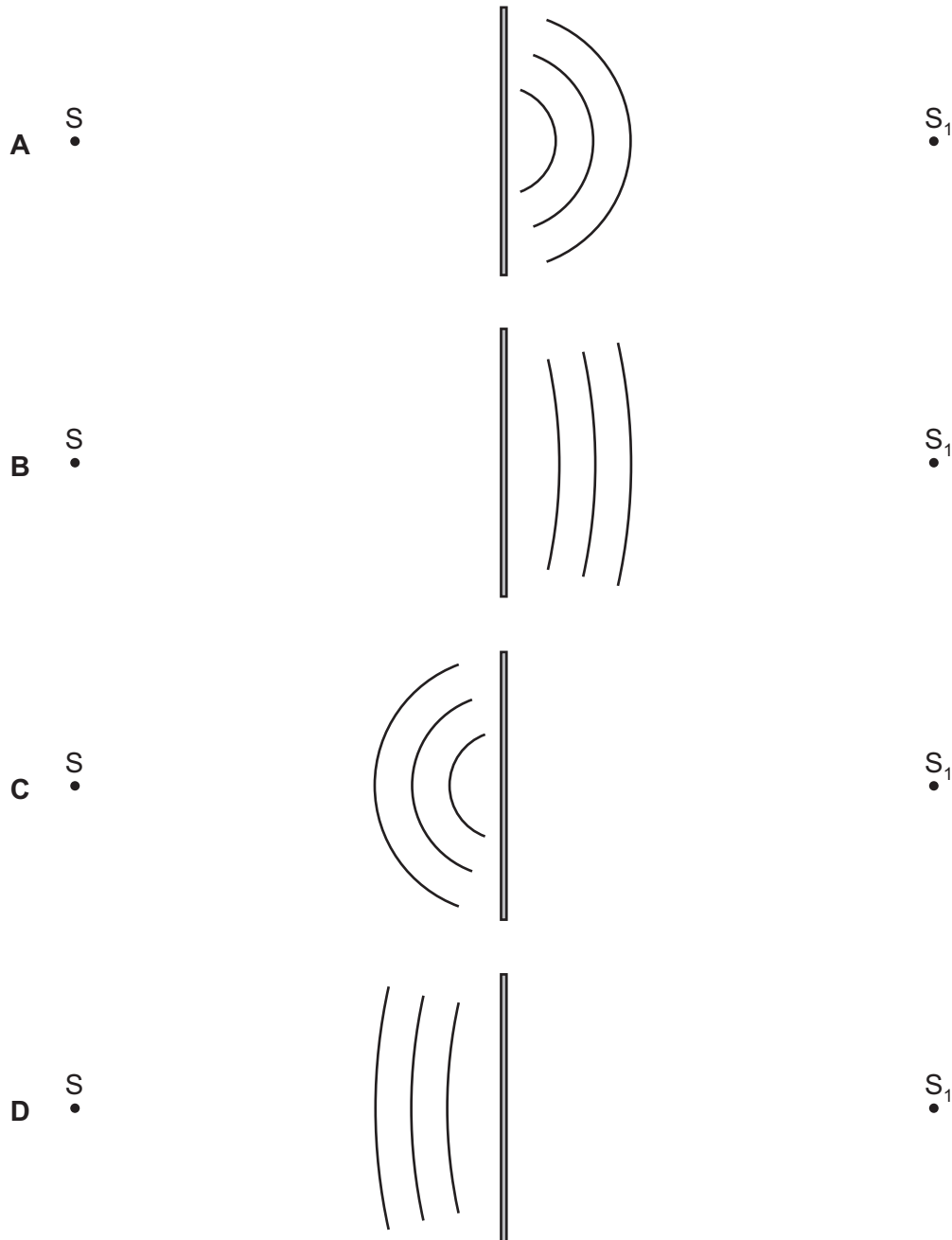
Which row shows the amplitude and the wavelength of the wave?

	amplitude / m	wavelength / m
A	1	2
B	1	4
C	2	2
D	2	4

- 19 The diagram represents circular wavefronts coming from point S. The wavefronts strike a barrier and are reflected so that they appear to come from point S_1 .



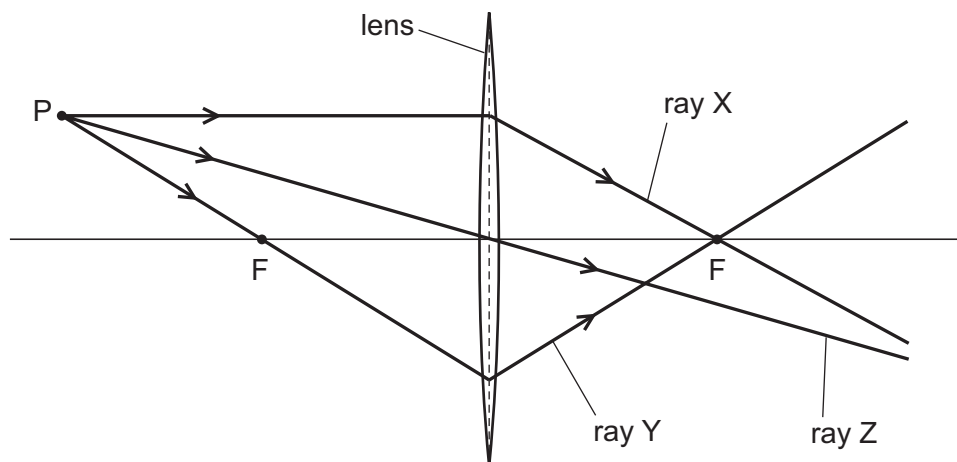
Which diagram shows the reflected wavefronts?



- 20 A plane mirror is fitted to a wall.

Which statement about the image formed by the mirror is correct?

- A The image is real.
 - B The image is left to right (laterally inverted).
 - C The image is smaller than the object.
 - D The image is upside down.
- 21 A student draws a diagram representing three rays of light from point P passing through a converging lens. Each point labelled F is a principal focus of the lens.



Which of the rays has the student drawn correctly?

- A ray X and ray Y
 - B ray X and ray Z
 - C ray Y only
 - D ray Z only
- 22 Which group contains only electromagnetic waves?
- A light waves, radio waves, γ -rays
 - B light waves, radio waves, sound waves
 - C light waves, sound waves, γ -rays
 - D radio waves, sound waves, γ -rays

- 23** Two sounds X and Y are produced by loudspeakers.

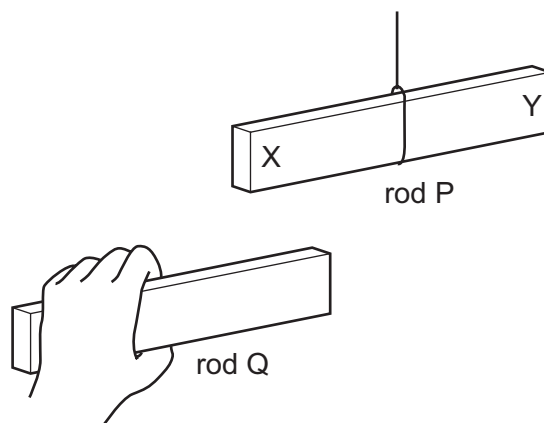
The amplitude and frequency of each sound wave is given in the table.

	amplitude / mm	frequency / Hz
X	1.3	475
Y	2.0	235

How does sound Y compare with sound X?

- A** Y is louder and has a higher pitch.
 - B** Y is louder and has a lower pitch.
 - C** Y is quieter and has a higher pitch.
 - D** Y is quieter and has a lower pitch.
- 24** Sounds are produced by vibrating objects. A certain object vibrates but a person nearby cannot hear any sound.
- Which statement could explain why nothing is heard?
- A** The amplitude of the sound waves is too large.
 - B** The frequency of the vibration is too high.
 - C** The sound waves are transverse.
 - D** The speed of the sound waves is too high.
- 25** Which statement about a permanent bar magnet is correct?
- A** It is made from a soft magnetic material.
 - B** It repels a non-magnetic material.
 - C** Its field lines cross each other where the magnetic field is strong.
 - D** Its N-pole repels the N-pole of another magnet.
- 26** Which procedure may be used to demagnetise a steel bar?
- A** cooling it in a freezer for several hours
 - B** earthing it with a copper wire for several seconds
 - C** removing it slowly from a coil carrying an alternating current (a.c.)
 - D** rubbing it in one direction with a woollen cloth

27 Two plastic rods P and Q are both negatively charged. Rod P hangs freely.

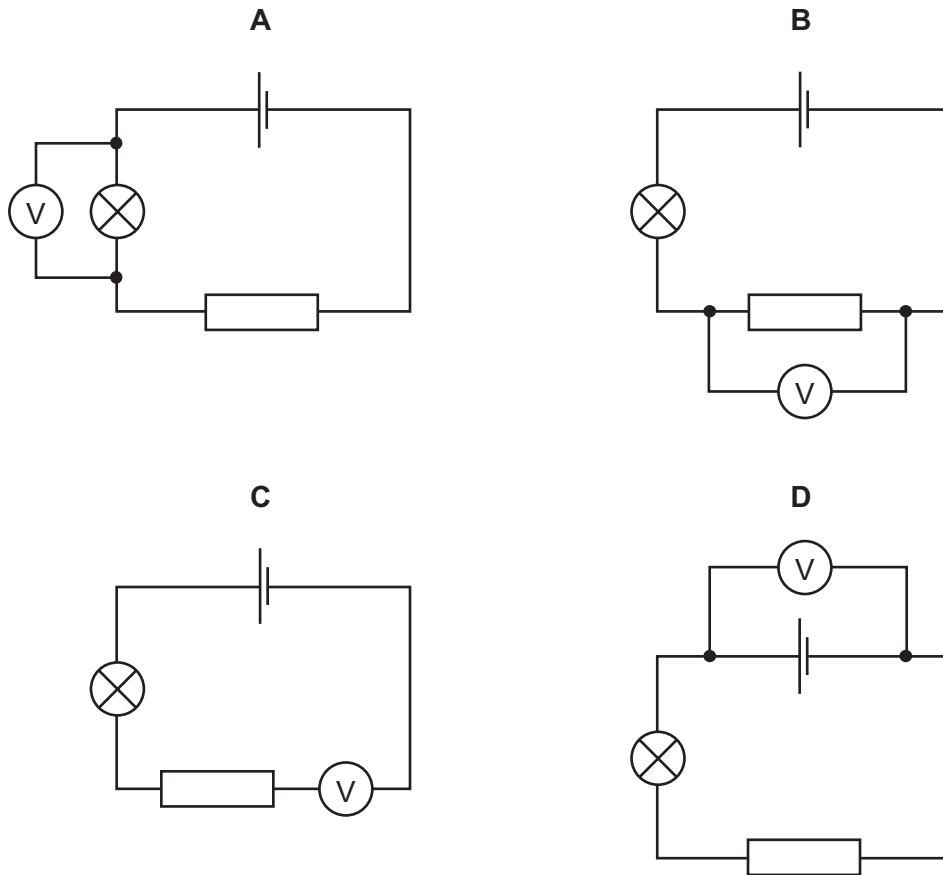


The end of rod Q is brought near to end X of rod P, and then near to end Y of rod P.

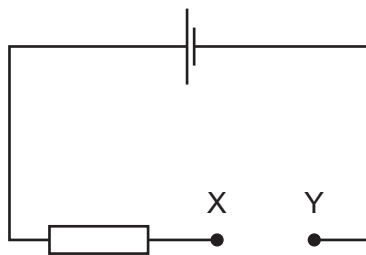
What happens to the rods in each position?

	near end X	near end Y
A	they attract	they attract
B	they attract	they repel
C	they repel	they attract
D	they repel	they repel

28 Which circuit shows a voltmeter measuring the p.d. across a resistor?



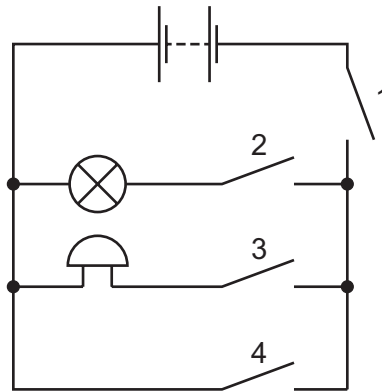
29 A student has four pieces of resistance wire made of the same material. Each piece is connected in turn between the terminals X and Y in the circuit.



In which wire is the current the largest?

	length of wire / m	diameter of wire / mm
A	0.5	0.5
B	0.5	1.0
C	1.0	0.5
D	1.0	1.0

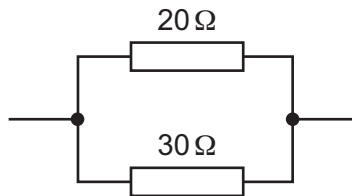
30 A student connects the circuit shown.



Which switches must be closed for both the bell to ring and the lamp to light?

- A 1 and 4 only
- B 2 and 3 only
- C 1, 2 and 3 only
- D 1, 2, 3 and 4

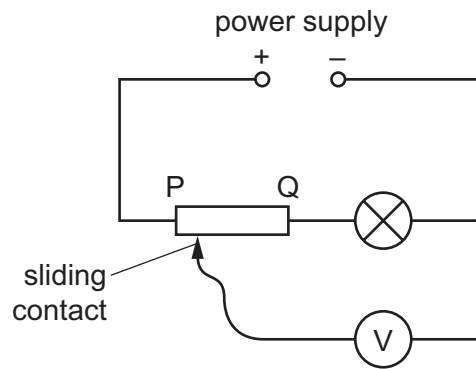
31 Two resistors are connected in parallel.



Which value could be the resistance of the combination?

- A $12\ \Omega$
- B $20\ \Omega$
- C $25\ \Omega$
- D $50\ \Omega$

32 The circuit contains a variable potential divider PQ, a lamp and a voltmeter.



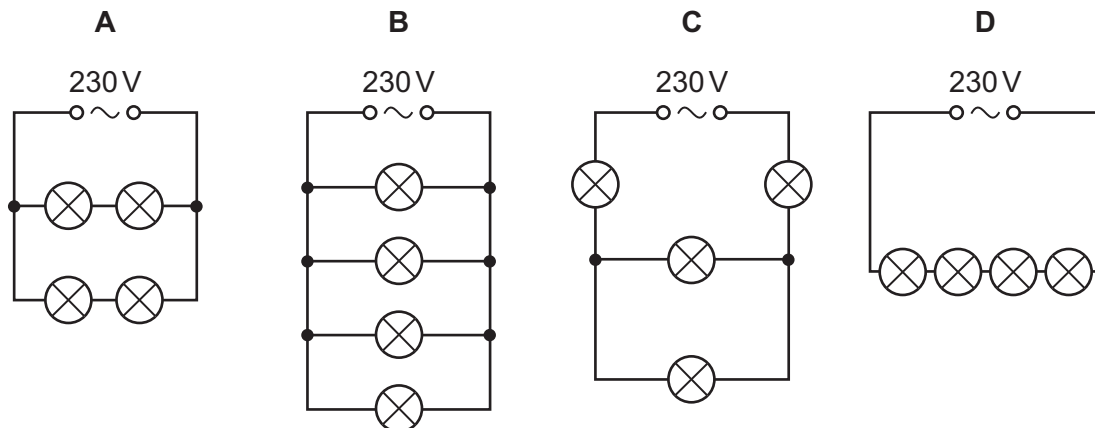
The sliding contact of the potential divider is moved towards end Q.

What happens to the brightness of the lamp and what happens to the voltmeter reading?

	brightness of lamp	voltmeter reading
A	becomes brighter	decreases
B	becomes brighter	increases
C	does not change	decreases
D	does not change	increases

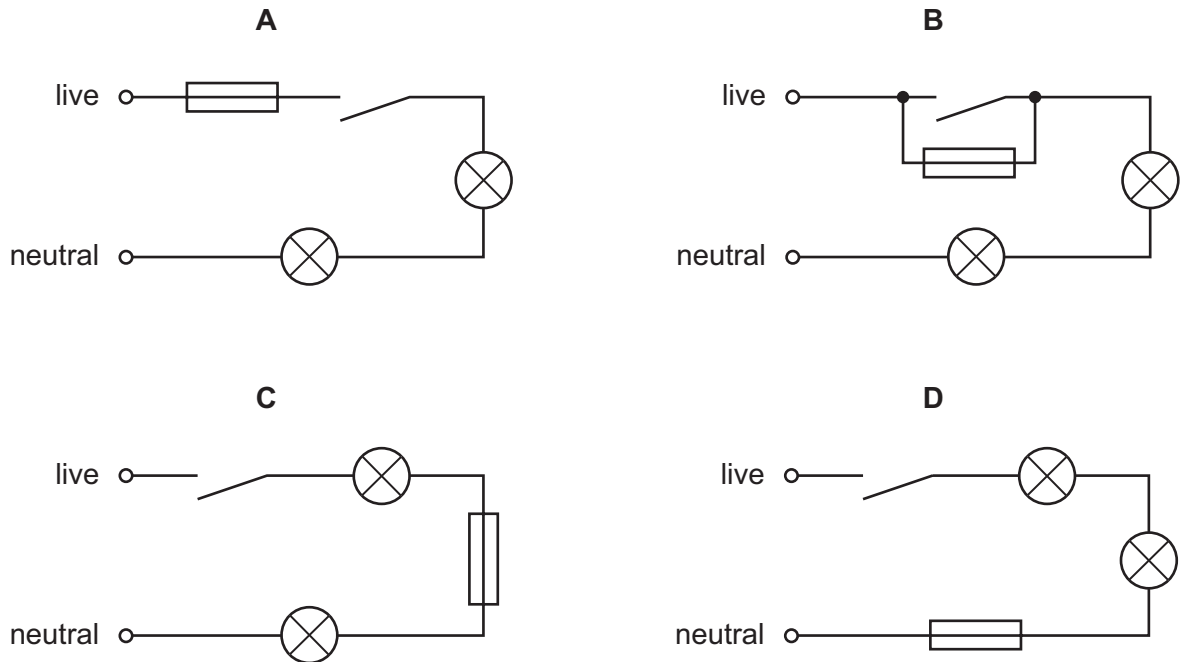
33 Four lamps are each labelled '60 W 230 V'.

In which circuit are the lamps connected so that they operate at normal brightness?

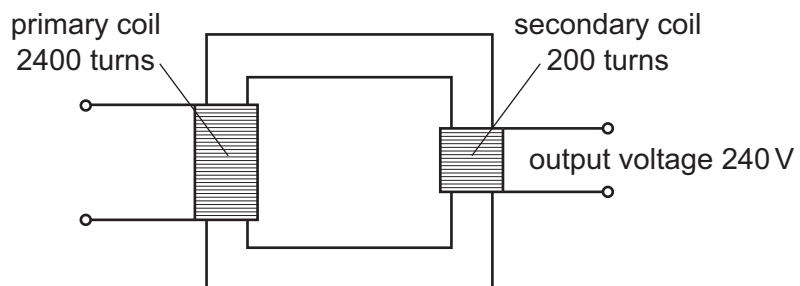


34 A fuse is used to protect an electric circuit.

Which diagram shows where the fuse should be connected?



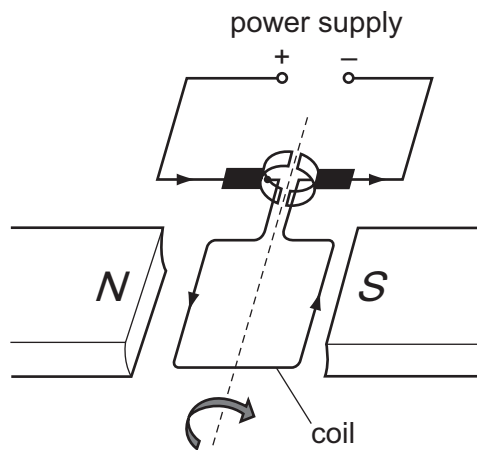
35 A transformer has 2400 turns on its primary coil and 200 turns on its secondary coil.



What input voltage is needed to give an output voltage of 240 V?

- A** 12 V **B** 20 V **C** 240 V **D** 2880 V

36 A current-carrying coil in a magnetic field experiences a turning effect.

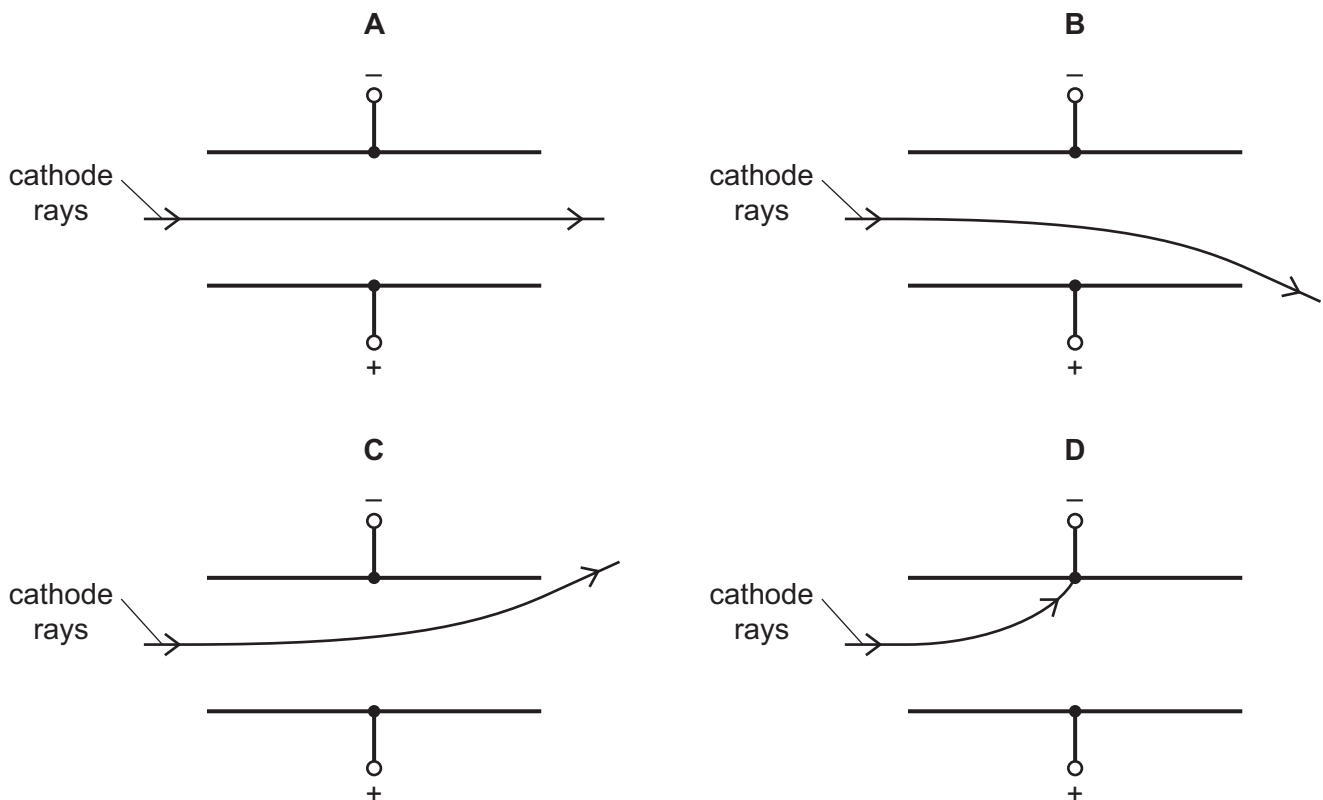


How can the turning effect be increased?

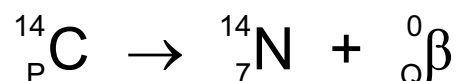
- A** Increase the number of turns on the coil.
- B** Reduce the size of the current.
- C** Reverse the direction of the magnetic field.
- D** Use thinner wire for the coil.

37 A beam of cathode rays enters an electric field between two charged plates.

Which diagram shows a path that the rays could take?



- 38 Radioactive carbon-14 decays into nitrogen by emitting a β -particle. The equation below represents the decay.



What are the values of P and Q?

	P	Q
A	6	1
B	6	-1
C	8	1
D	8	-1

- 39 A student investigates how the radiation from a radioactive source changes with time.

The table shows the results from the detector used by the student.

time / minutes	count rate / counts per minute
0	340
2.0	180
4.0	100
6.0	60
8.0	40

The experiment is repeated by many other students, who also measure the count rate every two minutes.

The half-life of the source is known to be exactly 2.0 minutes.

Why is the measured count rate **always greater** than half the previous value?

- A** Radioactive emissions occur randomly with time.
 - B** The detector used is very close to the source.
 - C** There is background radiation present.
 - D** The radioactive source is decaying.
- 40 The nuclide notation for radium-226 is ${}^{226}_{88}\text{Ra}$.

How many electrons orbit the nucleus of a neutral atom of radium-226?

- A** 0
- B** 88
- C** 138
- D** 226

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Cambridge International Examinations
Cambridge International General Certificate of Secondary Education

PHYSICS

0625/11

Paper 1 Multiple Choice

October/November 2015

45 minutes

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB recommended)



READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, glue or correction fluid.

Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

DO NOT WRITE IN ANY BARCODES.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

Electronic calculators may be used.

The syllabus is approved for use in England, Wales and Northern Ireland as a Cambridge International Level 1/Level 2 Certificate.

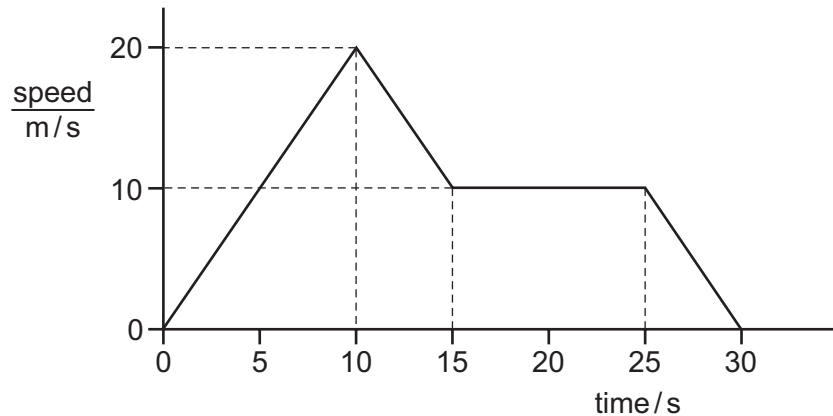
This document consists of **16** printed pages.



- 1 Which option contains **only** apparatus that could be used to determine the volume of a small block of unknown material?

A measuring cylinder, metre rule
 B measuring cylinder, stopwatch
 C metre rule, balance
 D metre rule, stopwatch

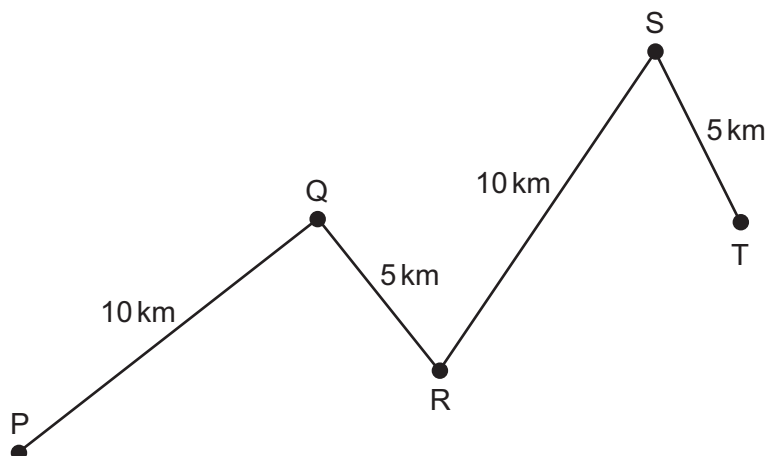
- 2 The graph represents the motion of a car.



What is the distance travelled by the car while it is moving at a constant speed?

A 100 m B 150 m C 250 m D 300 m

- 3 A car travels along the route PQRST in 30 minutes.



What is the average speed of the car?

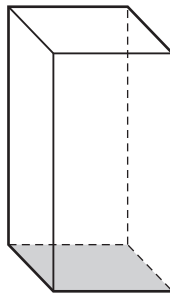
A 10 km/hour B 20 km/hour C 30 km/hour D 60 km/hour

- 4 The mass of an object is measured on Earth. The mass is 5.0 kg.

The object is taken to the Moon. The mass of the object is measured on the Moon.

What is the mass of the object on the Moon?

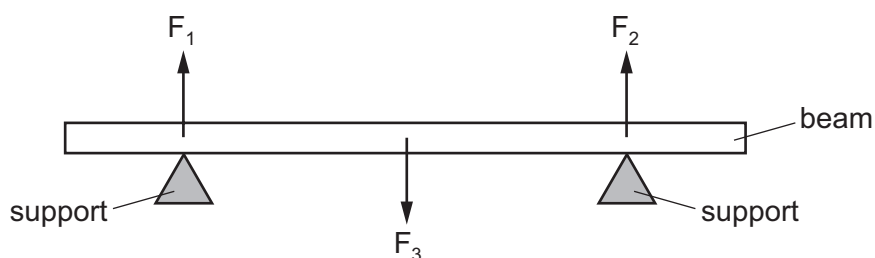
- A 0 kg
 - B more than 0 kg, but less than 5.0 kg
 - C 5.0 kg
 - D more than 5.0 kg
- 5 A student wishes to determine the density of the solid block shown.



Which quantities must be known?

- A the area of the shaded face and the volume of the block
- B the area of the shaded face and the weight of the block
- C the mass of the block and the height of the block
- D the mass of the block and the volume of the block

- 6 A heavy beam rests on two supports. The diagram shows the only three forces F_1 , F_2 and F_3 acting on the beam.



The beam is in equilibrium.

Which statement is correct?

- A** All the forces are equal in size.
- B** The resultant force on the beam is in the opposite direction to the resultant turning effect.
- C** The resultant force on the beam is zero and the resultant turning effect on the beam is zero.
- D** The total upward force is twice the total downward force.
- 7 Which list contains only properties of an object that can be changed by a force?
- A** direction of motion, mass, shape
- B** direction of motion, mass, speed
- C** direction of motion, shape, speed
- D** mass, shape, speed
- 8 What needs to be known to calculate the work done by a force acting on an object?

	the size of the force	the distance the force moves the object	the time for which the force acts
A	✓	✓	✓
B	✓	✓	✗
C	✓	✗	✓
D	✓	✗	✗

key

✓ = needed

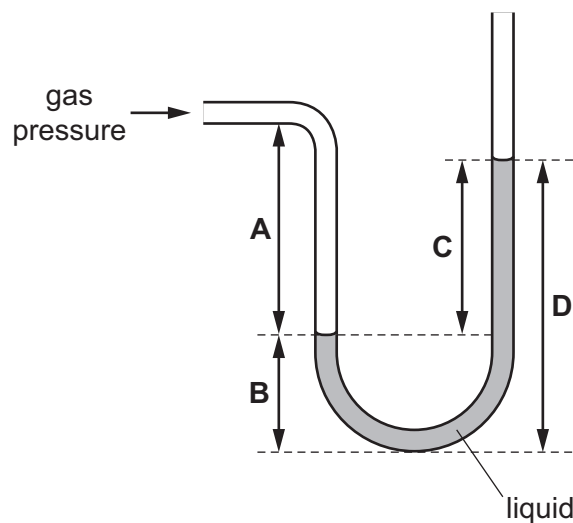
✗ = not needed

- 9 Electrical energy may be obtained from nuclear fission.

In which order is the energy transferred in this process?

- A** nuclear fuel → generator → reactor and boiler → turbines
B nuclear fuel → generator → turbines → reactor and boiler
C nuclear fuel → reactor and boiler → generator → turbines
D nuclear fuel → reactor and boiler → turbines → generator
- 10 The diagram shows a manometer containing a liquid. The manometer is used to find the difference between the pressure of a gas and atmospheric pressure.

Which distance represents this pressure difference?



- 11 Four physics teachers investigate pressure. They wear identical clothes and lie on different beds of nails.

The table gives the weight of each teacher and the total area of contact between the teacher and the nails.

Which teacher experiences the least pressure from the nails?

	weight of teacher / N	total area of contact / cm ²
A	700	13
B	800	20
C	900	14
D	1000	21

- 12** A cylinder of constant volume contains a fixed mass of gas. The gas is cooled.

What happens to the pressure of the gas and what happens to the kinetic energy of the gas molecules?

	pressure of gas	kinetic energy of molecules
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 13** A swimmer feels cold after leaving warm water on a warm, windy day.

Why does she feel cold even though the air is warm?

- A** The less energetic water molecules on her skin escape quickly.
- B** The more energetic water molecules on her skin do not escape quickly.
- C** The water on her skin does not evaporate quickly enough to keep her warm.
- D** The water on her skin evaporates quickly and cools her skin.

- 14** A circular metal disc is heated.

Which quantity decreases?

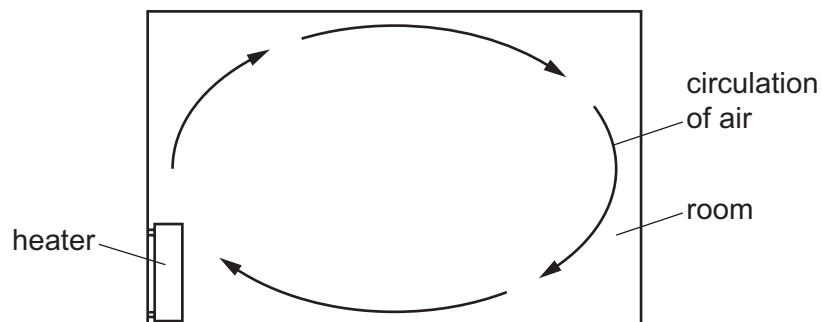
- A** its density
- B** its diameter
- C** its thickness
- D** its volume

- 15** The same quantity of thermal (heat) energy is given to two objects X and Y. The temperature rise of object X is less than the temperature rise of object Y.

What accounts for this difference?

- A** X has a larger thermal capacity than Y.
- B** X is a better thermal conductor than Y.
- C** Y has a larger thermal capacity than X.
- D** Y is a better thermal conductor than X.

- 16 The air in a room is heated by a heater. The diagram shows the circulation of the air in the room.



Which statement about the air that is heated is correct?

- A The air contracts and becomes less dense.
 - B The air contracts and becomes more dense.
 - C The air expands and becomes less dense.
 - D The air expands and becomes more dense.
- 17 Four rods are made from different metals P, Q, R and S. The rods have equal lengths and equal diameters. The rods are heated at one end, in the same way.

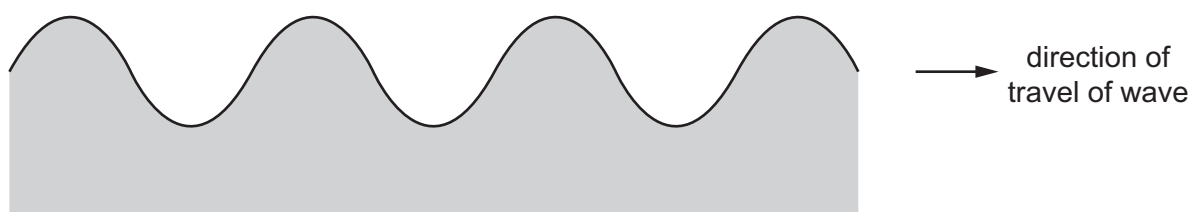
The table shows the time taken for the temperature at the other end of each rod to rise by 1.0°C .

Which metal is the best conductor of thermal energy (heat)?

metal	time taken/s
P	35
Q	30
R	45
S	40

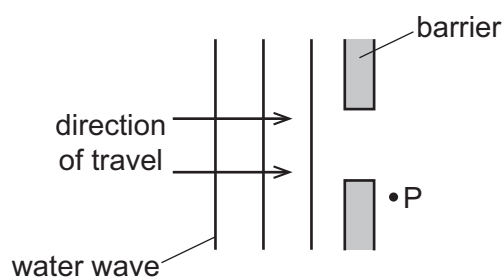
- A metal P B metal Q C metal R D metal S

- 18 The diagram shows a side view of a water wave at a particular time. The diagram is drawn full size.



Which statement about the wave is correct?

- A The wave is longitudinal and the frequency can be measured from the diagram.
 - B The wave is longitudinal and the wavelength can be measured from the diagram.
 - C The wave is transverse and the frequency can be measured from the diagram.
 - D The wave is transverse and the wavelength can be measured from the diagram.
- 19 The diagram shows a water wave approaching a barrier with a gap.



The wave reaches point P.

What is the name of the effect that causes the wave to reach point P?

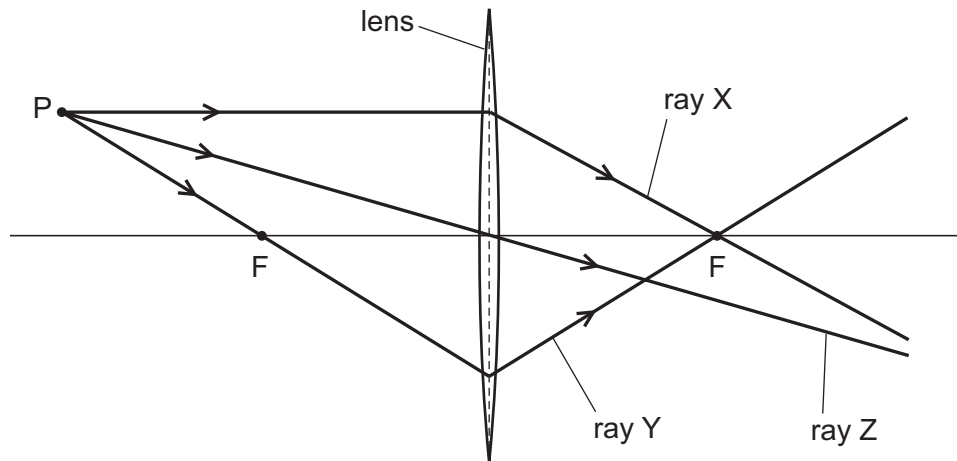
- A diffraction
 - B dispersion
 - C reflection
 - D refraction
- 20 The diagram represents the electromagnetic spectrum.

Some regions have been labelled, and some labels are missing.

Which region should be labelled as infra-red waves?

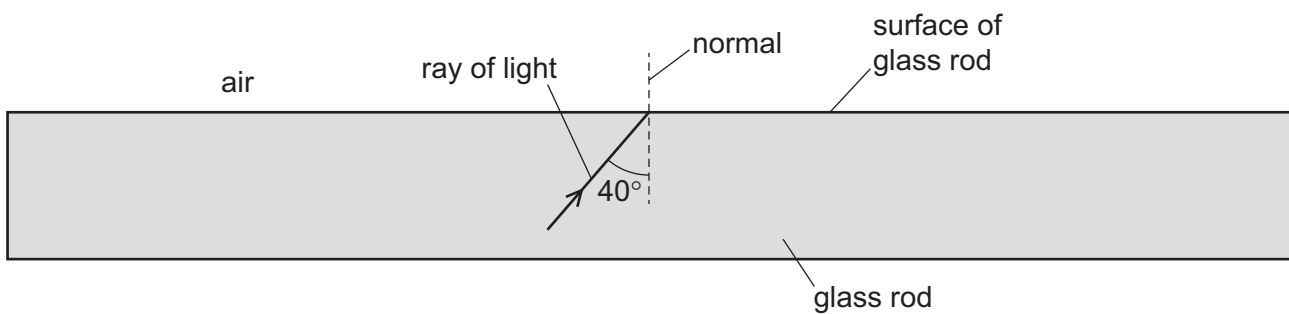
radio waves	A	B	visible light	C	D	γ -rays
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- 21 A student draws a diagram representing three rays of light from point P passing through a converging lens. Each point labelled F is a principal focus of the lens.



Which of the rays has the student drawn correctly?

- A** ray X and ray Y
B ray X and ray Z
C ray Y only
D ray Z only
- 22 The diagram shows a ray of light inside a glass rod. The critical angle for the light in the glass is 42° .



Which row shows what happens to the light when it reaches the surface of the glass rod?

	any light reflected?	any light refracted?
A	no	no
B	no	yes
C	yes	no
D	yes	yes

23 Which row states whether light waves and whether sound waves can travel in a vacuum?

	sound waves	light waves
A	no	no
B	no	yes
C	yes	no
D	yes	yes

24 Sounds are produced by vibrating objects. A certain object vibrates but a person nearby cannot hear any sound.

Which statement could explain why nothing is heard?

- A** The amplitude of the sound waves is too large.
- B** The frequency of the vibration is too high.
- C** The sound waves are transverse.
- D** The speed of the sound waves is too high.

25 Which row states whether each metal is ferrous or non-ferrous?

	ferrous	non-ferrous
A	aluminium	copper
B	copper	iron
C	iron	steel
D	steel	aluminium

26 Which procedure may be used to demagnetise a steel bar?

- A** cooling it in a freezer for several hours
- B** earthing it with a copper wire for several seconds
- C** removing it slowly from a coil carrying an alternating current (a.c.)
- D** rubbing it in one direction with a woollen cloth

27 What is the unit of electrical power?

- A** ampere
- B** joule
- C** volt
- D** watt

- 28** Some resistors are made using one type of wire. Two different lengths of wire are available. Each length is available in two different diameters.

Which wire has the highest resistance?

- A** the wire with the greater length and the larger diameter
- B** the wire with the greater length and the smaller diameter
- C** the wire with the smaller length and the larger diameter
- D** the wire with the smaller length and the smaller diameter

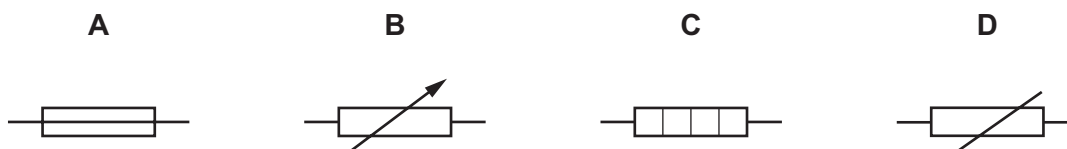
- 29** Four students are each given an identical resistor and asked to find its resistance. They each measure the potential difference across the resistor and the current in it.

One student makes a mistake.

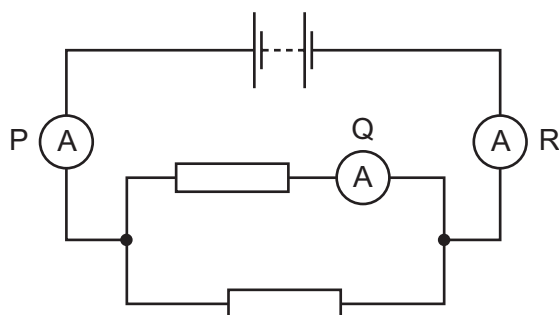
Which row shows the results of the student that makes a mistake?

	potential difference / V	current / A
A	1.2	0.500
B	2.4	1.100
C	1.5	0.625
D	3.0	1.250

- 30** What is the circuit symbol for a variable resistor?

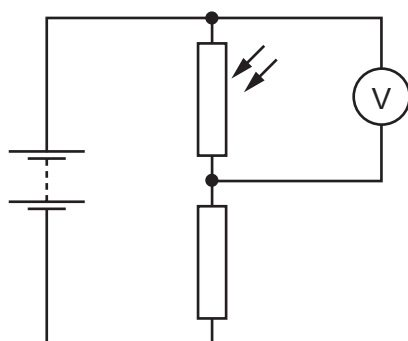


- 31 The diagram shows a circuit containing three ammeters P, Q and R.



Which statement about the readings on the ammeters is correct?

- A** The reading on P is equal to the reading on Q.
 - B** The reading on P is equal to the reading on R.
 - C** The reading on Q is greater than the reading on P.
 - D** The reading on Q is greater than the reading on R.
- 32 The diagram shows a light-dependent resistor (LDR) connected in a potential divider circuit.



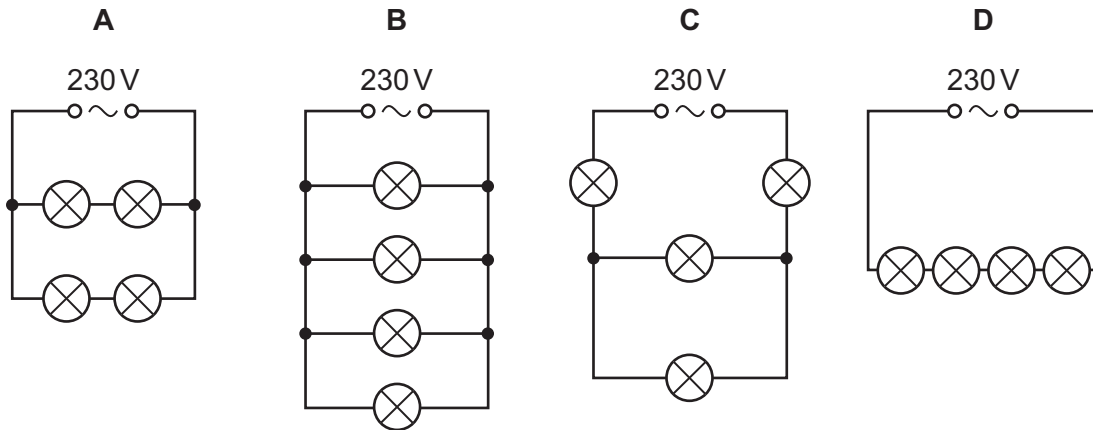
The brightness of the light falling on the LDR is increased.

Which row shows what happens to the resistance of the LDR, and what happens to the reading on the voltmeter?

	resistance of LDR	reading on voltmeter
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

33 Four lamps are each labelled '60 W 230 V'.

In which circuit are the lamps connected so that they operate at normal brightness?

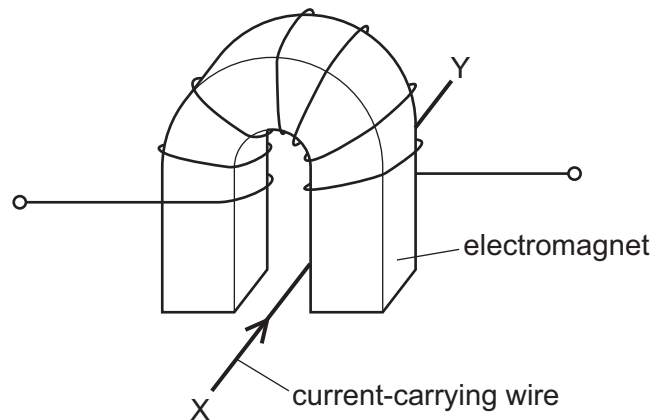


34 A step-up transformer is used before electricity is transmitted by overhead cables.

Which statement explains why the step-up transformer is used?

- A** It increases the current to increase the speed at which the electricity travels.
- B** It increases the current to reduce energy loss in the cables.
- C** It increases the voltage to increase the speed at which the electricity travels.
- D** It increases the voltage to reduce energy loss in the cables.

- 35 A current-carrying wire XY lies in the magnetic field between the two poles of a U-shaped electromagnet. A force acts on the wire XY because of the magnetic field.



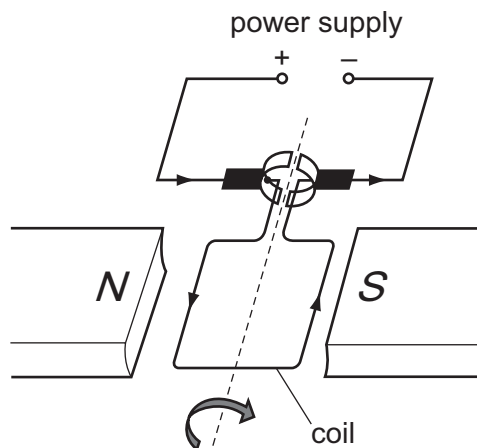
Each of the following actions is carried out separately.

- The current in the wire XY is reversed.
- The magnetic field is reversed.
- Both the current in the wire XY and the magnetic field are reversed at the same time.

How many of these actions cause the direction of the force on the wire XY to be reversed?

- A** 0 **B** 1 **C** 2 **D** 3

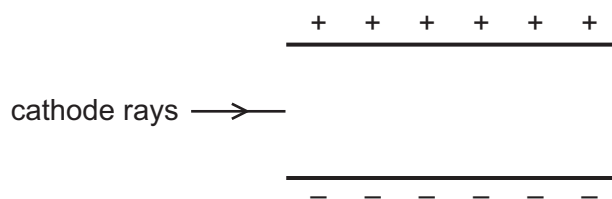
- 36 A current-carrying coil in a magnetic field experiences a turning effect.



How can the turning effect be increased?

- A** Increase the number of turns on the coil.
B Reduce the size of the current.
C Reverse the direction of the magnetic field.
D Use thinner wire for the coil.

- 37 The diagram shows cathode rays entering an electric field between two charged plates.



How does the path of the cathode rays change and why?

- A They move towards the negative plate because cathode rays have a negative charge.
 - B They move towards the negative plate because cathode rays have a positive charge.
 - C They move towards the positive plate because cathode rays have a negative charge.
 - D They move towards the positive plate because cathode rays have a positive charge.
- 38 The nuclide symbol for radioactive polonium is $^{210}_{84}\text{Po}$.

A nucleus of this type of polonium emits an α -particle.

What is the proton number (atomic number) of the nucleus after it has emitted the α -particle?

- A 82 B 83 C 84 D 85

- 39 A student investigates how the radiation from a radioactive source changes with time.

The table shows the results from the detector used by the student.

time / minutes	count rate / counts per minute
0	340
2.0	180
4.0	100
6.0	60
8.0	40

The experiment is repeated by many other students, who also measure the count rate every two minutes.

The half-life of the source is known to be exactly 2.0 minutes.

Why is the measured count rate **always greater** than half the previous value?

- A Radioactive emissions occur randomly with time.
- B The detector used is very close to the source.
- C There is background radiation present.
- D The radioactive source is decaying.

- 40 The nuclide notation for radium-226 is ${}^{226}_{88}\text{Ra}$.

How many electrons orbit the nucleus of a neutral atom of radium-226?

- A 0
- B 88
- C 138
- D 226

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Cambridge International General Certificate of Secondary Education

MARK SCHEME for the October/November 2015 series

0625 PHYSICS

0625/13

Paper 1 (Multiple Choice), maximum raw mark 40

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Page 2	Mark Scheme	Syllabus	Paper
	Cambridge IGCSE – October/November 2015	0625	13

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	21	B
2	A	22	C
3	D	23	B
4	A	24	B
5	D	25	D
6	C	26	C
7	B	27	A
8	B	28	C
9	D	29	A
10	B	30	B
11	C	31	D
12	C	32	C
13	D	33	B
14	A	34	B
15	C	35	D
16	D	36	A
17	B	37	D
18	D	38	C
19	C	39	C
20	D	40	B

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<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	A	21	B
2	D	22	A
3	D	23	B
4	C	24	B
5	A	25	D
6	C	26	C
7	D	27	D
8	C	28	B
9	D	29	B
10	B	30	C
11	C	31	A
12	B	32	C
13	D	33	B
14	A	34	A
15	A	35	D
16	C	36	A
17	D	37	B
18	A	38	B
19	D	39	C
20	B	40	B

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9	D	29	B
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11	B	31	B
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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/13

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
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Soft pencil (type B or HB is recommended)

INSTRUCTIONS

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The mass of each ball is shown.

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Which balls will float in a liquid of density 0.95 g/cm^3 ?

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Which statement is correct?

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Why does it take a few seconds before his hand starts to feel hot?

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Which statement is correct?

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- 17 Twelve crests of a water wave pass a boat anchored at sea in a time of 180 s.

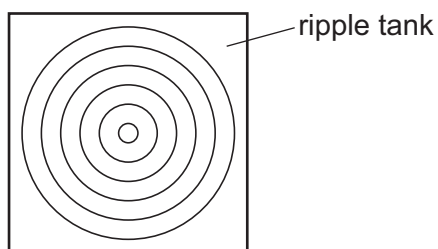
The distance between consecutive crests of the wave is 10 m.

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- A 0.056 m/s B 0.67 m/s C 15 m/s D 18 m/s

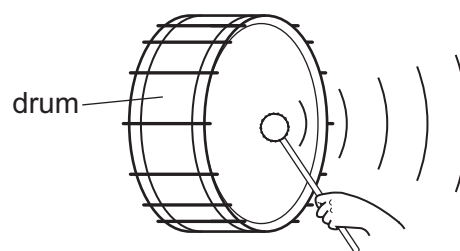
18 The diagrams show examples of wave motion.

waves on water



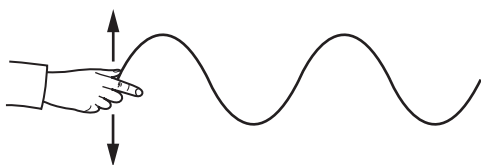
1

waves in air



2

waves on a rope



3

waves in a spring



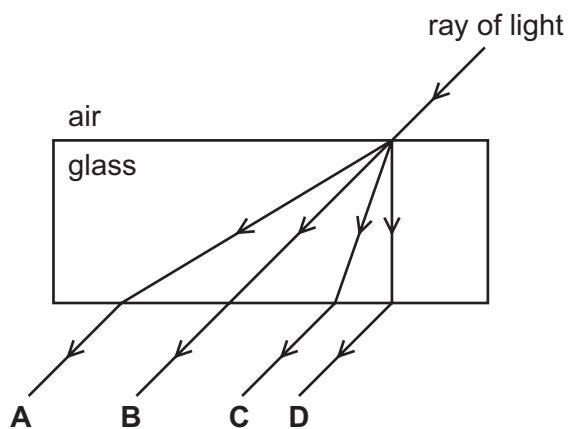
4

Which waves are longitudinal?

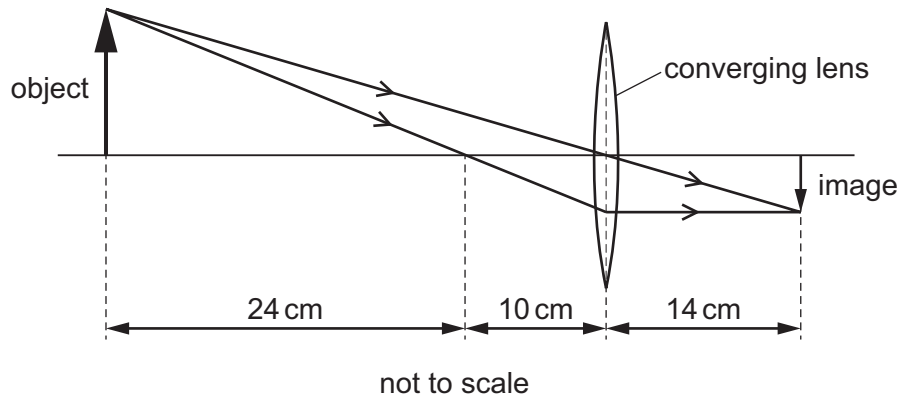
- A** 1 only **B** 2, 3 and 4 **C** 2 and 3 only **D** 2 and 4 only

19 A ray of light passes through a glass block.

What is the path of the light?



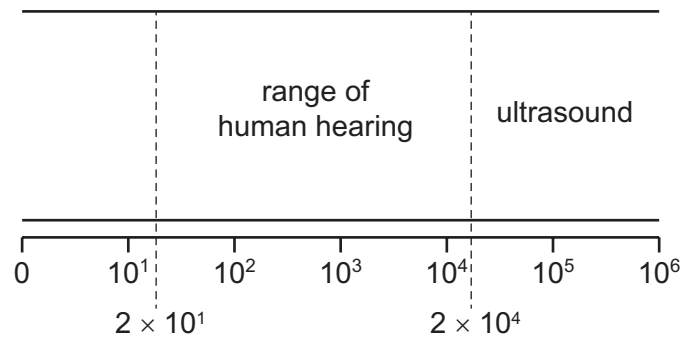
- 20 The diagram shows a converging lens forming an image.



What is the focal length of the converging lens?

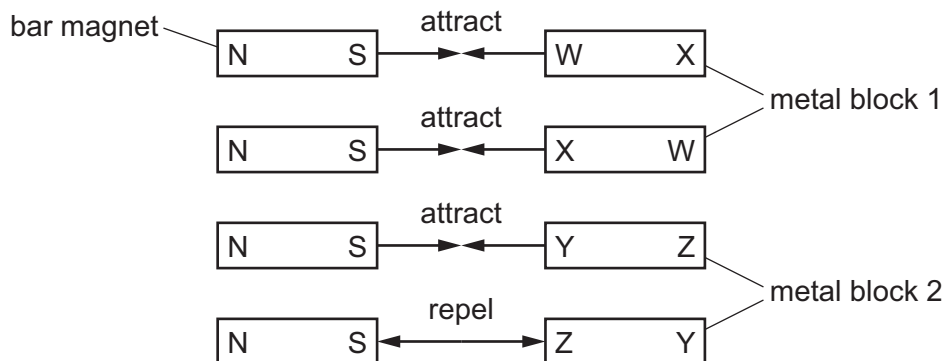
- A** 10 cm **B** 14 cm **C** 24 cm **D** 34 cm
- 21 Which type of radiation does a remote controller use to send its instructions to a television (TV)?
- A** gamma rays
B infrared radiation
C ultraviolet radiation
D X-rays
- 22 A person exposed to excessive radiation from certain parts of the electromagnetic spectrum is likely to suffer cell damage.
- Which radiations cause cell damage?
- A** gamma rays and ultraviolet
B gamma rays only
C green light and red light
D radio waves and green light

- 23 The diagram shows the ranges of human hearing and of ultrasound waves.



Which characteristic of sound waves do the numbers on the diagram refer to?

- A amplitude in cm
 - B frequency in Hz
 - C speed in metres/second
 - D wavelength in metres
- 24 A student has a bar magnet and two metal blocks.
- The student places the bar magnet next to end W and end X of metal block 1.
- The student then places the bar magnet next to end Y and end Z of metal block 2.
- The diagram shows which ends of the metal blocks are attracted or repelled by the bar magnet.



Which metal blocks are bar magnets?

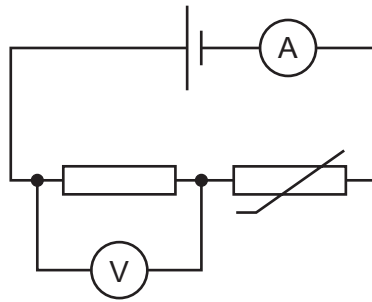
- A both metal block 1 and metal block 2
- B metal block 1 only
- C metal block 2 only
- D neither metal block 1 nor metal block 2

- 25** A glass rod is rubbed with a silk cloth. The glass rod becomes positively charged.

Which statement explains how this happens?

- A** Negative charges move from the glass rod to the silk cloth.
 - B** Negative charges move from the silk cloth to the glass rod.
 - C** Positive charges move from the glass rod to the silk cloth.
 - D** Positive charges move from the silk cloth to the glass rod.
- 26** Which statement describes what is happening when there is an electric current in a metal wire?
- A** Electrons in the wire vibrate.
 - B** Electrons move along the wire.
 - C** Positive charges in the wire vibrate.
 - D** Positive charges move along the wire.
- 27** Which **two** electrical quantities have the same unit?
- A** current and charge
 - B** electromotive force (e.m.f.) and potential difference (p.d.)
 - C** potential difference (p.d.) and current
 - D** resistance and electromotive force (e.m.f.)
- 28** A 10 V d.c. power supply is connected across a resistor.
- The current in the resistor is 0.050 A.
- What is the resistance of the resistor?
- A** $0.0050\ \Omega$ **B** $0.50\ \Omega$ **C** $2.0\ \Omega$ **D** $200\ \Omega$

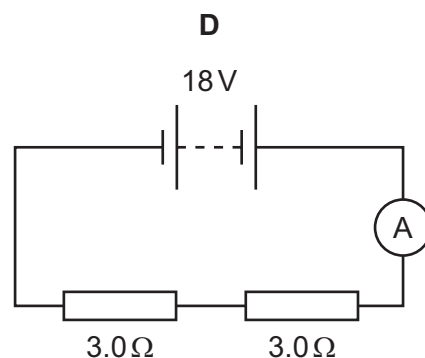
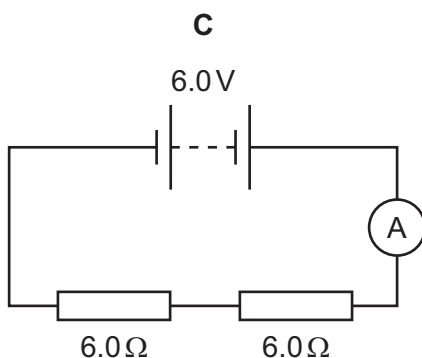
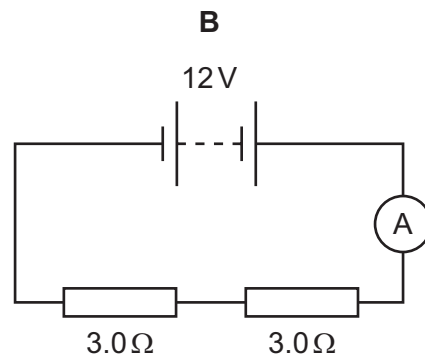
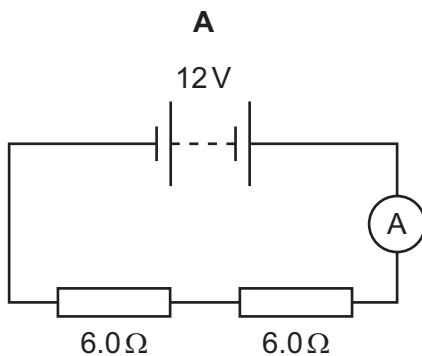
- 29 A resistor and a thermistor are connected in a series circuit, as shown.



What happens to the readings on the ammeter and the voltmeter when the thermistor is heated?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 30 In which circuit does the ammeter read 2.0 A ?



- 31 A 240 V mains-powered vacuum cleaner has a power output of 2700 W .

Which fuse is suitable for use in the plug of the vacuum cleaner?

- A** 3A **B** 5A **C** 10A **D** 13A

- 32 The transformer in a laptop power supply is used to change mains voltage from 240 V to 20 V.

The transformer has 600 turns on the secondary coil.

Which type of transformer is it and how many turns does the transformer have on the primary coil?

	transformer type	number of turns on primary coil
A	step-down	50
B	step-down	7200
C	step-up	50
D	step-up	7200

- 33 An ion of the element oxygen is negatively charged.

Which change causes the ion to become a neutral oxygen atom?

- A** gain of electrons
- B** gain of protons
- C** loss of electrons
- D** loss of protons

- 34 A nuclide of nitrogen is represented as $^{15}_7\text{N}$.

What is the number of neutrons in this nuclide?

- A** 7 **B** 8 **C** 15 **D** 22

- 35 Which list **only** includes sources of naturally occurring background radiation?

- A** cosmic rays, radon gas, nuclear power stations
- B** food and drink, mobile phones, cosmic rays
- C** rocks and buildings, food and drink, radon gas
- D** rocks and buildings, mobile phones, nuclear power stations

- 36** A student compares alpha emissions, beta emissions and gamma emissions.

Which statement is correct?

- A** Alpha emissions are the most ionising.
- B** Beta emissions are the most penetrating.
- C** Beta emissions are part of the electromagnetic spectrum.
- D** Gamma emissions are streams of charged particles.

- 37** In school laboratories, radioactive sources are stored in lead-lined boxes.

Why is lead used?

- A** Lead neutralises radiation.
- B** Lead absorbs radiation.
- C** Lead repels radiation.
- D** Lead dissolves radiation.

- 38** The Sun, the Earth and the Moon are all bodies in the Solar System.

The Earth has a periodic cycle of day and night and a periodic cycle of seasons.

Which row relates the periods of these cycles to the motions of these bodies?

	period of day and night	period of the seasons
A	the Earth orbits the Sun	the Earth rotates about its axis
B	the Earth orbits the Sun	the Moon orbits the Earth
C	the Earth rotates about its axis	the Earth orbits the Sun
D	the Earth rotates about its axis	the Moon orbits the Earth

- 39** The strength of the gravitational field at the surface of a planet depends on the1..... .

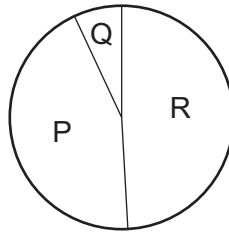
The strength of the gravitational field2..... as the distance from the planet increases.

Which words correctly complete the sentences?

	1	2
A	atmospheric pressure	decreases
B	mass of the planet	decreases
C	atmospheric pressure	increases
D	mass of the planet	increases

- 40** The Sun radiates most of its energy in three regions of the electromagnetic spectrum.

The diagram shows the proportion of the total energy radiated in each of these regions.



Region P is visible light and region Q is ultraviolet.

What is region R?

- A** infrared
- B** microwaves
- C** radio waves
- D** X-rays

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Cambridge IGCSE™

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0625/12

May/June 2025

45 minutes

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B mass
C volume
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The mass of each ball is shown.

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R	35
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Which balls will float in a liquid of density 0.95 g/cm^3 ?

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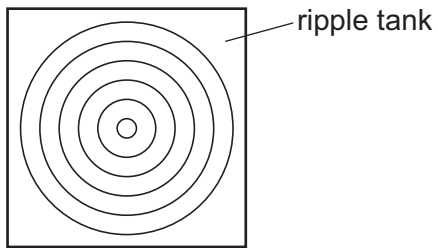
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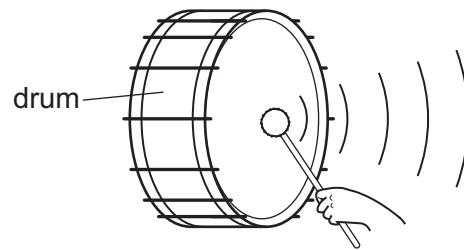
18 The diagrams show examples of wave motion.

waves on water



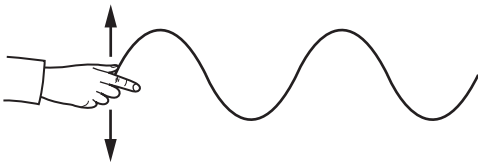
1

waves in air



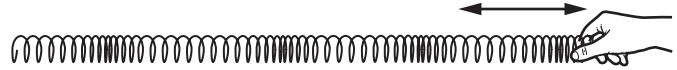
2

waves on a rope



3

waves in a spring



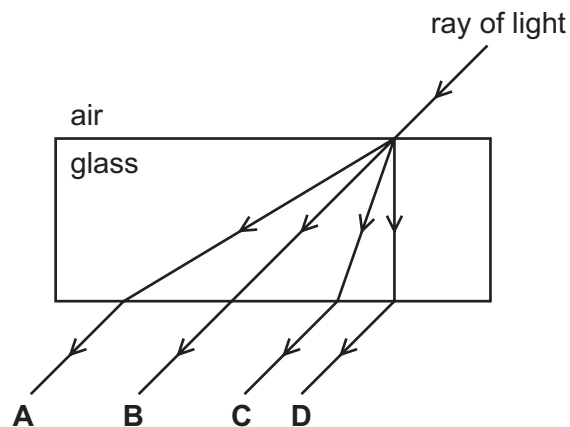
4

Which waves are longitudinal?

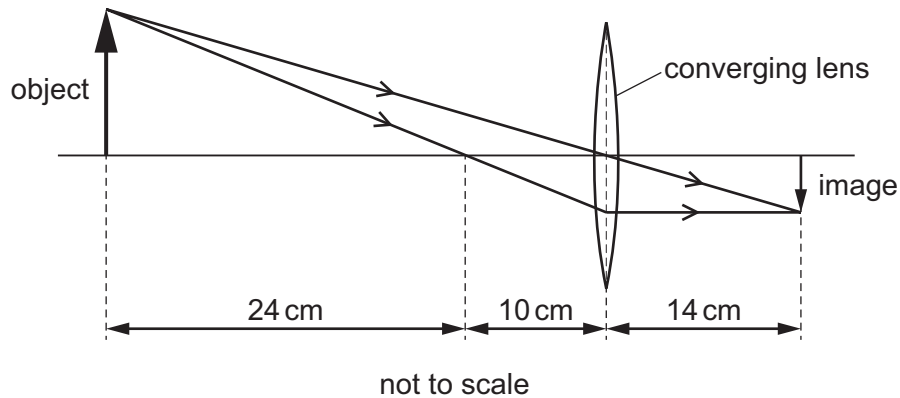
- A** 1 only **B** 2, 3 and 4 **C** 2 and 3 only **D** 2 and 4 only

19 A ray of light passes through a glass block.

What is the path of the light?



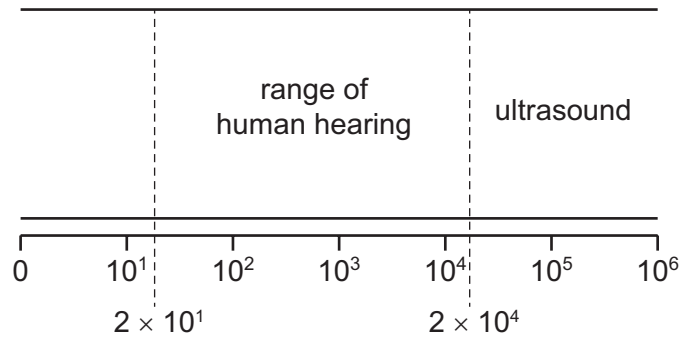
- 20 The diagram shows a converging lens forming an image.



What is the focal length of the converging lens?

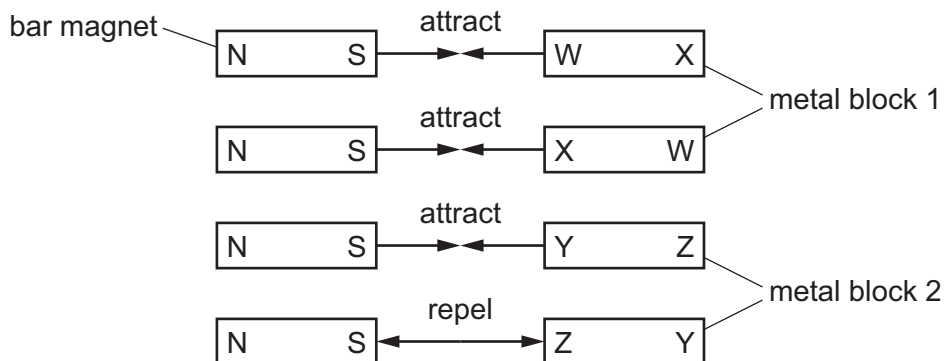
- A** 10 cm **B** 14 cm **C** 24 cm **D** 34 cm
- 21 Which type of radiation does a remote controller use to send its instructions to a television (TV)?
- A** gamma rays
B infrared radiation
C ultraviolet radiation
D X-rays
- 22 A person exposed to excessive radiation from certain parts of the electromagnetic spectrum is likely to suffer cell damage.
- Which radiations cause cell damage?
- A** gamma rays and ultraviolet
B gamma rays only
C green light and red light
D radio waves and green light

- 23 The diagram shows the ranges of human hearing and of ultrasound waves.



Which characteristic of sound waves do the numbers on the diagram refer to?

- A amplitude in cm
 - B frequency in Hz
 - C speed in metres/second
 - D wavelength in metres
- 24 A student has a bar magnet and two metal blocks.
- The student places the bar magnet next to end W and end X of metal block 1.
- The student then places the bar magnet next to end Y and end Z of metal block 2.
- The diagram shows which ends of the metal blocks are attracted or repelled by the bar magnet.



Which metal blocks are bar magnets?

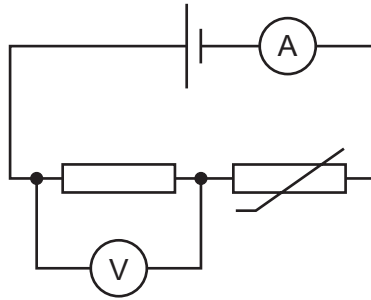
- A both metal block 1 and metal block 2
- B metal block 1 only
- C metal block 2 only
- D neither metal block 1 nor metal block 2

- 25** A glass rod is rubbed with a silk cloth. The glass rod becomes positively charged.

Which statement explains how this happens?

- A** Negative charges move from the glass rod to the silk cloth.
 - B** Negative charges move from the silk cloth to the glass rod.
 - C** Positive charges move from the glass rod to the silk cloth.
 - D** Positive charges move from the silk cloth to the glass rod.
- 26** Which statement describes what is happening when there is an electric current in a metal wire?
- A** Electrons in the wire vibrate.
 - B** Electrons move along the wire.
 - C** Positive charges in the wire vibrate.
 - D** Positive charges move along the wire.
- 27** Which **two** electrical quantities have the same unit?
- A** current and charge
 - B** electromotive force (e.m.f.) and potential difference (p.d.)
 - C** potential difference (p.d.) and current
 - D** resistance and electromotive force (e.m.f.)
- 28** A 10 V d.c. power supply is connected across a resistor.
- The current in the resistor is 0.050 A.
- What is the resistance of the resistor?
- A** $0.0050\ \Omega$ **B** $0.50\ \Omega$ **C** $2.0\ \Omega$ **D** $200\ \Omega$

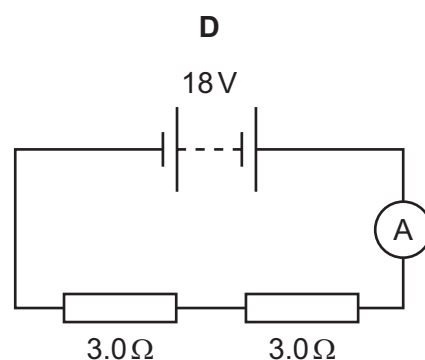
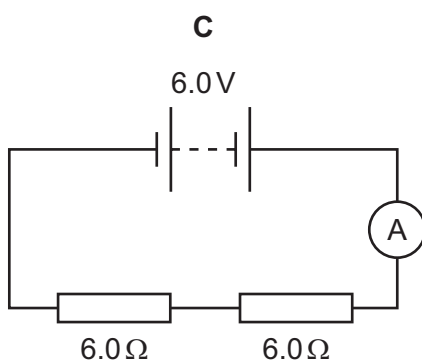
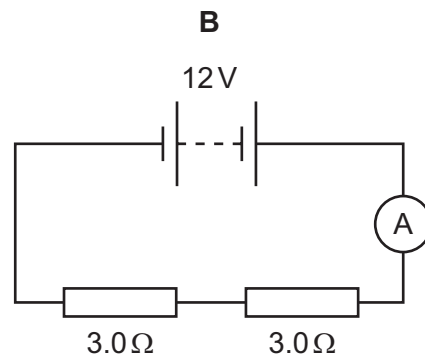
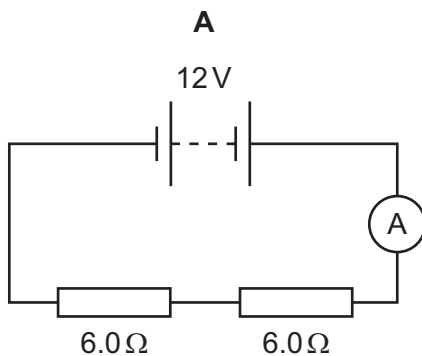
- 29 A resistor and a thermistor are connected in a series circuit, as shown.



What happens to the readings on the ammeter and the voltmeter when the thermistor is heated?

	ammeter reading	voltmeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 30 In which circuit does the ammeter read 2.0 A?



- 31 A 240 V mains-powered vacuum cleaner has a power output of 2700 W.

Which fuse is suitable for use in the plug of the vacuum cleaner?

- A** 3 A **B** 5 A **C** 10 A **D** 13 A

- 32 The transformer in a laptop power supply is used to change mains voltage from 240 V to 20 V.

The transformer has 600 turns on the secondary coil.

Which type of transformer is it and how many turns does the transformer have on the primary coil?

	transformer type	number of turns on primary coil
A	step-down	50
B	step-down	7200
C	step-up	50
D	step-up	7200

- 33 An ion of the element oxygen is negatively charged.

Which change causes the ion to become a neutral oxygen atom?

- A** gain of electrons
- B** gain of protons
- C** loss of electrons
- D** loss of protons

- 34 A nuclide of nitrogen is represented as $^{15}_7\text{N}$.

What is the number of neutrons in this nuclide?

- A** 7 **B** 8 **C** 15 **D** 22

- 35 Which list **only** includes sources of naturally occurring background radiation?

- A** cosmic rays, radon gas, nuclear power stations
- B** food and drink, mobile phones, cosmic rays
- C** rocks and buildings, food and drink, radon gas
- D** rocks and buildings, mobile phones, nuclear power stations

- 36** A student compares alpha emissions, beta emissions and gamma emissions.

Which statement is correct?

- A** Alpha emissions are the most ionising.
- B** Beta emissions are the most penetrating.
- C** Beta emissions are part of the electromagnetic spectrum.
- D** Gamma emissions are streams of charged particles.

- 37** In school laboratories, radioactive sources are stored in lead-lined boxes.

Why is lead used?

- A** Lead neutralises radiation.
- B** Lead absorbs radiation.
- C** Lead repels radiation.
- D** Lead dissolves radiation.

- 38** The Sun, the Earth and the Moon are all bodies in the Solar System.

The Earth has a periodic cycle of day and night and a periodic cycle of seasons.

Which row relates the periods of these cycles to the motions of these bodies?

	period of day and night	period of the seasons
A	the Earth orbits the Sun	the Earth rotates about its axis
B	the Earth orbits the Sun	the Moon orbits the Earth
C	the Earth rotates about its axis	the Earth orbits the Sun
D	the Earth rotates about its axis	the Moon orbits the Earth

- 39** The strength of the gravitational field at the surface of a planet depends on the1..... .

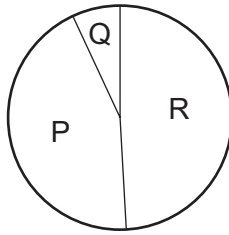
The strength of the gravitational field2..... as the distance from the planet increases.

Which words correctly complete the sentences?

	1	2
A	atmospheric pressure	decreases
B	mass of the planet	decreases
C	atmospheric pressure	increases
D	mass of the planet	increases

- 40** The Sun radiates most of its energy in three regions of the electromagnetic spectrum.

The diagram shows the proportion of the total energy radiated in each of these regions.



Region P is visible light and region Q is ultraviolet.

What is region R?

- A** infrared
- B** microwaves
- C** radio waves
- D** X-rays

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/11

May/June 2025

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

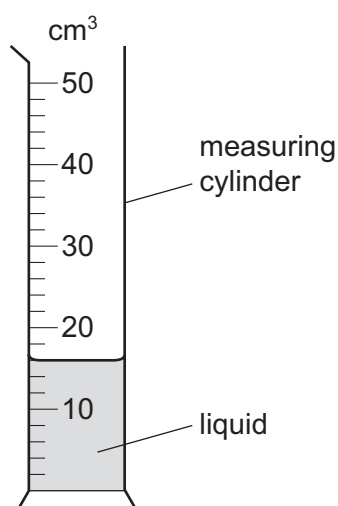
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 The diagram shows a measuring cylinder containing liquid.



What is the volume of liquid in the measuring cylinder?

- A** 13.0 cm³ **B** 13.5 cm³ **C** 16.0 cm³ **D** 17.0 cm³
- 2 Which statement about the speed and velocity of an object is correct?
- A** The speed and velocity of an object are always greater than zero.
- B** The speed of an object is always greater than its velocity.
- C** The velocity of an object is equal to its speed in a given direction.
- D** Speed can be measured in m/s but velocity can only be measured in km/h.
- 3 The table shows a student's times in three races.

race	distance / m	time / s
1	100	12
2	150	17
3	200	25

What is the order of the student's average speeds in each race, from lowest to highest?

- A** 1 → 2 → 3 **B** 1 → 3 → 2 **C** 2 → 1 → 3 **D** 3 → 1 → 2

- 4 Which row gives the correct weight for the mass shown when it is at the Earth's surface?

	mass / kg	weight / N
A	2.00	19.6
B	10.0	1.02
C	10.0	9.80
D	20.0	2.04

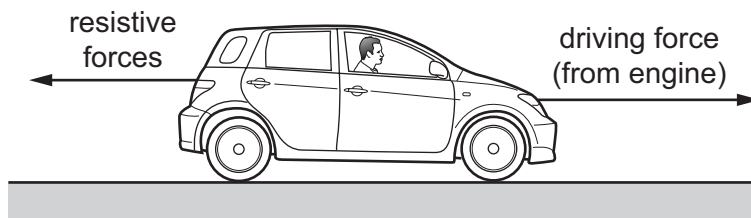
- 5 The masses and volumes of four objects are shown.

Which object floats in a liquid of density 1.4 g/cm^3 ?

	mass of object / g	volume of object / cm^3
A	1.0	0.5
B	2.0	3.0
C	6.0	4.0
D	9.0	6.0

- 6 A car is driven from rest on a long straight road. The car engine exerts a constant driving force.

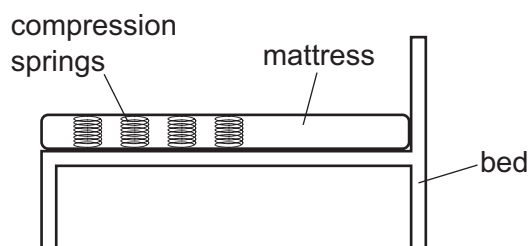
The diagram shows the horizontal forces acting on the car. The resistive forces are proportional to the speed of the car.



Why does the car eventually reach a maximum speed?

- A** The resistive forces decrease to make the acceleration of the car negative.
- B** The resistive forces decrease to make the acceleration of the car zero.
- C** The resistive forces increase to make the acceleration of the car negative.
- D** The resistive forces increase to make the acceleration of the car zero.

- 7 A mattress placed on a bed contains many compression springs.



Which energy store of the mattress **increases** when a person lies on the bed?

- A gravitational potential
 - B chemical
 - C elastic
 - D kinetic
- 8 Which energy resource does **not** contain a store of chemical energy?
- A biofuel
 - B coal
 - C nuclear fuel
 - D oil
- 9 A person carries 20 tiles from the ground to the roof of a house. Each tile has a mass of 1.2 kg. The roof of the house is 15 m above the ground.

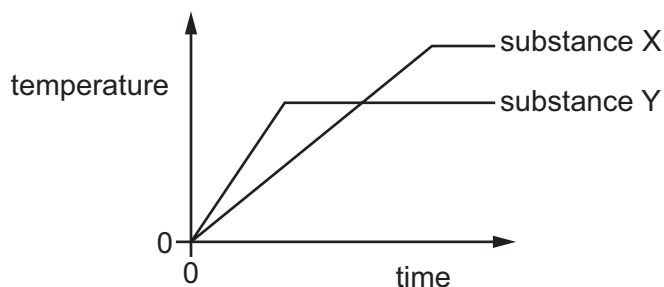
How much work does the person do against gravity on the tiles in carrying them to the roof?

- A 37 J B 180 J C 360 J D 3500 J
- 10 Which pair of changes, applied at the same time, always increases the pressure that a brick exerts on the ground?

	weight of brick	area of brick in contact with ground
A	increase	increase
B	increase	decrease
C	decrease	increase
D	decrease	decrease

- 11** Two different pure substances, X and Y, are heated. Both substance X and substance Y are initially in the solid state.

The graph shows how the temperature of each substance changes with time.



What does the graph tell us about the substances?

- A** Substance X has a lower boiling point than substance Y.
 - B** Substance X has a lower melting point than substance Y.
 - C** Substance Y has a lower boiling point than substance X.
 - D** Substance Y has a lower melting point than substance X.
- 12** A liquid substance changes into a gas.

Which property of the substance remains the same?

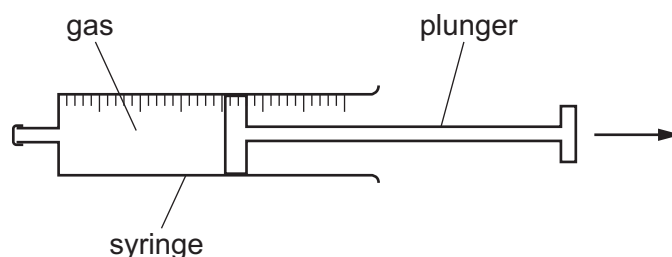
- A** ability to flow
 - B** compressibility
 - C** shape
 - D** volume
- 13** At room temperature, iron is difficult to compress.

At the same temperature, oxygen is much easier to compress.

Which comparison of the particles in iron and the particles in oxygen explains this?

- A** The iron particles are closer together.
- B** The iron particles have a greater mass.
- C** The iron particles can be magnetised.
- D** The iron particles have less average kinetic energy.

- 14 A teacher has a gas in a sealed syringe with a movable plunger.



The teacher slowly moves the plunger out of the syringe. The temperature of the gas remains constant.

Which row explains what happens to the pressure of the gas in the syringe?

	pressure of gas	explanation
A	decreases	gas particles collide with the syringe wall less frequently
B	decreases	gas particles collide with the syringe wall more frequently
C	increases	gas particles collide with the syringe wall less frequently
D	increases	gas particles collide with the syringe wall more frequently

- 15 Evaporation occurs from the surface of a pool of water.

Which statement describes this change of state?

- A** Electrons move from the liquid and become a gas.
- B** Molecules that move from the liquid have the same energy as the molecules that stay in the liquid.
- C** The more-energetic molecules escape the liquid.
- D** The more-energetic molecules remain in the liquid.

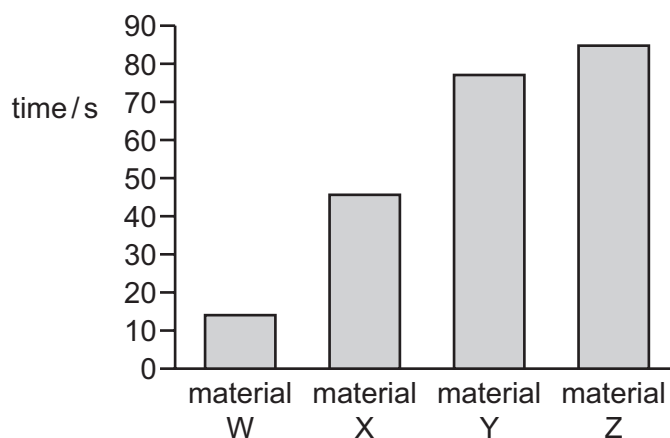
- 16** Students test four rods of different materials, W, X, Y and Z, to compare how well the materials conduct thermal energy.

The rods are all the same size and shape.

The students use wax to stick a pin to one end of the first rod. The wax melts when it is warm. The other end of the rod is heated and the time taken for the pin to fall is recorded.

The experiment is repeated for each rod.

The students plot a chart of the time taken for each pin to fall.



Which statement is correct?

- A** X is a better thermal conductor than Y.
 - B** Y is a better thermal conductor than W.
 - C** Z is a better thermal conductor than W.
 - D** Z is a better thermal conductor than X.
- 17** A student has four metal containers filled with the same volume of water at a temperature of 80 °C.

The metal containers are put in a room at a temperature of 20 °C.

The metal containers are identical except for the texture and colour of their surfaces.

Which surface will emit energy at the greatest rate?

- A** dull black
- B** shiny black
- C** dull white
- D** shiny white

- 18 A water wave has a speed of 2.0 m/s.

4.0 complete waves pass a point every 10 seconds.

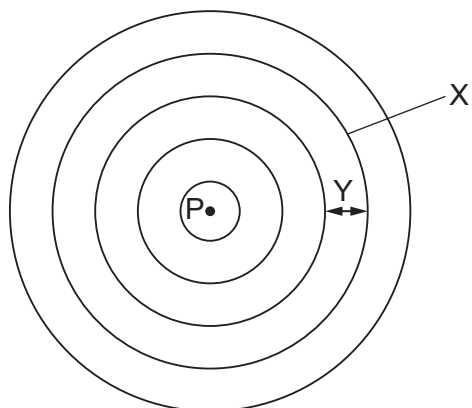
What is the wavelength of the wave?

- A** 0.50 m **B** 0.80 m **C** 5.0 m **D** 8.0 m

- 19 Which row correctly describes the vibrations of a transverse wave and also gives a correct example of a transverse wave?

	description of vibrations	example of a transverse wave
A	at right angles to the wave direction	sound
B	at right angles to the wave direction	water wave
C	parallel to the wave direction	sound
D	parallel to the wave direction	water wave

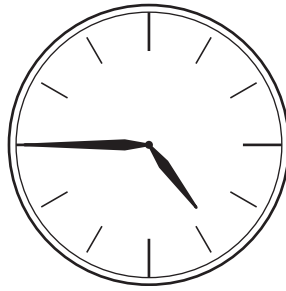
- 20 The diagram shows a view from above of waves on the surface of a pond after the water has been disturbed at point P.



What are the correct labels for X and Y?

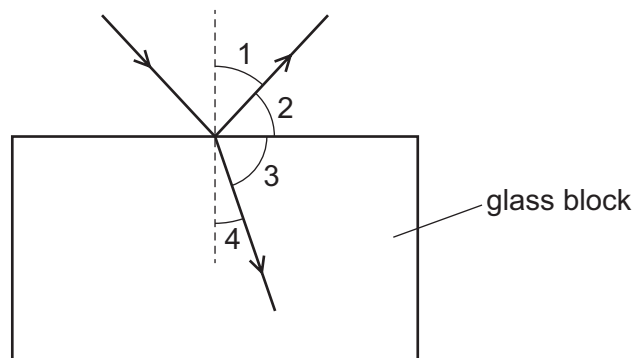
	X	Y
A	amplitude	wavelength
B	wavefront	wavelength
C	amplitude	wavefront
D	wavefront	amplitude

- 21 The diagram shows the image of a clock as seen by a person standing in front of a plane mirror.



What is the actual time?

- A** 4:15 **B** 4:45 **C** 7:15 **D** 7:45
- 22 The diagram shows a ray of light incident on the surface of a glass block.



The ray of light is partially reflected back into the air and partially refracted into the glass block.

Which row correctly identifies the angle of reflection and the angle of refraction?

	angle of reflection	angle of refraction
A	1	3
B	1	4
C	2	3
D	2	4

- 23** The amplitude and the frequency of a sound wave both increase.

What happens to the loudness and what happens to the pitch of the sound?

	loudness	pitch
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 24** Which statements about sound and ultrasound are correct?

	sound	ultrasound
A	humans can hear frequencies between 20 Hz and 20 000 Hz	the speed in air is 330 m/s
B	humans can hear frequencies less than 20 000 Hz	the speed in air is 100 m/s
C	the speed in air is 100 m/s	has a frequency greater than 20 000 Hz
D	the speed in air is 330 m/s	has a frequency less than 20 Hz

- 25** Which row describes the magnetic properties of soft iron and steel?

	soft iron	steel
A	easy to demagnetise	easy to demagnetise
B	easy to demagnetise	hard to demagnetise
C	hard to demagnetise	easy to demagnetise
D	hard to demagnetise	hard to demagnetise

- 26 Two charged rods, X and Y, are brought, in turn, near to a negatively charged rod.

X is attracted by the negatively charged rod, and Y is repelled.

What are the charges on X and Y?

	charge on X	charge on Y
A	negative	negative
B	negative	positive
C	positive	negative
D	positive	positive

- 27 The table gives data for four different electrical devices.

Which device develops the greatest power?

	device	voltage	current
A	car headlight	12 V	3.0 A
B	cooling fan	110 V	0.40 A
C	electric spark generator	400 kV	0.10 mA
D	mains lamp	240 V	0.20 A

- 28 A mains-operated radio is a double-insulated appliance.

Which statement is correct?

- A** The radio does **not** need a neutral connection.
- B** The radio **must** be earthed.
- C** The radio circuit is safe **without** a fuse or trip switch.
- D** The outer casing of the radio **cannot** become live.

- 29 Which statement describes a magnetic field?

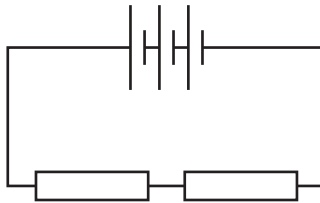
- A** It is a region where all objects experience a force.
- B** It is a region where all electric charges experience a force.
- C** It is a region where all magnetic poles experience a force.
- D** It is a region where all metals experience a force.

- 30 A plastic rod is charged by friction. The plastic rod becomes positively charged.

Which particles are transferred in this process?

- A atoms
- B electrons
- C protons
- D neutrons

- 31 A teacher makes a circuit using three 1.5 V cells and two $20\ \Omega$ resistors connected in series.



Which row gives the combined electromotive force (e.m.f.) of the cells and the combined resistance of the resistors?

	combined e.m.f. / V	combined resistance / Ω
A	1.5	10
B	1.5	40
C	4.5	10
D	4.5	40

- 32 A student investigates the turning effect on a current-carrying coil placed in a magnetic field.

The student changes the current in the coil and the number of turns on the coil.

Which two changes together **must** increase the turning effect?

- A decreasing the current and decreasing the number of turns
- B decreasing the current and increasing the number of turns
- C increasing the current and decreasing the number of turns
- D increasing the current and increasing the number of turns

- 33 There is a current of 3.0 A in a conductor.

The current is changed to 6.0 A in the opposite direction.

Which effect does this have on the magnetic field around the conductor?

- A It decreases and it acts in the opposite direction.
- B It decreases and it acts in the same direction.
- C It increases and it acts in the opposite direction.
- D It increases and it acts in the same direction.

- 34 The table shows the contribution to background radiation from the four most common sources.

source	percentage of total background radiation / %
radon gas (in the air)	48
rocks and buildings	13
food and drink	10
cosmic rays	11

Which percentage of background radiation comes from all other sources?

- A 14% B 18% C 21% D 82%

- 35 The unstable nucleus of an element undergoes radioactive decay. The proton number remains unchanged.

Which description of this radioactive decay event is correct?

- A The nucleus emits an α -particle.
- B The nucleus emits a β -particle.
- C The nucleus emits γ -radiation.
- D The nucleus emits an α -particle, a β -particle and γ -radiation together.

- 36 When they lose electrons,1..... form2..... charged3..... .

Which words correctly complete the sentence?

	1	2	3
A	atoms	negatively	ions
B	ions	negatively	atoms
C	atoms	positively	ions
D	ions	positively	atoms

- 37 An isotope of krypton has an atomic number of 36 and a mass number of 78.

What is the number of protons in a krypton nucleus?

- A** 36 **B** 42 **C** 78 **D** 114

- 38 What is the time taken for the Earth to spin once on its axis and the time taken for the Earth to orbit the Sun?

	spin on its axis	orbit the Sun
A	24 hours	1 month
B	24 hours	365 days
C	1 month	1 month
D	1 month	365 days

- 39 Electromagnetic radiation from distant galaxies is redshifted when observed from the Earth.

This is evidence that supports the Big Bang Theory.

Which statement is correct?

- A** Light from galaxies that are moving towards the Earth is redshifted.
B Redshift is an increase in the wavelength of electromagnetic radiation.
C The Big Bang Theory says that the Solar System is expanding out into the Universe.
D The Big Bang Theory says that, billions of years ago, the Universe expanded to its current size in a few seconds.

- 40** Which statement about the Milky Way is correct?
- A** It is a galaxy made up of billions of stars.
 - B** It is a group of galaxies in our Solar System.
 - C** It is all the stars that we can see in the night sky.
 - D** It is the path taken by a comet when it gets close to the Sun.

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PHYSICS**0625/13**

Paper 1 Multiple Choice (Core)

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	C	1
4	D	1
5	A	1
6	D	1
7	C	1
8	D	1
9	A	1
10	D	1
11	B	1
12	B	1
13	D	1
14	A	1
15	A	1
16	C	1
17	B	1
18	D	1
19	C	1
20	A	1
21	B	1
22	A	1
23	B	1
24	C	1
25	A	1
26	B	1
27	B	1
28	D	1

Question	Answer	Marks
29	D	1
30	B	1
31	D	1
32	B	1
33	C	1
34	B	1
35	C	1
36	A	1
37	B	1
38	C	1
39	B	1
40	A	1

Cambridge IGCSE™

PHYSICS**0625/12**

Paper 1 Multiple Choice (Core)

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2025 series for most Cambridge IGCSE, Cambridge International A and AS Level components, and some Cambridge O Level components.

This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	B	1
3	C	1
4	D	1
5	A	1
6	D	1
7	C	1
8	D	1
9	A	1
10	D	1
11	B	1
12	B	1
13	D	1
14	A	1
15	A	1
16	C	1
17	B	1
18	D	1
19	C	1
20	A	1
21	B	1
22	A	1
23	B	1
24	C	1
25	A	1
26	B	1
27	B	1
28	D	1

Question	Answer	Marks
29	D	1
30	B	1
31	D	1
32	B	1
33	C	1
34	B	1
35	C	1
36	A	1
37	B	1
38	C	1
39	B	1
40	A	1

Cambridge IGCSE™

PHYSICS**0625/11**

Paper 1 Multiple Choice (Core)

May/June 2025

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	C	1
2	C	1
3	D	1
4	A	1
5	B	1
6	D	1
7	C	1
8	C	1
9	D	1
10	B	1
11	D	1
12	A	1
13	A	1
14	A	1
15	C	1
16	A	1
17	A	1
18	C	1
19	B	1
20	B	1
21	C	1
22	B	1
23	D	1
24	A	1
25	B	1
26	C	1
27	D	1
28	D	1

Question	Answer	Marks
29	C	1
30	B	1
31	D	1
32	D	1
33	C	1
34	B	1
35	C	1
36	C	1
37	A	1
38	B	1
39	B	1
40	A	1



Cambridge IGCSE™

PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

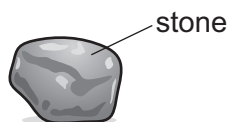
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 A student wishes to find the volume of a small, irregularly shaped stone.



A ruler and a measuring cylinder containing some water are available.

Which apparatus is needed?

- A neither the ruler nor the measuring cylinder
 - B the measuring cylinder only
 - C the ruler and the measuring cylinder
 - D the ruler only
- 2 A car travels along a road at a constant speed.

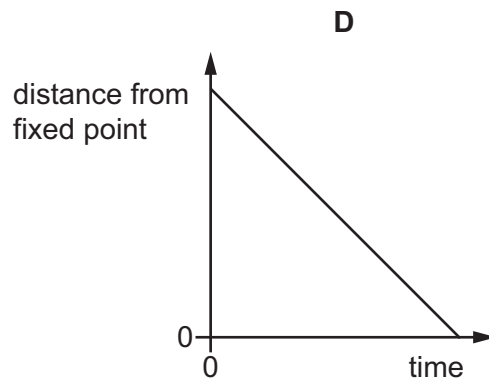
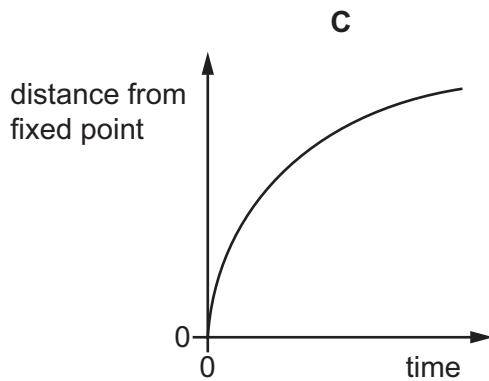
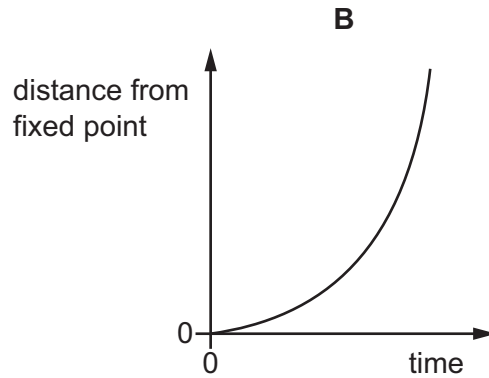
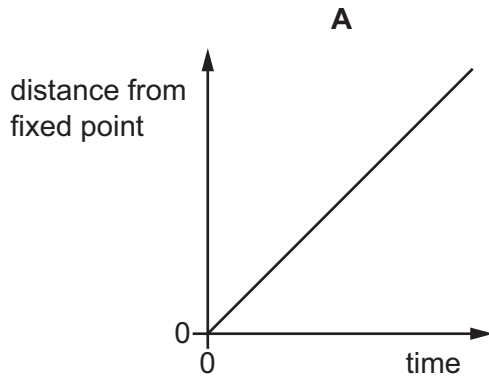
What correctly describes the speed–time graph of the car?

- A a horizontal line
- B a vertical line
- C a diagonal line upwards
- D a curved line with increasing gradient

3 Four objects are moving along a straight line.

The distance of each object from a fixed point on the line is plotted against time.

Which object is decelerating?



4 An object is taken from the Earth to the Moon.

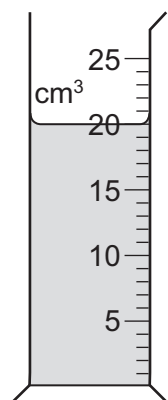
The acceleration of free fall is 9.8 m/s^2 on the Earth and is 1.6 m/s^2 on the Moon.

Which row describes the mass and the weight of the object on the Moon compared to on the Earth?

	mass	weight
A	greater than on the Earth	same as on the Earth
B	less than on the Earth	same as on the Earth
C	same as on the Earth	greater than on the Earth
D	same as on the Earth	less than on the Earth

- 5 The diagram shows some liquid in a measuring cylinder.

The mass of the liquid is 16 g.



What is the density of the liquid?

- A** 0.80 g/cm^3 **B** 1.3 g/cm^3 **C** 36 g/cm^3 **D** 320 g/cm^3
- 6 A single force acts upon an object and the volume of the object does **not** change.
- Which statement is correct?
- A** The density of the object changes.
B The velocity of the object changes.
C The mass of the object changes.
D The weight of the object changes.
- 7 Four electric motors do the same quantity of work when they lift a weight of 10 N through the same distance.

The times taken by each motor to lift the weight are shown.

What is the time taken by the motor which produces the most power?

- A** 1.5 s **B** 6.0 s **C** 15 s **D** 60 s

- 8 Which row describes how the pressure beneath the surface of a liquid depends on the depth below the surface and on the density of the liquid?

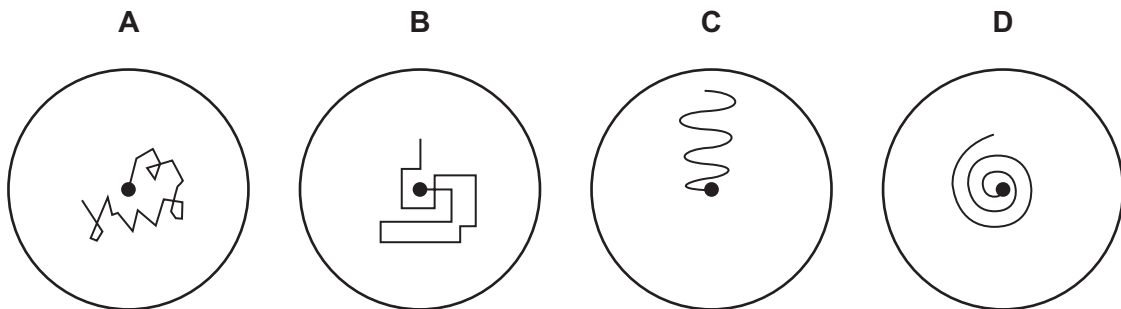
	depth below surface	density of liquid
A	pressure decreases with increasing depth	pressure decreases with increasing density
B	pressure decreases with increasing depth	pressure increases with increasing density
C	pressure increases with increasing depth	pressure decreases with increasing density
D	pressure increases with increasing depth	pressure increases with increasing density

- 9 In which process does a gas turn into a liquid?

- A** boiling
B condensing
C freezing
D melting

- 10 A pollen grain in a beaker of still water is viewed through a microscope.

Which diagram shows the most likely movement of the pollen grain?



- 11 What is the relationship between temperatures in kelvin (T) and temperatures in degrees Celsius (θ)?

- A** $\theta = T + 273$ **B** $\theta = T - 273$ **C** $\theta = T + 373$ **D** $\theta = T - 373$

- 12 In a famous experiment, a scientist showed that, when water in an insulated tub is stirred, its temperature increases.

What can be concluded from this experiment?

- A Temperature is a store of energy.
 - B The water absorbs heat from the surroundings.
 - C Work done by stirring increases the internal energy of the water.
 - D Energy is not conserved.
- 13 Solids expand when heated and contract when cooled.

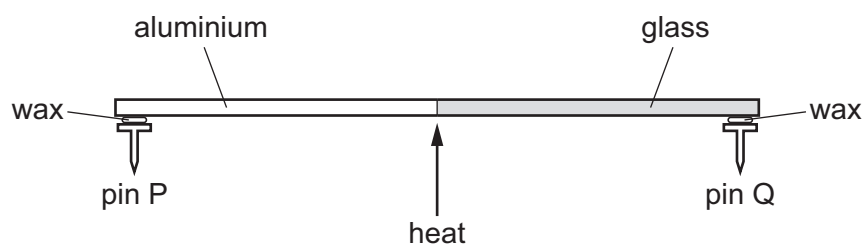
In which example is there a useful application of expansion and **not** an unwanted effect?

- A bridges
 - B railway lines
 - C telephone cables
 - D thermostats
- 14 A rod is made half of aluminium and half of glass.

A pin is attached to each end of the rod by wax.

The rod is heated at the centre.

A pin falls off when the wax melts.



Which statement is correct?

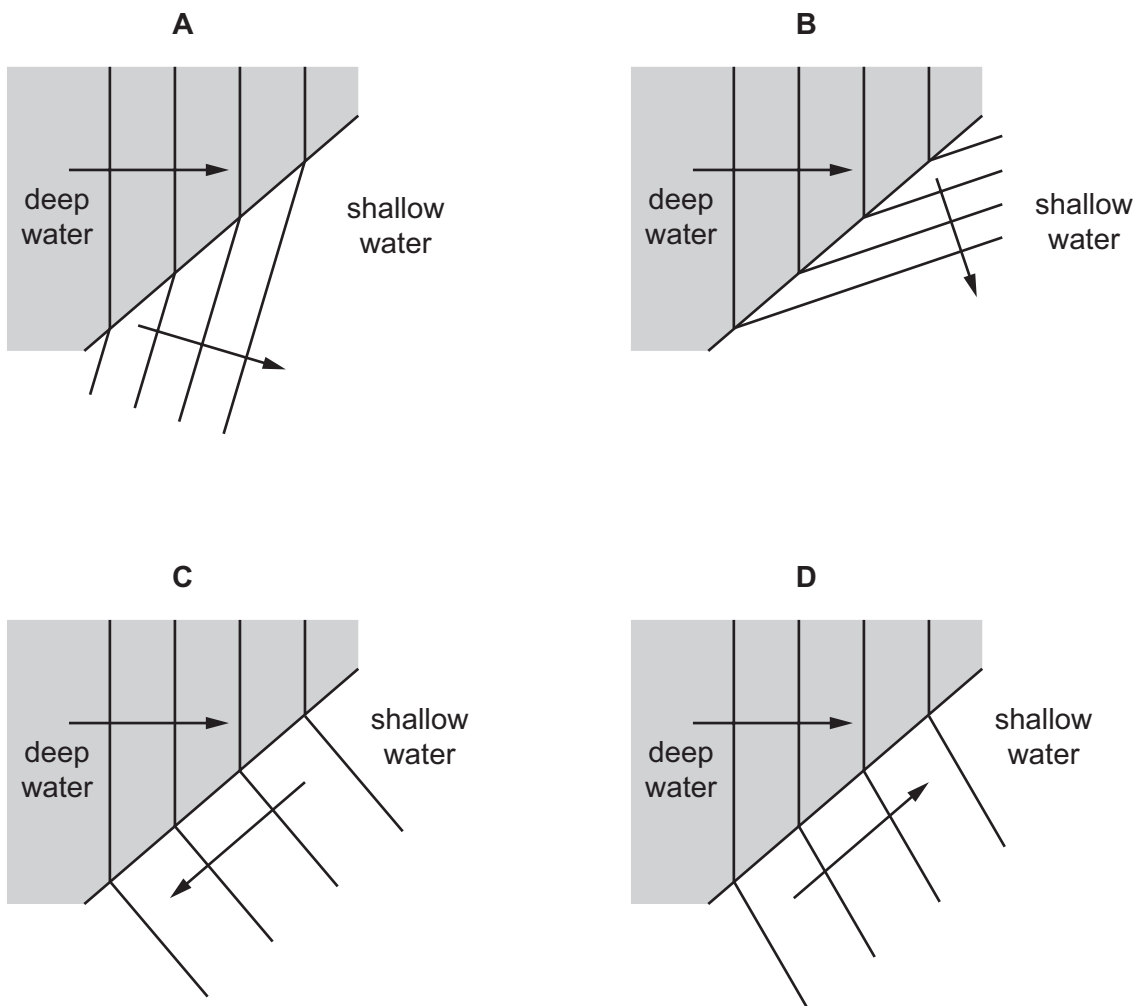
- A Pin P falls off first because aluminium is a better thermal conductor than glass.
- B Pin P falls off first because aluminium is a worse thermal conductor than glass.
- C Pin Q falls off first because glass is a better thermal conductor than aluminium.
- D Pin Q falls off first because glass is a worse thermal conductor than aluminium.

- 15 In countries where it is usually hot, houses are often painted white.

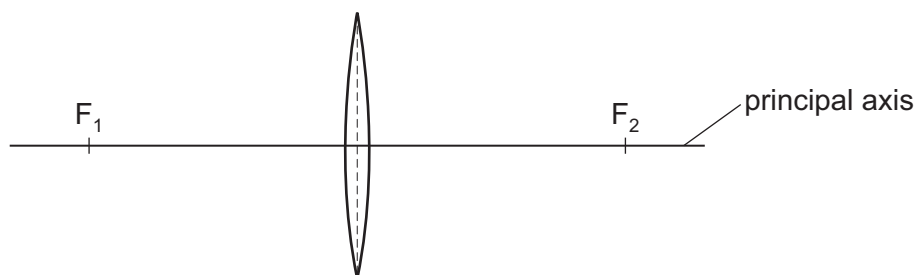
What is the reason for this?

- A White surfaces are good reflectors of infrared radiation.
 - B White surfaces are good conductors of infrared radiation.
 - C White surfaces are good absorbers of infrared radiation.
 - D White surfaces are good emitters of infrared radiation.
- 16 What is the name of the distance from one crest in a transverse wave to the next crest?
- A amplitude
 - B period
 - C wavefront
 - D wavelength
- 17 Waves on the surface of water travel from deep water to shallow water.

Which diagram shows the correct path of the waves in the shallow water?



- 18 The diagram shows a thin converging lens with principal focuses at F_1 and F_2 .



A small light source is placed at F_1 . A beam of light from the source passes through the lens.

Which statement correctly describes the beam of light emerging from the lens?

- A The beam converges to F_2 .
 - B The beam converges to a point to the right of F_2 .
 - C The beam diverges from a point to the left of F_1 .
 - D The beam travels parallel to the principal axis.
- 19 A student reads the following relationship in his physics book.

$$i = r$$

What is the student reading about?

- A diffraction due to a gap
- B dispersion of light by a prism
- C reflection in a plane mirror
- D refraction as light enters glass

- 20 The table shows information about different colours of light.

colour of light	frequency / Hz
violet	7.2×10^{14}
blue	6.3×10^{14}
yellow	5.2×10^{14}
red	4.5×10^{14}

Using the data, what is the frequency of orange light?

- A** 4.0×10^{14} Hz
B 5.0×10^{14} Hz
C 6.0×10^{14} Hz
D 7.0×10^{14} Hz

- 21 The table gives the wavelengths of three electromagnetic waves.

electromagnetic wave	wavelength / m
P	3.2×10^{-3}
Q	5.0×10^{-9}
R	2.6×10^3

Which row correctly identifies P, Q and R?

	P	Q	R
A	microwave	radio wave	X-ray
B	microwave	X-ray	radio wave
C	radio wave	microwave	X-ray
D	radio wave	X-ray	microwave

- 22 A healthy human ear is able to hear a range of frequencies.

What is this approximate range?

- A** 10 Hz to 1000 Hz
B 20 Hz to 2000 Hz
C 20 Hz to 20 000 Hz
D 200 Hz to 200 000 Hz

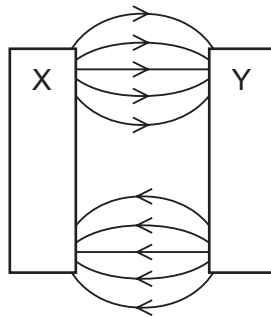
- 23** A student hits two wooden blocks together in front of a wall and calculates the speed of sound to be 340 m/s .

The time between the student hitting the blocks and hearing the echo is 0.59 s .



What is the distance between the student and the wall?

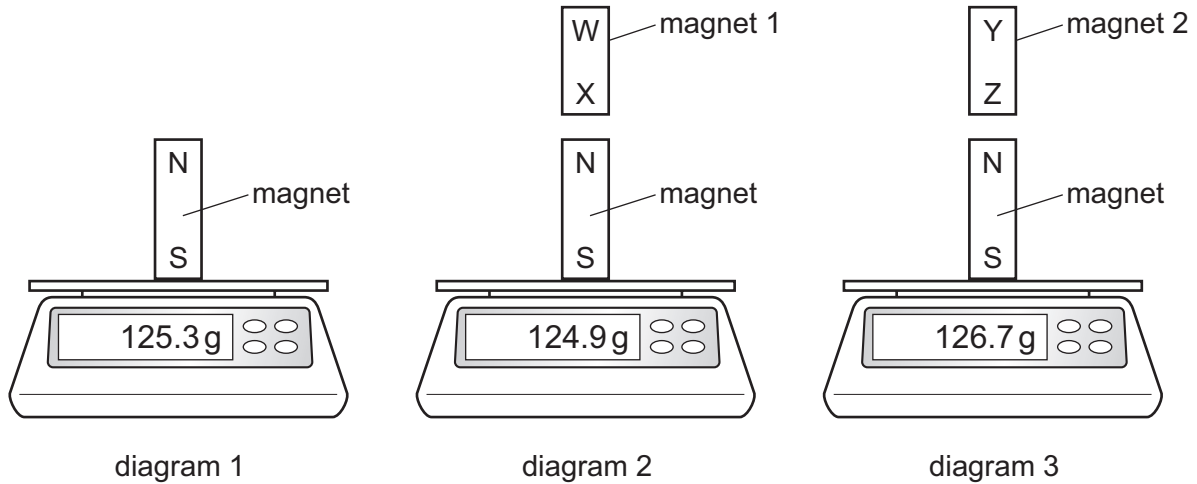
- A** 100 m **B** 200 m **C** 290 m **D** 570 m
- 24** The diagram shows part of the magnetic field between two bar magnets.



Which statement about the ends of the magnets is correct?

- A** X and Y are both N poles.
B X and Y are both S poles.
C X is an N pole and Y is an S pole.
D X is an S pole and Y is an N pole.

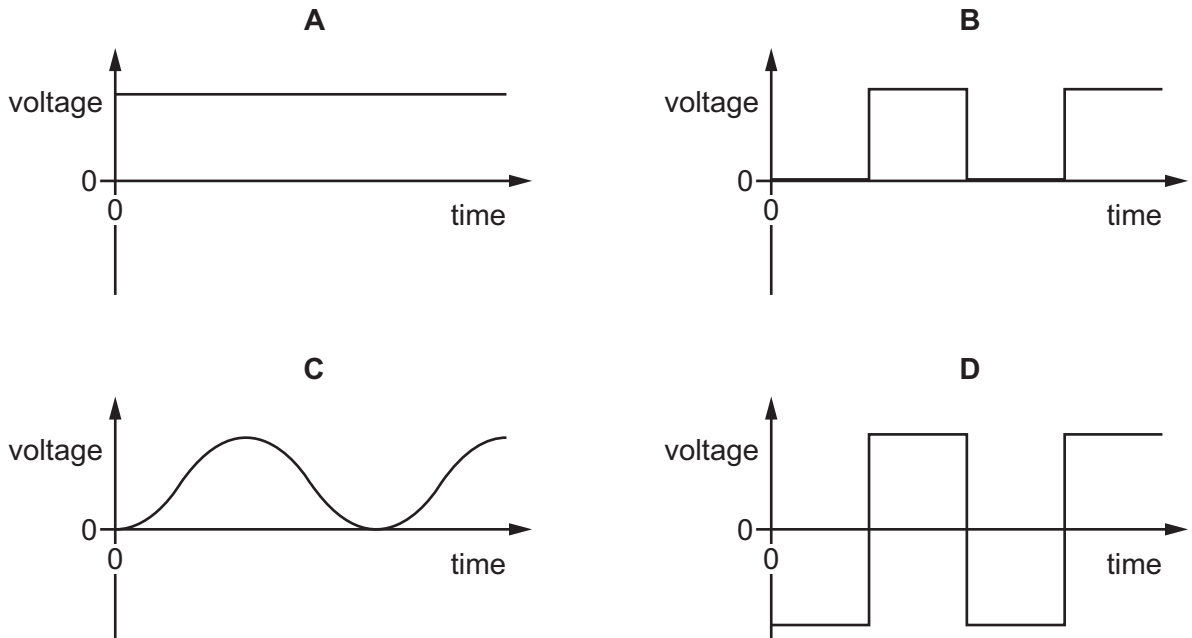
25 Diagrams 1, 2 and 3 show an experiment to compare two magnets 1 and 2.



Which row explains the readings on the balances?

	polarity of magnet 1	polarity of magnet 2
A	X is an N pole	Z is an N pole
B	X is an N pole	Z is an S pole
C	X is an S pole	Z is an N pole
D	X is an S pole	Z is an S pole

26 Which graph of voltage against time shows an a.c. voltage?



- 27** There is a current I in a resistor for a time t . The potential difference (p.d) across the resistor is V .

What does the product IVt represent?

- A** the charge flowing through the resistor
- B** the energy transferred by the resistor
- C** the power dissipated by the resistor
- D** the resistance of the resistor

- 28** A student designs a circuit to turn on a fan when the temperature increases.

Which component does the student need to use in her circuit?



- 29** Four statements about safety when designing electrical products are listed.

- 1 If the outer casing is a conductor, it should be earthed.
- 2 If the outer casing is a conductor, it should **not** be earthed.
- 3 If the outer casing is an insulator, it should be earthed.
- 4 If the outer casing is an insulator, it should **not** be earthed.

Which statements are correct?

- A** 1 and 3
- B** 1 and 4
- C** 2 and 3
- D** 2 and 4

- 30** A transformer has 400 turns on the primary coil connected to the 240 V mains supply.

There are two secondary coils to serve different parts of a television. Secondary coil 1 has 2500 turns and secondary coil 2 has 20 turns.

Which row is correct?

	p.d. across secondary coil 1/V	p.d. across secondary coil 2/V
A	12	3.2
B	1500	12
C	4200	12
D	4200	33

31 A class is designing a d.c. motor. To achieve a greater turning effect, three suggestions are made.

- 1 Have a larger current in the coil of the motor.
- 2 Have a stronger magnet in the motor.
- 3 Put a larger number of turns on the coil.

Which suggestions will help to increase the turning effect?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

32 Which statement about the structure of an atom is correct?

- A** It contains positively charged particles only.
B It contains negatively charged particles only.
C It contains no charged particles.
D It contains positively charged particles and negatively charged particles.

33 A nuclide of strontium is represented by ${}^{90}_{38}\text{Sr}$.

Which row gives the number and location of the electrons in an atom of ${}^{90}_{38}\text{Sr}$?

	number of electrons	location of electrons
A	38	in the nucleus
B	38	surrounding the nucleus
C	52	in the nucleus
D	52	surrounding the nucleus

34 A radiation detector in a laboratory is measuring background radiation.

Which row describes the readings and the cause?

	readings	cause
A	vary with no pattern	background radiation is random
B	vary with no pattern	radiation detectors are unstable
C	slowly increase during the day	background radiation increases as temperature increases
D	slowly reduce during the day	background radiation decreases as temperature increases

- 35** The nuclei of the atoms in a substance are changing randomly and emitting radiation.

What is happening to the substance?

- A** It is undergoing electromagnetic induction.
- B** It is undergoing magnetisation.
- C** It is undergoing solidification.
- D** It is undergoing radioactive decay.

- 36** What is the safest way to store a radioactive source?

- A** in a lead-lined box in a metal cabinet
- B** on a shelf away from other radioactive materials
- C** in a glass bottle containing oil
- D** in a fume cupboard

- 37** Which row explains the apparent daily motion of the Sun across the sky and the cycle of phases of the Moon?

	the daily motion of the Sun across the sky	the cycle of phases of the Moon
A	the Earth rotates on its axis once every 24 hours	the Earth orbits the Sun once approximately every 365 days
B	the Earth rotates on its axis once every 24 hours	the Moon orbits the Earth approximately once every month
C	the Moon orbits the Earth approximately once every month	the Earth orbits the Sun once approximately every 365 days
D	the Moon orbits the Earth approximately once every month	the Earth rotates on its axis once every 24 hours

- 38** The nearest star to the Sun is 4 light-years away.

A spaceship travels at 0.1% of the speed of light.

What is the total time taken for a journey from the Sun to the star and back again?

- A** 400 years
- B** 800 years
- C** 4000 years
- D** 8000 years

- 39** In which regions of the electromagnetic spectrum does the Sun radiate most of its energy?
- A** microwave to infrared
 - B** ultraviolet to X-ray
 - C** infrared to ultraviolet
 - D** radio to infrared
- 40** What is the approximate diameter of the Milky Way?
- A** 100 000 kilometres
 - B** 1 000 000 kilometres
 - C** 100 000 light-years
 - D** 1 000 000 light-years

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Four athletes run twice around a track. The table shows their times at the end of each lap.

Which athlete runs the second lap the fastest?

	time at end of first lap / s	time at end of second lap / s
A	22.99	47.04
B	23.04	47.00
C	23.16	47.18
D	23.39	47.24

- 2 A racing car is fitted with an on-board computer. Every time the car passes the starting line, the computer records the distance travelled in the following two seconds.

Which set of data shows that the car is increasing in speed during the two seconds?

A

time / s	distance travelled / m
0	0
1	100
2	200

B

time / s	distance travelled / m
0	0
1	90
2	180

C

time / s	distance travelled / m
0	0
1	80
2	190

D

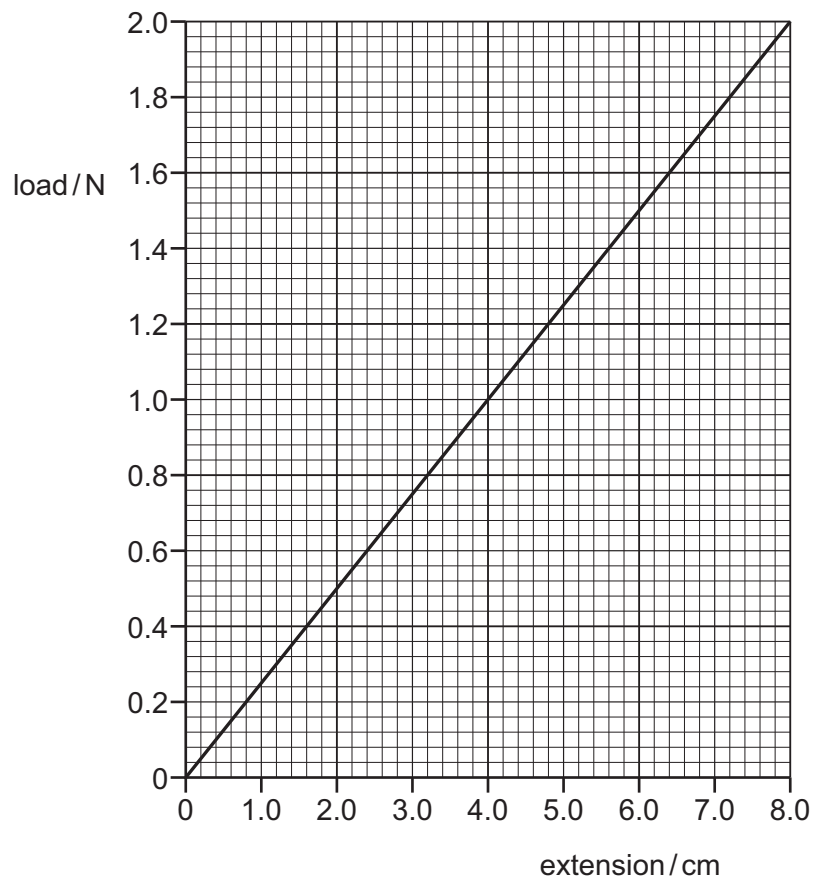
time / s	distance travelled / m
0	0
1	100
2	180

- 3 Which row shows the gravitational field strength on the Earth and the definition of velocity?

	gravitational field strength on the Earth	definition of velocity
A	9.8 kg / N	the change in the speed
B	9.8 N / kg	the change in the speed
C	9.8 kg / N	the speed in a given direction
D	9.8 N / kg	the speed in a given direction

- 4 Which statement about the weight of an object is correct?
- A The weight of an object is the gravitational force per unit mass.
 - B The weight of an object is the gravitational force on an object.
 - C The weight is zero when the object is falling at constant speed in the Earth's atmosphere.
 - D The weight is zero when the object is in orbit around the Earth.
- 5 The diagram shows a load–extension graph for a steel spring.

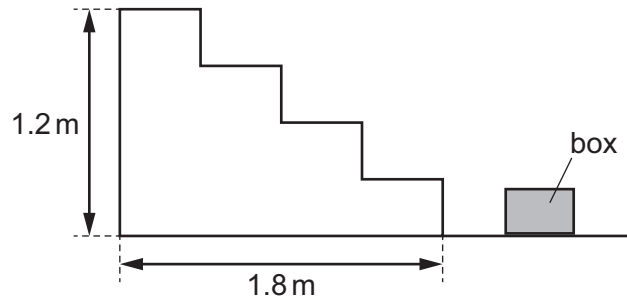
The length of the spring with no load attached is 2.0 cm.



Which load increases the length of the spring to 6.0 cm?

- A 0.5 N B 1.0 N C 1.5 N D 2.0 N
- 6 Which force between two solid surfaces opposes motion?
- A friction
 - B gravity
 - C kinetic
 - D weight

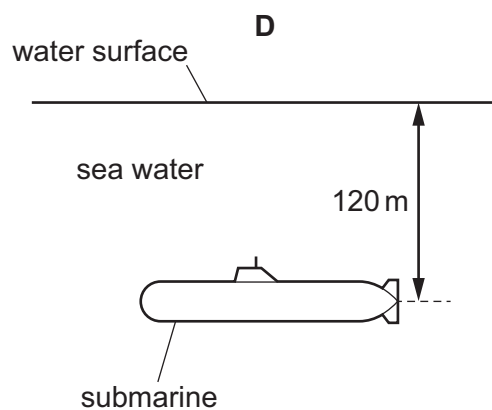
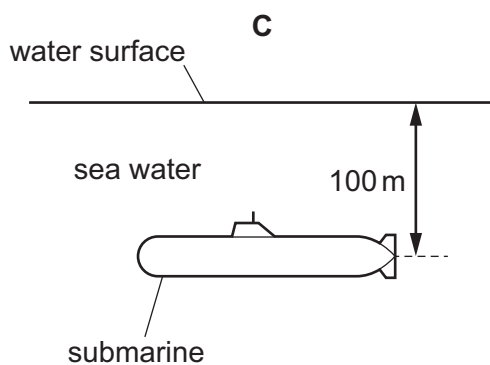
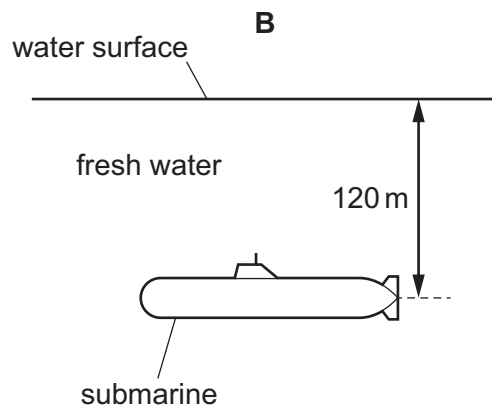
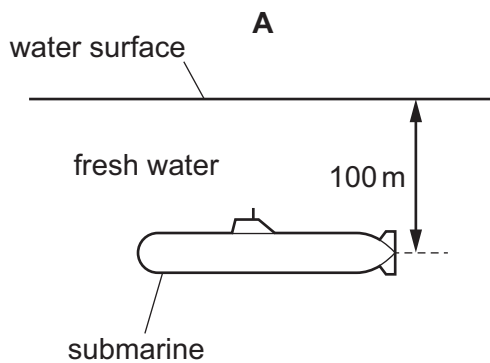
- 7 A box of mass 2.0 kg and weight 20 N is carried to the top of some stairs.



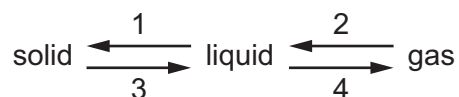
How is the work done against gravity on the box calculated?

- A** 2.0×1.8 **B** 20×1.2 **C** 2.0×1.2 **D** $20 \times 1.2 \times 1.8$
- 8 Four submarines are submerged. The density of fresh water is 1000 kg/m^3 and the density of sea water is 1020 kg/m^3 .

Which submarine experiences the greatest pressure due to the water?



- 9 The diagram shows the changes in state, 1, 2, 3 and 4, between solids, liquids and gases.



What are processes 2 and 3?

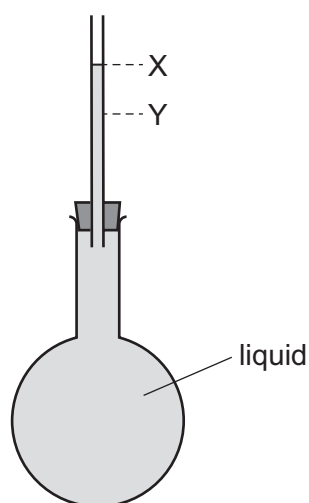
	2	3
A	condensation	melting
B	condensation	freezing
C	evaporation	melting
D	evaporation	freezing

- 10 Copper has a melting point of 1100 °C.

What is the melting point in kelvin?

- A** 727 K **B** 827 K **C** 1373 K **D** 1473 K

- 11 A liquid at room temperature fills a flask and a glass tube to level X.



The flask is placed in ice and the liquid level in the tube changes to level Y.

Why does the level change to level Y?

- A** The flask contracts.
B The flask expands.
C The liquid contracts.
D The liquid expands.

12 Which row correctly describes the state of water at the temperatures shown?

	250 K	300 K	350 K	400 K
A	liquid	liquid	gas	gas
B	solid	solid	liquid	liquid
C	solid	liquid	liquid	gas
D	solid	liquid	gas	gas

13 A liquid is evaporating. The liquid is **not** boiling.

Which statement about the liquid at an instant in time is correct?

- A** Any molecule can escape and from any part of the liquid.
- B** Any molecule can escape, but only from the liquid's surface.
- C** Only molecules with enough energy can escape and only from the liquid's surface.
- D** Only molecules with enough energy can escape, but from any part of the liquid.

14 A student stirs a hot liquid in a pan with a spoon.

Which row explains which material the spoon should be made from so that the student does **not** burn their hand?

	material of spoon	explanation
A	metal	it is a good conductor
B	metal	it is a good insulator
C	wood	it is a good conductor
D	wood	it is a good insulator

15 Thermal radiation is emitted from all objects.

Which mediums can thermal radiation travel through?

	glass	water	air	vacuum
A	✓	✓	✓	✓
B	✓	✓	✓	✗
C	✗	✓	✓	✓
D	✗	✗	✗	✓

key

✓ = can travel through this medium

✗ = cannot travel through this medium

- 16** The radius of a circular pond is 3.0 m.

Circular waves are made at the centre of the pond.

There are 10 whole waves between the centre and the edge of the pond and it takes 5.0 s for all 10 waves to reach the edge.

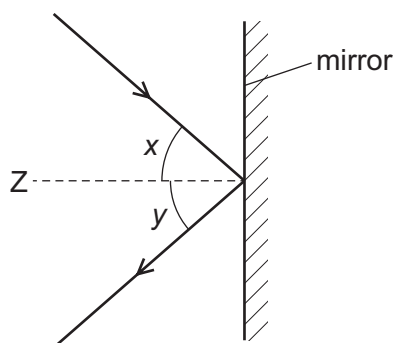
What is the wavelength and the frequency of the waves?

	wavelength / m	frequency / Hz
A	0.30	0.50
B	0.30	2.0
C	0.60	0.50
D	0.60	2.0

- 17** How does the direction of vibration of a longitudinal wave compare to its direction of propagation and which wave example is longitudinal?

	direction of vibration compared to direction of propagation	wave example
A	at right angles	ultrasound waves
B	at right angles	ultraviolet waves
C	parallel	ultrasound waves
D	parallel	ultraviolet waves

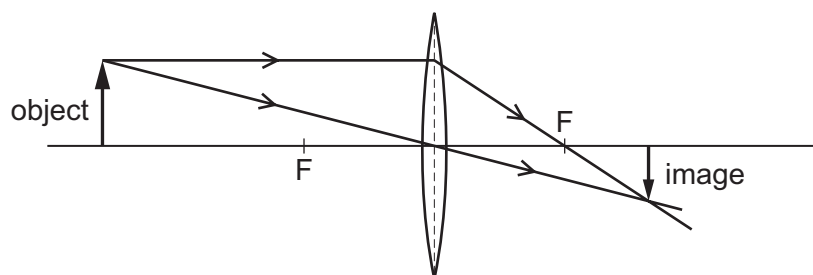
- 18 The diagram shows a ray of light reflecting from a mirror.



Which row shows the correct names for x , y and Z ?

	x	y	Z
A	angle of incidence	angle of reflection	normal
B	angle of incidence	angle of reflection	principal focus
C	angle of reflection	angle of refraction	normal
D	angle of reflection	angle of refraction	principal focus

- 19 A converging lens forms an image of an object placed in front of it.



What are the characteristics of the image?

- A** real, inverted, diminished
- B** real, upright, enlarged
- C** virtual, inverted, diminished
- D** virtual, upright, enlarged

- 20** White light can be split into different colours by passing it through a prism.

What is the name of this process?

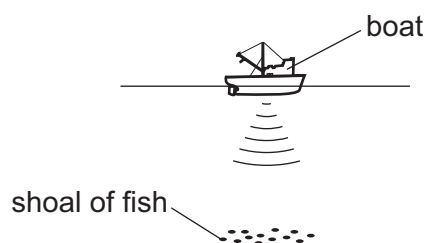
- A** diffraction
- B** dispersion
- C** reflection
- D** refraction

- 21** Excessive exposure to some regions of the electromagnetic spectrum can cause harmful effects to people.

Which regions cause mutation of cells in the body, skin burns and internal heating of body cells?

	mutation of cells	skin burns	internal heating of body cells
A	radio waves	infrared	visible light
B	radio waves	ultraviolet	microwaves
C	X-rays	infrared	microwaves
D	X-rays	ultraviolet	visible light

- 22** A pulse of sound is produced at the bottom of a boat. The sound travels through the water and is reflected from a shoal of fish. The sound reaches the boat again 1.2 s after it is produced. The speed of sound in the water is 1500 m/s.



How far below the bottom of the boat is the shoal of fish?

- A** 450 m
- B** 900 m
- C** 1800 m
- D** 3600 m

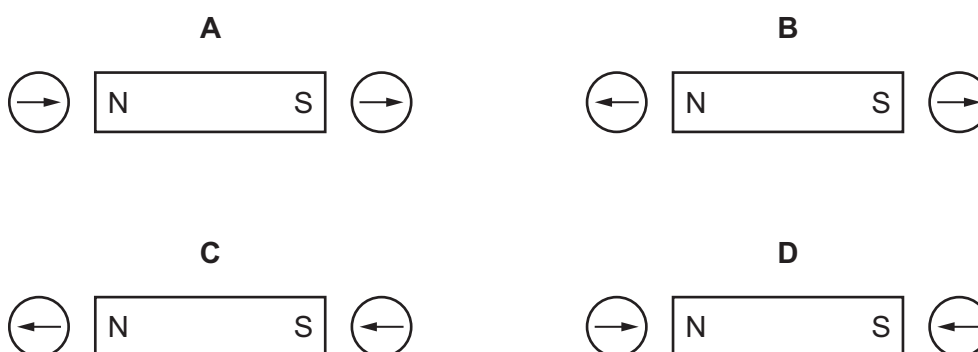
- 23** The frequency of a sound wave increases from 31 000 Hz to 32 000 Hz.

What is the nature of the wave and how does it sound to a human with normal hearing?

- A** It is a longitudinal wave with a decreasing pitch.
- B** It is a transverse wave with an increasing pitch.
- C** It is a longitudinal wave that cannot be heard by a human with normal hearing.
- D** It is a transverse wave that cannot be heard by a human with normal hearing.

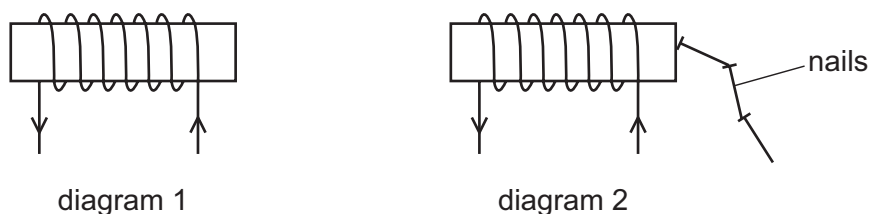
- 24** Small compasses are used to determine the direction of a magnetic field.

Which diagram shows the direction that the compass needles point when placed near the N and S poles of a bar magnet?



- 25** A metal bar is placed inside a current-carrying coil, as shown in diagram 1.

There is a small current in the coil. The bar holds a few nails, as shown in diagram 2.



When there is no current in the coil, the nails drop off.

Which row is correct?

	metal from which the bar is made	effect of a larger current in the coil
A	soft iron	it makes no difference
B	soft iron	the bar holds more nails
C	steel	it makes no difference
D	steel	the bar holds more nails

- 26** There is a current I in a resistor of resistance R . The potential difference (p.d.) across the resistor is V .

Which equation gives the power P dissipated by the resistor?

A $P = IR$
 B $P = IV$
 C $P = \frac{I}{R}$
 D $P = \frac{I}{V}$

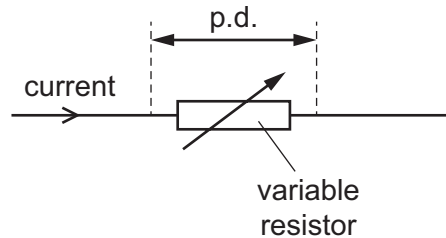
27 A student tests the electrical conduction of four materials.

- aluminium
- iron
- plastic
- silver

Which materials are good conductors of electricity?

- A** aluminium, iron and silver
- B** aluminium and silver only
- C** iron, silver and plastic
- D** plastic only

28 This question is about the potential difference (p.d.) across, and the current in, a variable resistor.



The resistance of the variable resistor is increased.

Which statement is correct?

- A** The current can remain constant if the p.d. is decreased.
- B** The current can remain constant if the p.d. is increased.
- C** The p.d. can remain constant if the current is kept constant.
- D** The p.d. can remain constant if the current is increased.

- 29** Diagram 1 shows a resistor connected in a circuit. Diagram 2 shows an identical resistor connected in parallel with the first resistor.

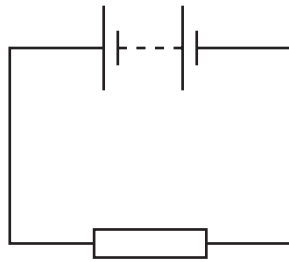


diagram 1

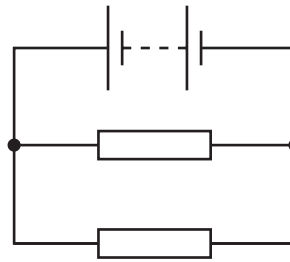
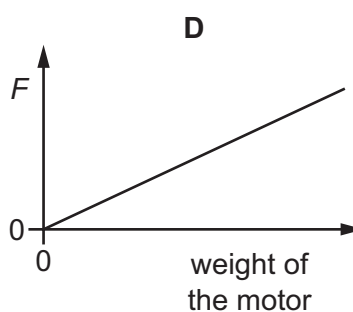
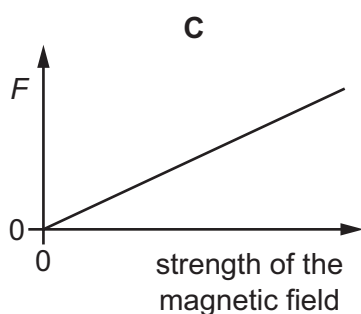
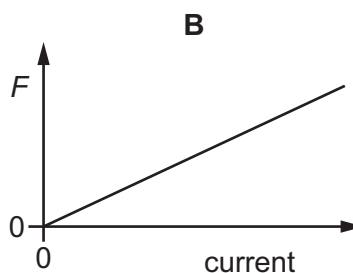
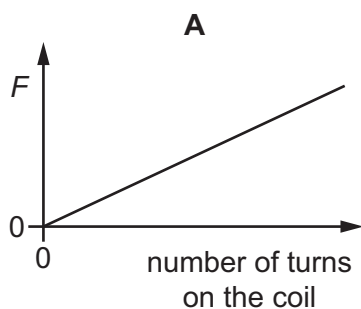


diagram 2

What is the combined resistance of the two resistors in diagram 2?

- A** greater than in the circuit of diagram 1
 - B** less than in the circuit of diagram 1, but greater than zero
 - C** the same as in the circuit of diagram 1
 - D** zero
- 30** Where must a fuse be connected in a mains electric circuit?
- A** the earth wire only
 - B** the live wire only
 - C** the neutral wire only
 - D** the live wire and the earth wire

- 31 Which graph does **not** show the correct trend for the force F causing the turning effect on the coil in a d.c. motor?



- 32 Which statement about the structure of an atom is correct?

- A** Negative electrons surround a neutral nucleus.
- B** Negative electrons surround a positive nucleus.
- C** Positive electrons surround a neutral nucleus.
- D** Positive electrons surround a negative nucleus.

- 33 Four statements about isotopes of a certain element are listed.

Which statement about the isotopes **must** be correct?

- A** They are radioactive.
- B** They are unstable.
- C** They have the same number of neutrons.
- D** They have the same number of protons.

- 34 What makes a contribution to a person's annual dose of ionising radiation?

- A** food and drink
- B** greenhouse gases
- C** mobile phone masts
- D** radio and TV transmissions

- 35 The emissions from a radioactive source pass through a sheet of lead that is 10 mm thick.

Which type of radiation is emitted from the source and how is the radiation affected by an electric field?

	type of radiation	effect of electric field on radiation
A	α	radiation deflected
B	α	radiation not deflected
C	γ	radiation deflected
D	γ	radiation not deflected

- 36 Radioactive waste is often mixed with liquid glass. This mixture solidifies and is stored in steel drums.

Why is this process used to store radioactive waste?

- A** It reduces the decay rate.
 - B** It reduces the half-life of the waste.
 - C** The waste is contained in a solid block so it cannot leak out.
 - D** The glass is transparent so the waste can be seen.
- 37 Which planet is classed as a rocky planet?
- A** Jupiter
 - B** Saturn
 - C** Uranus
 - D** Venus
- 38 What is the cause of the phases of the Moon?
- A** the movement of the Earth around the Sun
 - B** the movement of the Moon around the Sun
 - C** the movement of the Moon around the Earth
 - D** the movement of the Sun around the Moon
- 39 What makes up the Universe?
- A** many billions of galaxies
 - B** Mercury, Venus, the Earth, Mars, Jupiter, Saturn, Uranus and Neptune only
 - C** the Sun, the Earth and the Moon only
 - D** the Solar System only

40 The Sun mostly consists of which gases?

- A** helium and hydrogen
- B** helium and nitrogen
- C** hydrogen and oxygen
- D** oxygen and nitrogen

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/11

May/June 2024

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

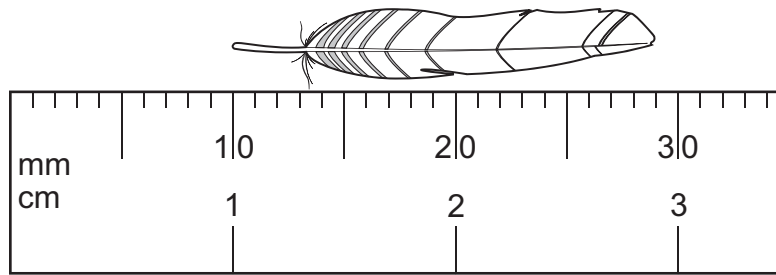
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.

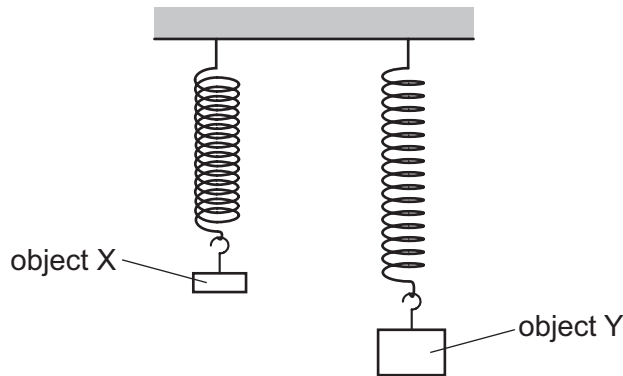


- 1 The diagram shows an enlarged drawing of the end of a metre ruler. It is being used to measure the length of a small feather.



What is the length of the feather?

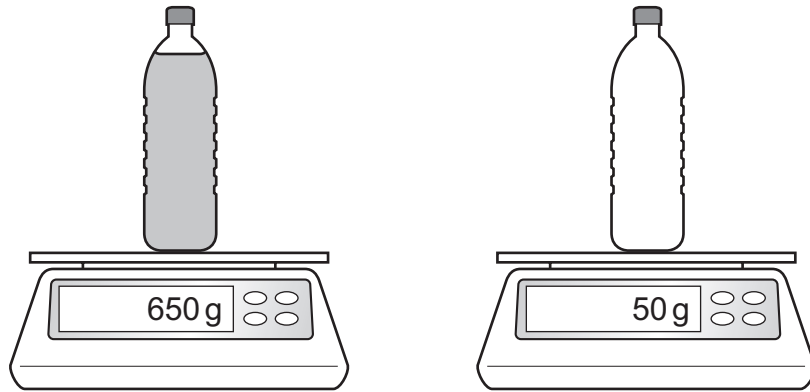
- A** 19 mm **B** 29 mm **C** 19 cm **D** 29 cm
- 2 The diagram shows two objects, X and Y, suspended from identical springs. The extension of each spring is different.



Which row compares the mass and the weight of the two objects?

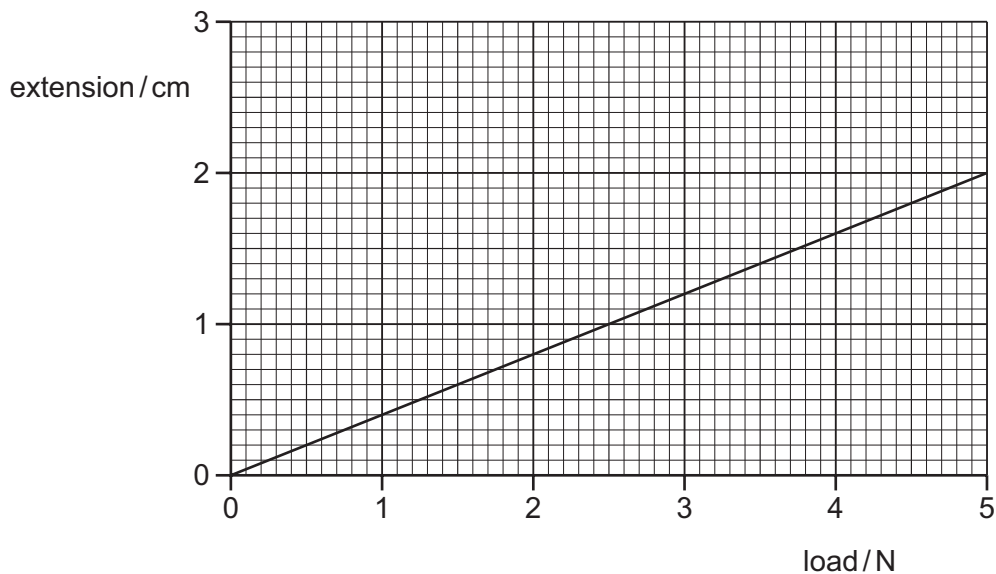
	mass	weight
A	both objects have the same mass	both objects have the same weight
B	both objects have the same mass	the weight of X is less than the weight of Y
C	the mass of X is less than the mass of Y	both objects have the same weight
D	the mass of X is less than the mass of Y	the weight of X is less than the weight of Y

- 3 A plastic bottle contains 750 cm^3 of oil. The diagram shows the mass of the bottle being measured when it is full and then when it is empty.



What is the density of the oil?

- A** 0.80 g/cm^3 **B** 0.93 g/cm^3 **C** 1.1 g/cm^3 **D** 1.3 g/cm^3
- 4 The extension–load graph for a spring is shown. The unstretched length of the spring is 15.0 cm .

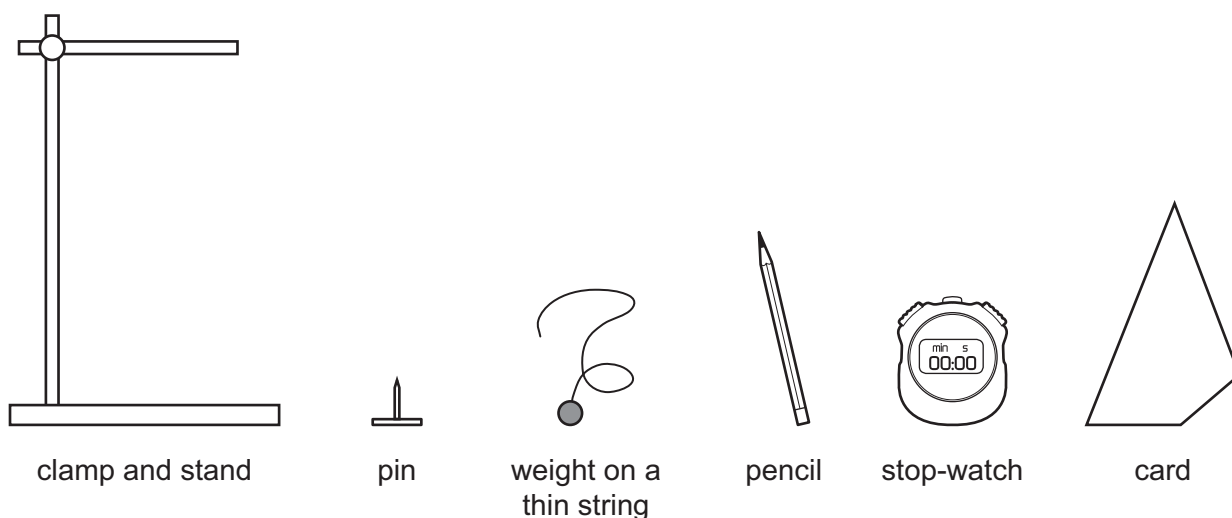


When an object of unknown weight is suspended on the spring, the length of the spring is 16.4 cm .

What is the weight of the object?

- A** 0.55 N **B** 0.67 N **C** 3.5 N **D** 4.1 N

- 5 A student determines the position of the centre of gravity of a piece of card. The diagram shows the equipment that is available. The equipment is not drawn to scale.



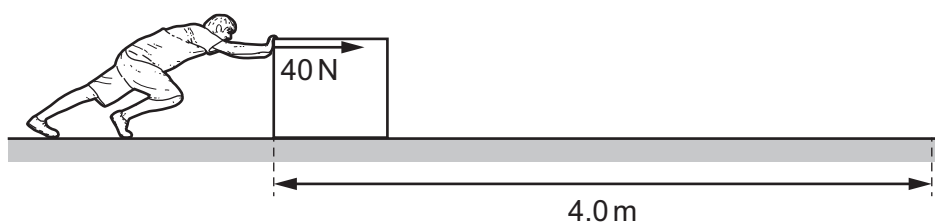
Which piece of equipment is **not** needed?

- A pencil
 - B pin
 - C weight on a thin string
 - D stop-watch
- 6 A cyclist stops his bicycle using the brakes.

Which row about energy stores is correct?

	decreases	increases
A	kinetic energy store of bicycle	chemical energy store of brakes
B	kinetic energy store of bicycle	thermal energy store of brakes
C	thermal energy store of brakes	kinetic energy store of bicycle
D	thermal energy store of brakes	chemical energy store of bicycle

- 7 A man applies a force of 40 N to push a box along the floor.

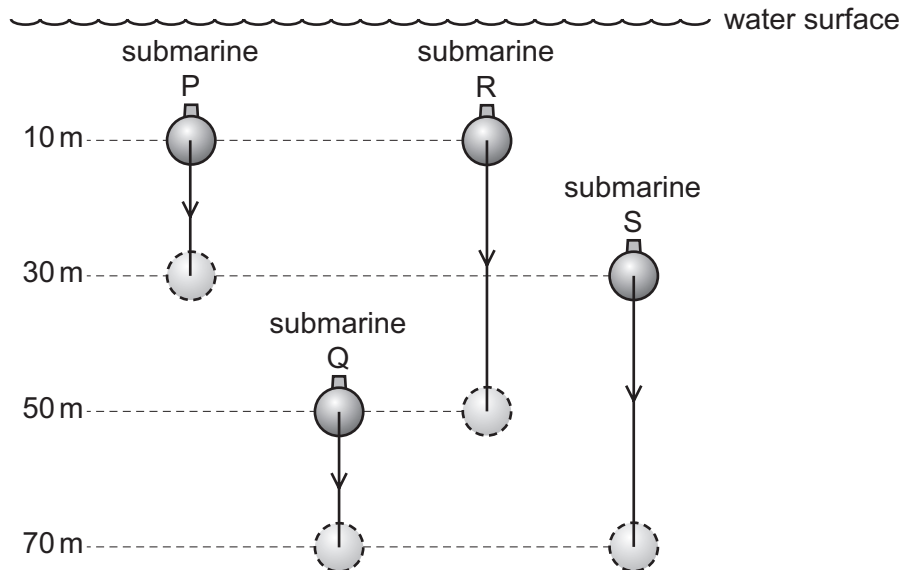


How much power is required to push the box 4.0 m in 3.0 seconds?

- A 3.3 W
- B 30 W
- C 53 W
- D 480 W

- 8 Four identical submarines, P, Q, R and S, are lowered from one depth to another in water of a constant density.

The initial and final depths of each submarine are shown.



Which row is correct for the difference in pressure from the initial to the final depth of each submarine?

	least change in pressure	greatest change in pressure
A	submarines P and Q	submarines R and S
B	submarines P and Q	submarine R only
C	submarine P only	submarines R and S
D	submarine P only	submarine R only

- 9 Some terms describing changes of state are listed.

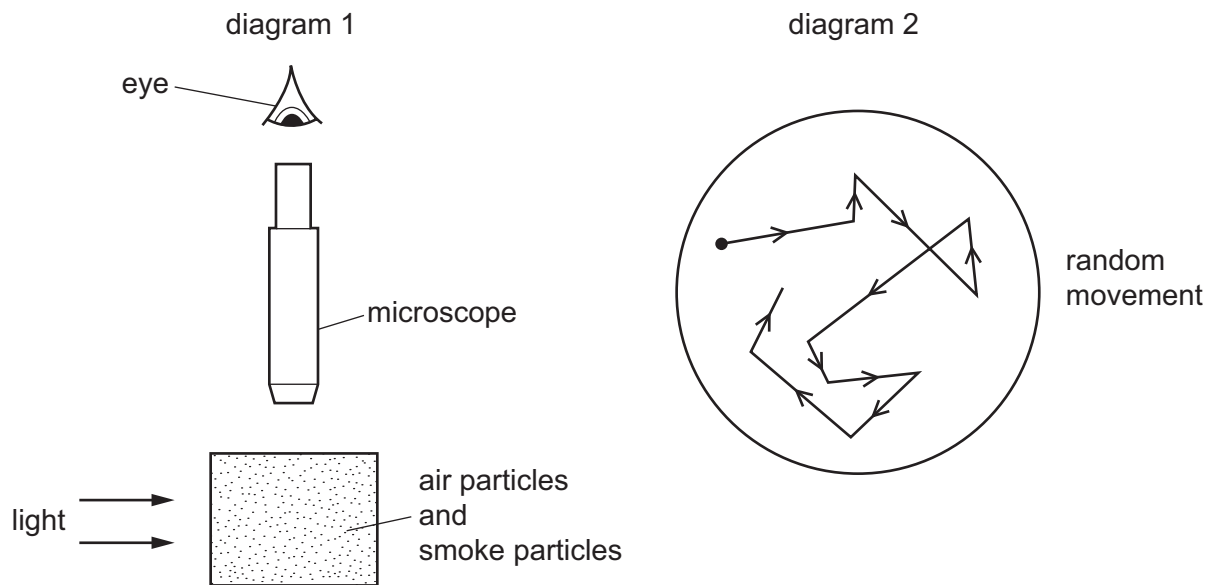
- 1 boiling
- 2 solidification
- 3 condensation
- 4 evaporation

Which two terms identify the same change of state?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4

10 Diagram 1 shows apparatus being used to observe smoke particles.

Diagram 2 shows how a smoke particle moves randomly.

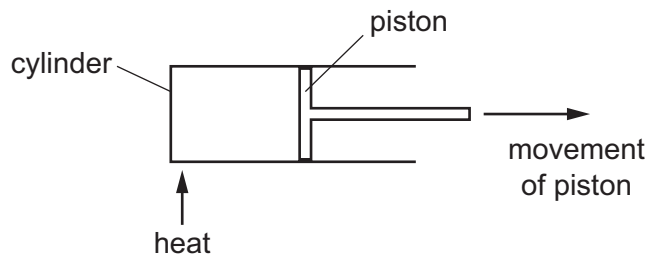


Why do the smoke particles move randomly?

- A** They are hit by air particles.
 - B** They are less dense than air.
 - C** They are moved by convection currents.
 - D** They gain energy from the light.
- 11** How is a temperature in degrees Celsius ($^{\circ}\text{C}$) converted to a temperature in kelvin (K)?
- A** Add 273 to the temperature in $^{\circ}\text{C}$.
 - B** Divide the temperature in $^{\circ}\text{C}$ by 273.
 - C** Multiply the temperature in $^{\circ}\text{C}$ by 273.
 - D** Subtract 273 from the temperature in $^{\circ}\text{C}$.

- 12 The diagram shows a frictionless piston inside a cylinder.

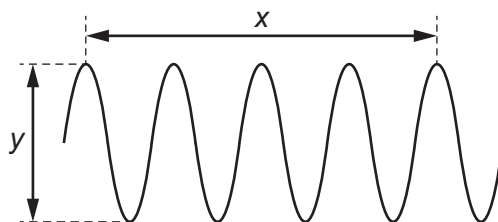
The air inside the cylinder is heated. The piston moves in the direction shown.



Which statement about the air inside the cylinder is correct?

- A The temperature and volume increase at constant pressure.
 - B The temperature and pressure increase at constant volume.
 - C The temperature, volume and pressure all increase.
 - D The volume and pressure increase at constant temperature.
- 13 What increases when the temperature of a copper block increases?
- A the number of copper atoms in the block
 - B the melting point of the block
 - C the internal energy of the block
 - D the density of the block
- 14 A textbook gives the description of a thermal process as 'more-energetic molecules escape from the surface of a liquid which causes the liquid to cool'.
- Which process is being described?
- A boiling
 - B Brownian motion
 - C condensation
 - D evaporation
- 15 In which situation is radiation the **main** method by which energy is transferred?
- A heating a pan of water using a gas camping stove
 - B energy reaching the Earth from the Sun
 - C heating the air in a room with a convection heater
 - D giving gravitational potential energy to a glider when it is lifted by thermal currents

16 The diagram represents a wave.



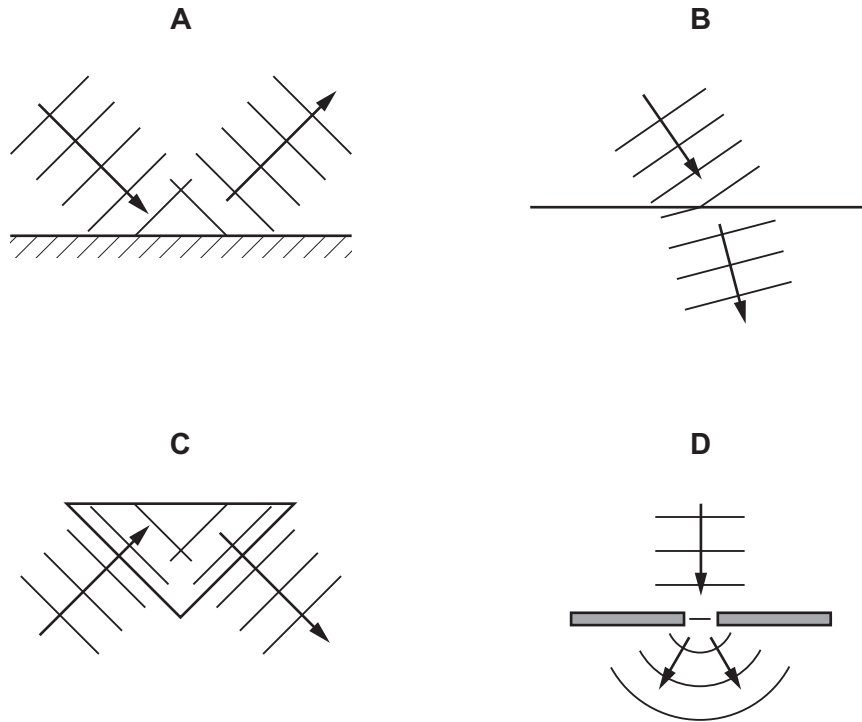
Which row gives the wavelength and the amplitude of the wave?

	wavelength	amplitude
A	x	y
B	y	x
C	x	$\frac{y}{2}$
D	$\frac{x}{4}$	$\frac{y}{2}$

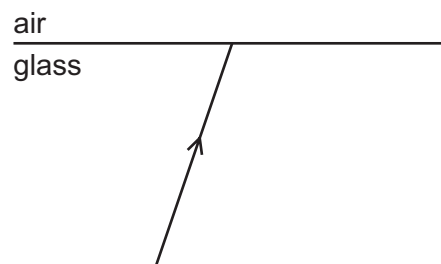
17 Which row describes a seismic P-wave?

	type of wave	direction of vibration
A	longitudinal	parallel to the direction of propagation of the wave
B	longitudinal	perpendicular to the direction of propagation of the wave
C	transverse	parallel to the direction of propagation of the wave
D	transverse	perpendicular to the direction of propagation of the wave

18 Which diagram shows waves diffracting?



19 The diagram shows a ray of light in glass incident on the surface between the glass and air.



What happens if the angle of incidence is made larger than the critical angle for the glass?

- A The angle of refraction becomes equal to 90° .
- B There is a refracted ray and a ray reflected inside the glass.
- C There is a refracted ray only.
- D There is only a ray reflected inside the glass.

20 Red light of frequency 430×10^{12} Hz has a wavelength of 700×10^{-9} m.

What is possible for blue light?

- A a frequency of 190×10^{12} Hz and a wavelength of 450×10^{-9} m
- B a frequency of 190×10^{12} Hz and a wavelength of 950×10^{-9} m
- C a frequency of 670×10^{12} Hz and a wavelength of 450×10^{-9} m
- D a frequency of 670×10^{12} Hz and a wavelength of 950×10^{-9} m

- 21** Infrared is a useful type of electromagnetic radiation. However, excessive exposure to infrared can cause harmful effects.

Which row gives a use and a harmful effect for infrared?

	use	harmful effect
A	detection of cancer	mutations
B	intruder alarms	skin burns
C	detection of fake bank notes	cell damage
D	thermal imaging	internal heating of body cells

- 22** The table shows data for the hearing ranges of different animals.

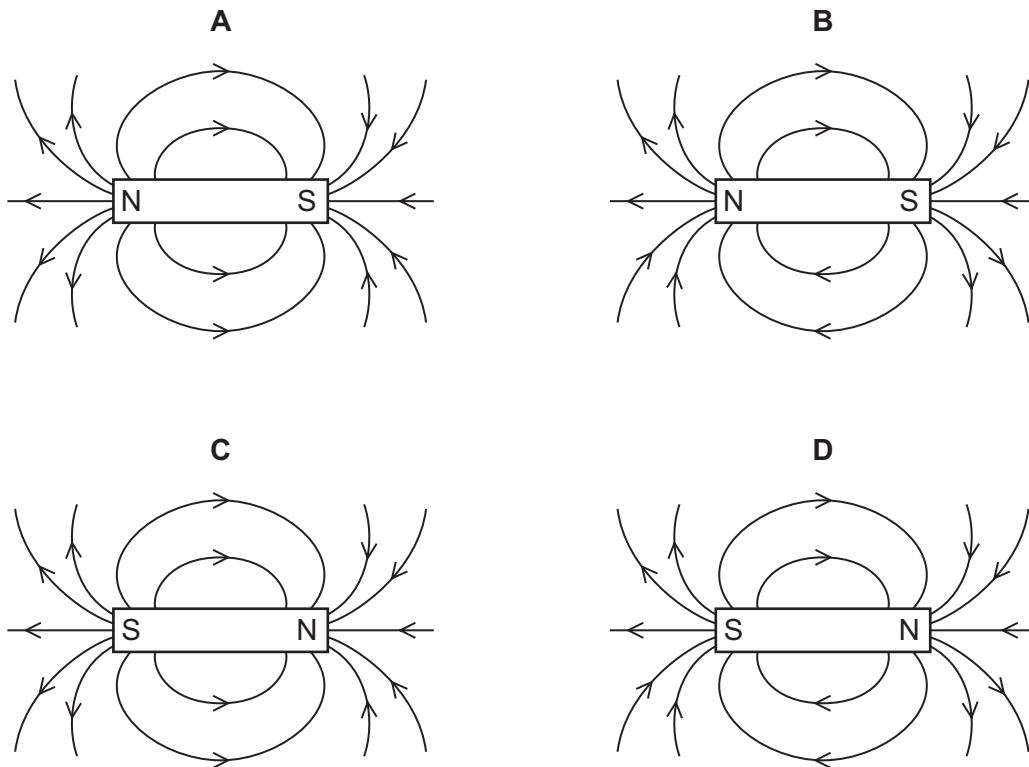
Which animal has the most similar hearing range to a human?

	animal	hearing range / Hz
A	turtle	20–1000
B	mouse	1000–1 000 000
C	moth	20 000–300 000
D	elephant	16–12 000

- 23** Which description of ultrasound is correct?

- A** longitudinal waves with a frequency greater than 20 000 Hz
- B** longitudinal waves with a frequency less than 20 Hz
- C** transverse waves with a frequency greater than 20 000 Hz
- D** transverse waves with a frequency less than 20 Hz

- 24** Which diagram shows the pattern and the direction of the magnetic field lines around a bar magnet?



- 25** Which material is a good electrical conductor?

- A** glass
- B** copper
- C** rubber
- D** wood

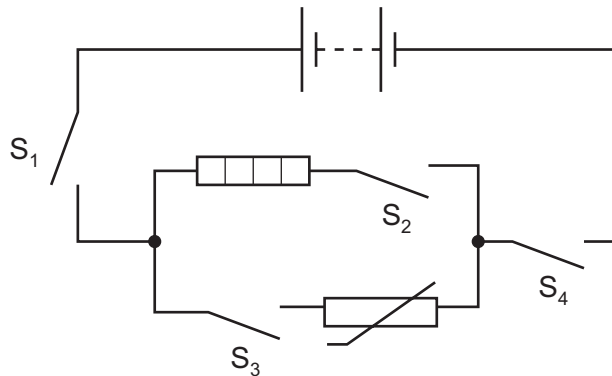
- 26** What is an electric current and how is an ammeter positioned in a circuit to measure the current in a resistor?

	what is an electric current?	positioning of an ammeter
A	a flow of charge	in parallel with the resistor
B	a flow of charge	in series with the resistor
C	a flow of voltage	in parallel with the resistor
D	a flow of voltage	in series with the resistor

27 Which unit is used for the potential difference (p.d.) between two points in an electrical circuit?

- A ampere (A)
- B joule (J)
- C ohm (Ω)
- D volt (V)

28 The circuit in the diagram contains four switches, S_1 , S_2 , S_3 and S_4 .



Which three switches must be closed for the heater to work?

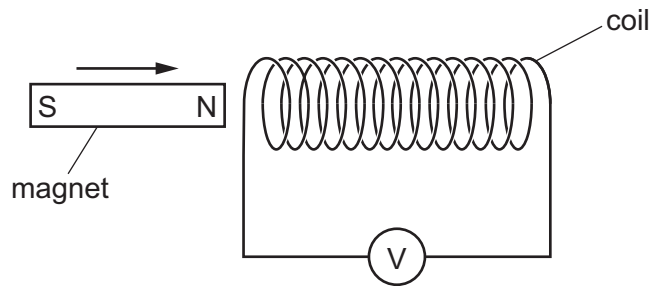
- A S_1 , S_2 and S_3
- B S_1 , S_2 and S_4
- C S_1 , S_3 and S_4
- D S_2 , S_3 and S_4

29 The metal case of an electrical appliance is fitted with an earth connection.

What is the purpose of this?

- A to complete the circuit supplying the appliance so that it works
- B to ensure that the cables supplying the appliance do not become damp
- C to prevent the cables supplying the appliance from overheating
- D to protect a user of the appliance from electric shock

- 30 A student moves a magnet into a coil, as shown. The voltmeter measures the magnitude of the electromotive force (e.m.f.) induced in the coil.



Which quantity does **not** affect the magnitude of the induced e.m.f.?

- A the number of turns per unit length in the coil
 - B the polarity of the magnet
 - C the speed of the magnet
 - D the strength of the magnet
- 31 A student makes a small d.c. motor. The teacher supplies the following equipment.
- battery
 - ammeter
 - voltmeter
 - coil of wire
 - magnets
 - resistor

The student is also able to use other equipment if he wishes.

Which equipment **must** be used?

- A ammeter, battery and resistor
- B ammeter, coil of wire and voltmeter
- C battery, coil of wire and magnets
- D magnets, resistor and voltmeter

- 32** A simple model of the atom consists of small particles orbiting a central nucleus.

Which row is correct?

	charge on nucleus	charge on orbiting particles
A	negative	negative
B	negative	positive
C	positive	negative
D	positive	positive

- 33** What is the charge on a proton and what is the charge on a neutron?

	proton	neutron
A	+1	-1
B	+1	0
C	-1	+1
D	-1	0

- 34** An isotope of radon is radioactive. It decays by emitting an α -particle.

What happens to the nucleus of a radon atom during the emission of the α -particle?

- A** It becomes the nucleus of a different isotope of radon with fewer neutrons.
- B** It becomes the nucleus of a different isotope of radon with more neutrons.
- C** It becomes the nucleus of an element with a higher proton number.
- D** It becomes the nucleus of an element with a lower proton number.

- 35** The nuclide notation of the isotope strontium-90 is $^{90}_{38}\text{Sr}$.

Which statement is correct?

- A** A nucleus of strontium-90 has 38 neutrons.
- B** A nucleus of strontium-90 has 52 neutrons.
- C** A nucleus of strontium-90 has 90 electrons.
- D** A nucleus of strontium-90 has 90 neutrons.

- 36 A box is used for storing radioactive sources.

What is the best material to use for lining the box to prevent radiation from escaping?

- A aluminium
- B lead
- C paper
- D plastic

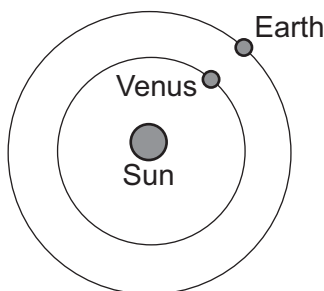
- 37 The Earth is a planet that orbits the Sun once in approximately 365 days.

What does this enable us to explain?

- A the cycle of day and night
- B the cycle of phases of the moon
- C the periodic nature of the seasons
- D the apparent daily motion of the Sun

- 38 The radius of the orbit of Venus around the Sun is 110×10^6 km.

The radius of the orbit of the Earth around the Sun is 150×10^6 km.



The speed of light is 0.30×10^6 km/s.

How long does light take to travel from Venus to the Earth when these planets are closest to each other?

- A 130 s B 370 s C 500 s D 870 s

- 39 What are the three main types of radiation emitted by the Sun?

- A infrared, radio waves and ultraviolet
- B infrared, visible light and ultraviolet
- C microwaves, visible light and X-rays
- D radio waves, X-rays and ultraviolet

40 What is a light-year?

- A** the time for light to travel 1 km in space
- B** the time for light to travel from the Sun to the Earth
- C** the distance travelled in space by light in one year
- D** the distance travelled by light from the Sun to the Earth

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PHYSICS**0625/13**

Paper 1 Multiple Choice (Core)

May/June 2024

MARK SCHEME

Maximum Mark: 40

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of **3** printed pages.

Question	Answer	Marks
1	B	1
2	A	1
3	C	1
4	D	1
5	A	1
6	B	1
7	A	1
8	D	1
9	B	1
10	A	1
11	B	1
12	C	1
13	D	1
14	A	1
15	A	1
16	D	1
17	A	1
18	D	1
19	C	1
20	B	1
21	B	1
22	C	1
23	A	1
24	C	1
25	C	1
26	D	1
27	B	1
28	D	1

Question	Answer	Marks
29	B	1
30	B	1
31	A	1
32	D	1
33	B	1
34	A	1
35	D	1
36	A	1
37	B	1
38	D	1
39	C	1
40	C	1

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PHYSICS**0625/12**

Paper 1 Multiple Choice (Core)

May/June 2024

MARK SCHEME

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Question	Answer	Marks
1	D	1
2	C	1
3	D	1
4	B	1
5	B	1
6	A	1
7	B	1
8	D	1
9	A	1
10	C	1
11	C	1
12	C	1
13	C	1
14	D	1
15	A	1
16	B	1
17	C	1
18	A	1
19	A	1
20	B	1
21	C	1
22	B	1
23	C	1
24	C	1
25	B	1
26	B	1
27	A	1
28	B	1

Question	Answer	Marks
29	B	1
30	B	1
31	D	1
32	B	1
33	D	1
34	A	1
35	D	1
36	C	1
37	D	1
38	C	1
39	A	1
40	A	1

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PHYSICS**0625/11**

Paper 1 Multiple Choice (Core)

May/June 2024

MARK SCHEME

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This document consists of **3** printed pages.

Question	Answer	Marks
1	A	1
2	D	1
3	A	1
4	C	1
5	D	1
6	B	1
7	C	1
8	A	1
9	B	1
10	A	1
11	A	1
12	A	1
13	C	1
14	D	1
15	B	1
16	D	1
17	A	1
18	D	1
19	D	1
20	C	1
21	B	1
22	D	1
23	A	1
24	A	1
25	B	1
26	B	1
27	D	1
28	B	1

Question	Answer	Marks
29	D	1
30	B	1
31	C	1
32	C	1
33	B	1
34	D	1
35	B	1
36	B	1
37	C	1
38	A	1
39	B	1
40	C	1



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PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

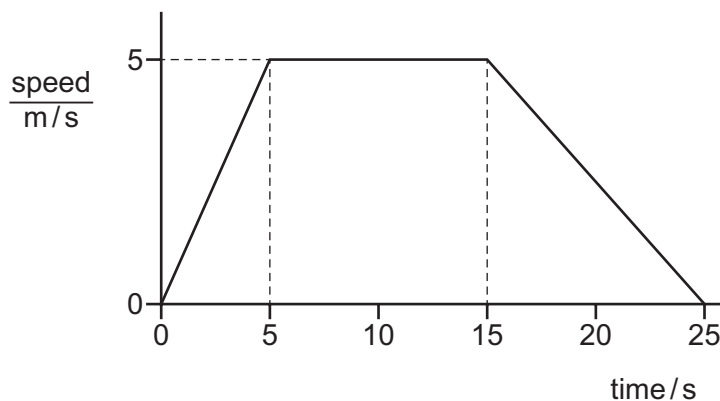
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages. Any blank pages are indicated.



- 1 Which piece of apparatus is used to measure the length of a copper rod of length approximately 2 cm?
- A** digital timer
- B** measuring cylinder
- C** ruler
- D** balance
- 2 The speed–time graph shows the motion of an object.



How far does the object travel at constant speed?

- A** 25 m **B** 50 m **C** 75 m **D** 125 m
- 3 A rock falls off a cliff onto a beach. The effect of air resistance on the rock is negligible.
- Which row describes the acceleration and speed of the rock as it falls?

	acceleration	speed
A	constant	constant
B	constant	increasing
C	increasing	constant
D	increasing	increasing

- 4 Two rectangular blocks consist of different materials.

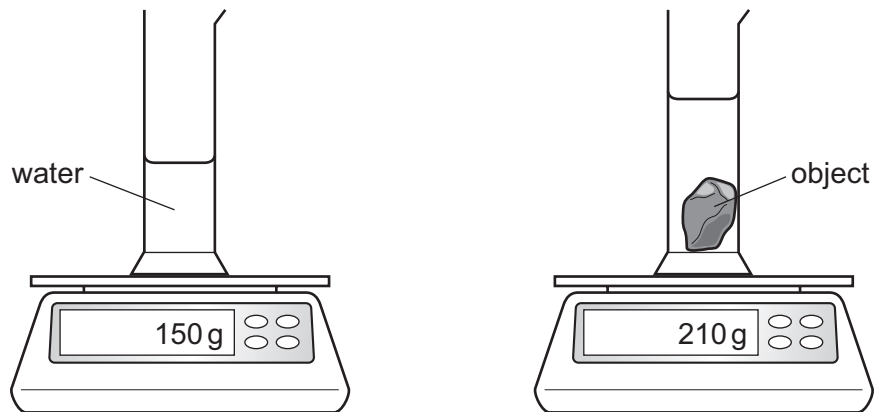
Four different methods are suggested to compare the two masses.

- 1 Compare the accelerations with which they fall freely.
- 2 Compare the values of their lengths $\times$ breadths $\times$ heights.
- 3 Hang each in turn from the same spring. Compare the extensions.
- 4 Place one in the right-hand pan of a beam balance and the other in the left-hand pan.

Which methods give a comparison of the two masses?

- A** 1, 2 and 3 **B** 1 and 2 only **C** 3 and 4 only **D** 4 only

- 5 A measuring cylinder containing 50 cm^3 of water is put on a balance.

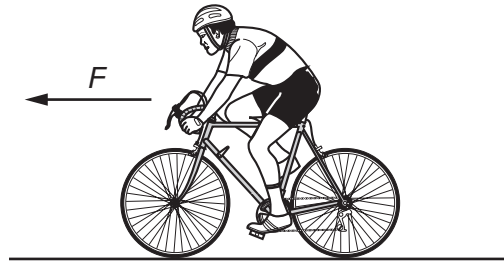


A solid object is put in the cylinder and the water level rises to 75 cm^3 .

What is the density of the object?

- A** 0.80 g/cm^3 **B** 2.4 g/cm^3 **C** 2.8 g/cm^3 **D** 8.4 g/cm^3

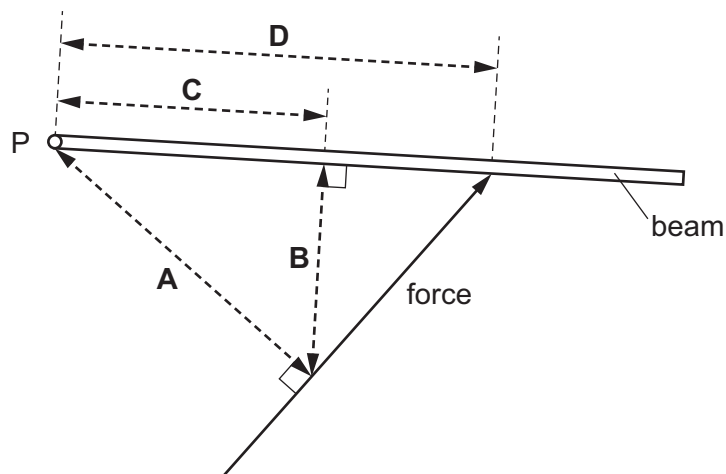
- 6 A cyclist is travelling in a straight line along a horizontal road at a constant speed.
A constant driving force F acts on the cyclist in the forward direction shown.



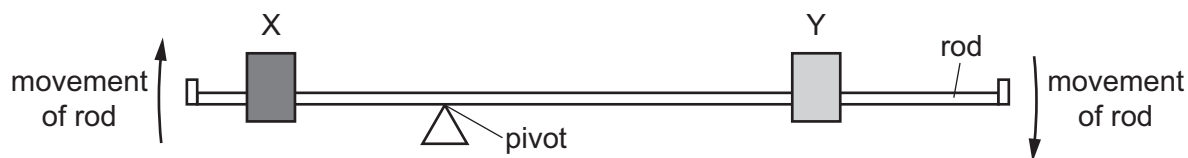
Which statement about the magnitude of the frictional forces acting on the cyclist is correct?

- A The magnitude is equal to F .
 - B The magnitude is smaller than F , but greater than zero.
 - C The magnitude is greater than F .
 - D The magnitude is zero.
- 7 A beam is pivoted at P. A force is applied to the beam.

Which distance is multiplied by the force to give the moment of the force about P?



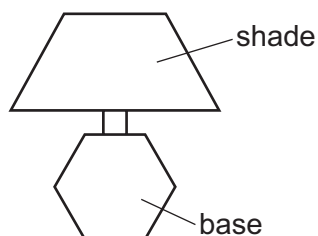
- 8 The diagram shows an unbalanced rod. Two loads X and Y can be moved along the rod.



The rod turns in a clockwise direction, as shown.

Which action could make the rod balance?

- A moving X to the left
 - B moving X to the right
 - C moving Y to the right
 - D moving the pivot to the left
- 9 It is important that an electric table lamp with a lamp shade does not get knocked over easily.



Which statement is correct?

- A The centre of mass must be as low as possible.
 - B The lamp must have a shade which is heavier than the base.
 - C The lamp must have a narrow base.
 - D The lamp shade must be wide.
- 10 A bicycle braking system transfers energy from a kinetic energy store to an internal energy store.
- A motor converts energy from a chemical energy store (battery) to a kinetic energy store.

What enables these energy transfers?

	braking system	motor
A	electrical work	mechanical work
B	electrical work	electrical work
C	mechanical work	mechanical work
D	mechanical work	electrical work

- 11 A toy car is pushed a distance of 2.4 m with a force equal to 460 N.

How much energy is transferred?

- A 190 J B 190 W C 1100 J D 1100 W

- 12 The engine of a motor vehicle develops a large power.

Which statement is correct?

- A The driving force acting on the vehicle must be large.
 B The engine must have a very large volume.
 C The engine must transfer large amounts of energy each second.
 D The vehicle must be very fast.

- 13 A solid cube has sides 0.50 m long and a mass of 120 kg. It stands on the ground on one face.

Which pressure does the cube exert on the ground?

- A 480 kg/m^3 B 960 kg/m^3 C 4700 N/m^2 D 9400 N/m^2

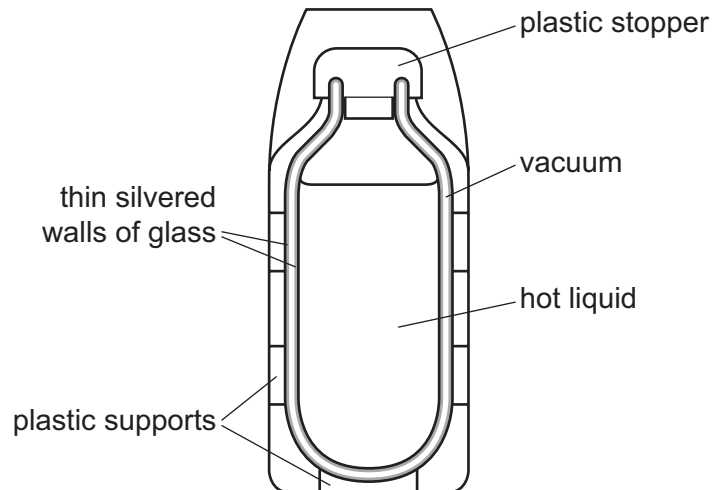
- 14 What is the lowest possible temperature (absolute zero) and what happens to the energy of particles at this temperature?

	lowest possible temperature / °C	particle energy
A	−273	particles have least kinetic energy
B	−273	particles have zero gravitational potential energy
C	0	particles have least kinetic energy
D	0	particles have zero gravitational potential energy

- 15 Which statement about the particles of a substance after condensation is correct?

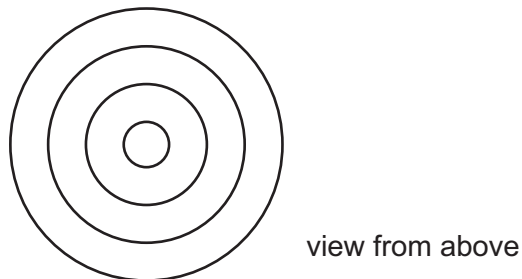
- A They are close to each other and slide over each other.
 B They are close to each other and vibrate about fixed points.
 C They are far apart from each other and vibrate about fixed points.
 D They are far apart from each other and move freely within the container.

- 16 The diagram shows a flask designed to reduce the loss of thermal energy from a hot liquid.



Which methods of thermal energy transfer are the silvered walls designed to reduce?

- A conduction, convection and radiation
 - B conduction and convection only
 - C conduction only
 - D radiation only
- 17 A drop of water from a tap falls onto the surface of some water of constant depth.



Water waves spread out on the surface of the water.

Which statement is correct?

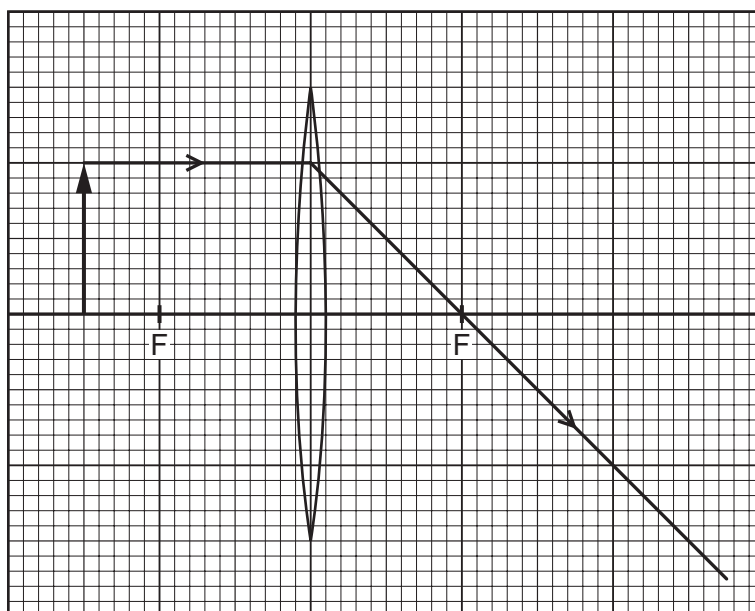
- A The waves are longitudinal and travel at the same speed in all directions.
- B The waves are longitudinal and travel more quickly in one direction than in others.
- C The waves are transverse and travel at the same speed in all directions.
- D The waves are transverse and travel more quickly in one direction than in others.

- 18 The optical image formed by a plane mirror is the same size as the object.

Which row describes the other characteristics of the optical image formed?

	distance from the mirror	real or virtual
A	same as object	virtual
B	smaller than object	virtual
C	larger than object	real
D	same as object	real

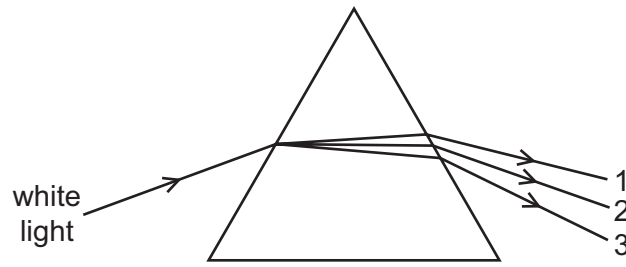
- 19 The diagram shows a partly completed scale drawing of an upright object placed 3 cm in front of a thin converging lens of focal length 2 cm.



What is the nature of the image formed by this lens?

- A** diminished, inverted and closer to the lens than the object
- B** diminished, upright and further from the lens than the object
- C** enlarged, inverted and closer to the lens than the object
- D** enlarged, inverted and further from the lens than the object

- 20 A narrow beam of white light passes through a prism and is dispersed into a spectrum.



Which row is correct?

	colour 1	colour 2	colour 3
A	blue	yellow	red
B	red	blue	yellow
C	red	yellow	blue
D	yellow	blue	red

- 21 Which row matches a region of the electromagnetic spectrum to one of its uses?

	region	use
A	gamma rays	intruder alarms
B	infrared	satellite television
C	microwaves	mobile (cell) phones
D	radio waves	sterilising food

- 22 A ship sounds its horn when it is 790 m from a cliff. A passenger on the ship hears the echo 4.8 s later.

What is the speed of the sound?

- A** 165 m/s **B** 330 m/s **C** 340 m/s **D** 1896 m/s

- 23 Which row gives the metal used to make the core of an electromagnet and one property of the electromagnet?

	metal	property
A	iron	permanent magnet
B	iron	temporary magnet
C	steel	permanent magnet
D	steel	temporary magnet

- 24** A plastic rod and a dry cloth are uncharged.

The rod is now rubbed with the cloth and they both become charged. The rod becomes negatively charged because some charged particles move from the cloth to the rod.

What is the charge on the cloth and which particles moved in the charging process?

	charge on cloth	particles that moved
A	negative	electrons
B	negative	neutrons
C	positive	electrons
D	positive	neutrons

- 25** A power supply of electromotive force (e.m.f.) 12 V is connected across a $3\ \Omega$ resistor.

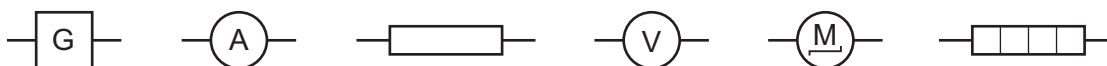
Which ammeter is most suitable to measure the current in the resistor?

- A** an ammeter with a range 0–0.5 A
B an ammeter with a range 0–1 A
C an ammeter with a range 0–5 A
D an ammeter with a range 0–50 A
- 26** The cost of electrical energy is \$0.25 for each unit of 1 kWh. A 2200 W heater is switched on for 48 minutes.

What is the cost of this use?

- A** \$0.44 **B** \$0.55 **C** \$26 **D** \$440

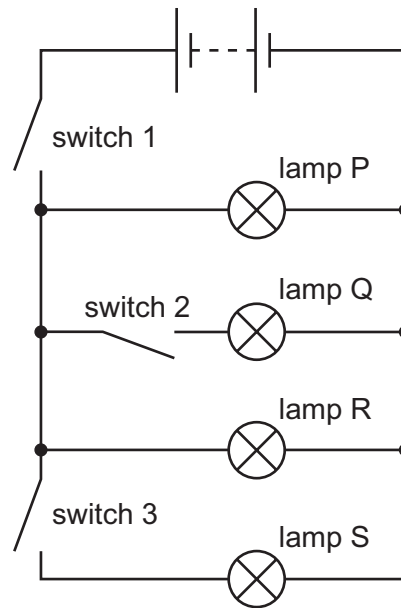
- 27** The diagram shows six different electrical circuit components.



Which circuit symbol is **not** present in the diagram?

- A** resistor
B voltmeter
C generator
D thermistor

- 28 The circuit shown contains three switches and four lamps P, Q, R and S.



Which switches must be closed to light only lamps P and R?

- A** switch 1 only
- B** switch 1 and switch 2
- C** switch 1 and switch 3
- D** switch 2 and switch 3

- 29 An electric heater has a metal frame.

The heating element is connected to the live and neutral wires of an a.c. supply. The metal frame is connected to the earth wire.

Which row gives the correct connections for the fuse and the switch?

	fuse	switch
A	in the earth	in the live
B	in the earth	in the neutral
C	in the live	in the live
D	in the live	in the neutral

30 Which electrical device uses the turning effect produced by a current-carrying coil in a magnetic field?

- A** a.c. generator
- B** d.c. motor
- C** relay
- D** transformer

31 Four appliances all use an electric current to operate.

Which appliance uses the magnetic effect of the current?

- A** a heater
- B** a light bulb
- C** a relay
- D** a remote controller

32 A student makes four different types of transformer. She counts the number of turns on the primary and secondary coils. She labels each transformer as either step-up or step-down.

Which row shows the correct labels?

	number of turns on the primary coil	number of turns on the secondary coil	step-up or step-down transformer
A	5	5	step-up
B	10	5	step-up
C	10	20	step-down
D	20	10	step-down

33 How is electricity transmitted over large distances and why is it transmitted in this way?

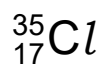
	how	why
A	at high voltage	for safety
B	at high voltage	to reduce energy loss
C	at low voltage	for safety
D	at low voltage	to reduce energy loss

- 34 An atom of an element contains electrons, neutrons and protons.

Which particles are found in the nucleus?

- A electrons and neutrons only
- B electrons and protons only
- C neutrons and protons only
- D electrons, neutrons and protons

- 35 A nuclide of chlorine has the symbol shown.



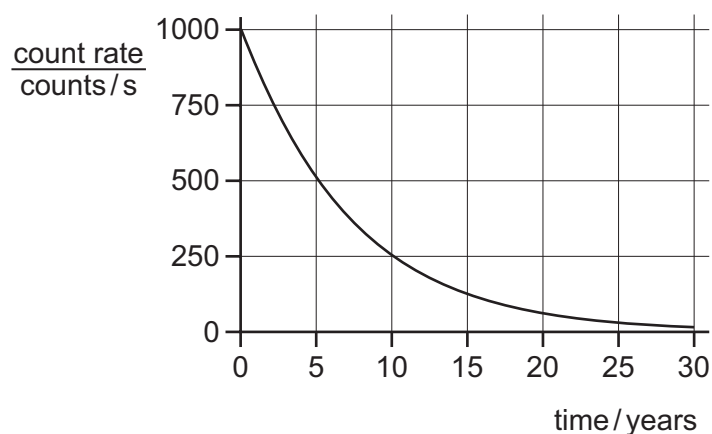
What is the nucleon number of this nuclide of chlorine?

- A 17 B 18 C 35 D 52

- 36 What is a major source of background radiation?

- A cosmic rays
- B microwaves from mobile (cell) phones
- C nuclear power stations
- D visible light from the Sun

- 37 The graph shows the radioactive decay curve of a substance.



What is the half-life of this substance?

- A 0.5 years B 5 years C 15 years D 30 years

38 Which row about the orbits of the Earth and the Moon is correct?

	approximate time for the Earth to orbit the Sun	approximate time for the Moon to orbit the Earth
A	1 day	30 days
B	30 days	1 day
C	365 days	1 day
D	365 days	30 days

39 Which description of a galaxy is correct?

- A** a collection of billions of stars
- B** a collection of gaseous planets orbiting a star
- C** a collection of rocky planets orbiting a star
- D** all of the stars that exist

40 Light from distant stars is redshifted.

What is redshift?

- A** a decrease in observed wavelength caused by a star moving away from the Earth
- B** a decrease in observed wavelength caused by a star moving towards the Earth
- C** an increase in observed wavelength caused by a star moving away from the Earth
- D** an increase in observed wavelength caused by a star moving towards the Earth

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

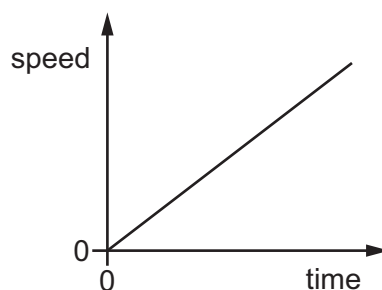
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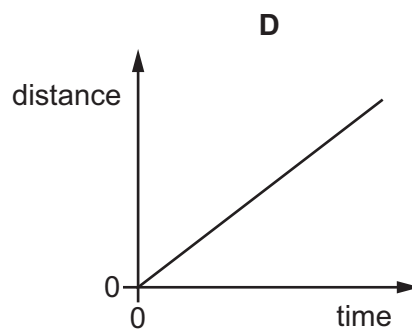
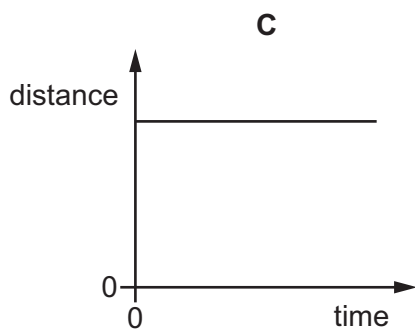
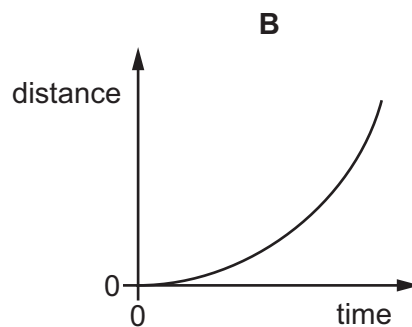
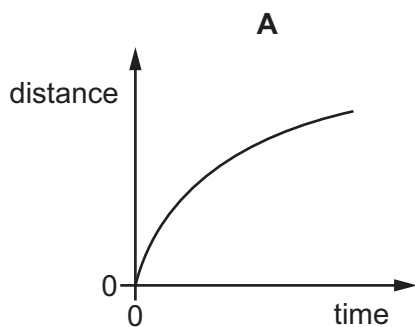
- 1 Which single apparatus is used to find the volume of a solid cube and which single apparatus is used to find the volume of a quantity of liquid?

	volume of solid cube	volume of liquid
A	balance	balance
B	balance	measuring cylinder
C	ruler	balance
D	ruler	measuring cylinder

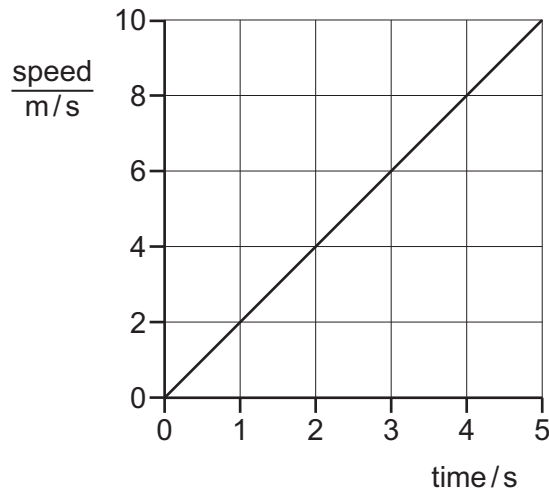
- 2 The speed–time graph represents a short journey.



Which distance–time graph represents the same journey?



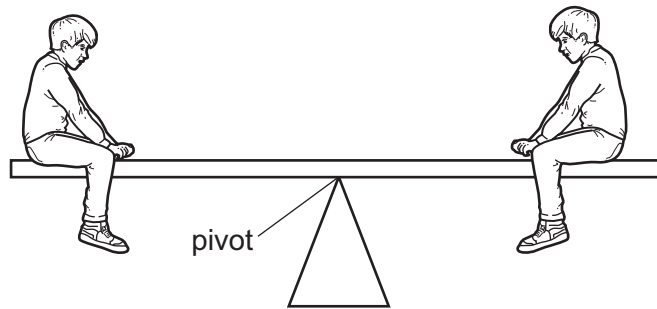
- 3 The graph represents the motion of a car.



How far has the car moved between 0 and 5 s?

- A** 2 m **B** 10 m **C** 25 m **D** 50 m
- 4 Which statement about mass or weight is **not** correct?
- A** Masses can be compared using a balance.
B Mass is a force.
C Weights can be compared using a balance.
D Weight is a force.
- 5 Which two quantities must be known to determine the density of a material?
- A** mass and area
B mass and volume
C weight and area
D weight and volume

- 6 Two boys are sitting on a see-saw. The see-saw is in equilibrium and remains horizontal.

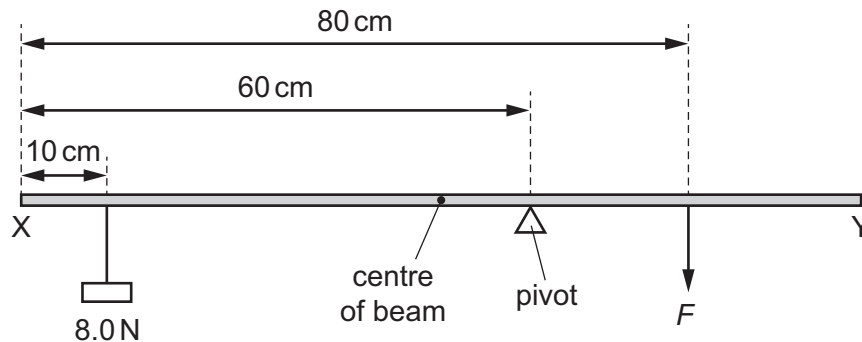


What affects the moment of each boy about the pivot?

- A** his distance from the pivot only
B his height above the ground and his weight
C his weight only
D his weight and distance from the pivot
- 7 A uniform metre rule is pivoted in equilibrium at the 50 cm mark. A mass of 25 g is placed at the 30 cm mark on the rule.

What is the smallest mass that can be placed on the rule to restore equilibrium?

- A** 5 g **B** 10 g **C** 15 g **D** 25 g
- 8 A uniform beam XY is 100 cm long and weighs 4.0 N.



The beam rests on a pivot 60 cm from end X.

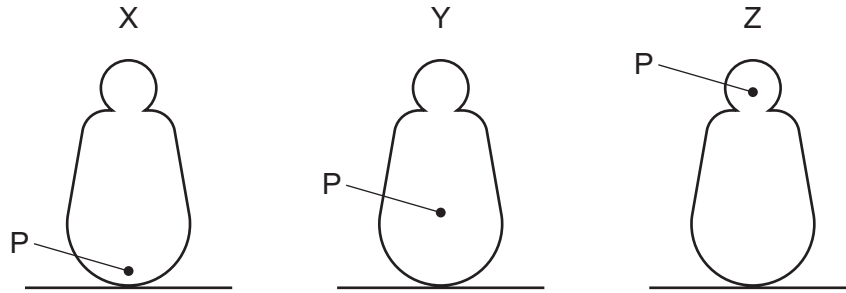
A load of 8.0 N hangs from the beam 10 cm from end X.

The beam is kept balanced by a force F acting on the beam 80 cm from end X.

What is the magnitude of force F ?

- A** 8.0 N **B** 18 N **C** 22 N **D** 44 N

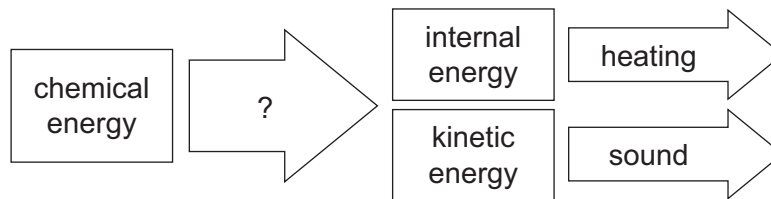
- 9 Three children's toys, X, Y and Z, are the same size and shape. They have weights at different positions inside so that the position of the centre of gravity of each toy is different. Each toy's centre of gravity is marked P.



Which toy is the most stable and which toy is the least stable when balanced in the positions shown?

	most stable	least stable
A	X	Y
B	X	Z
C	Y	X
D	Y	Z

- 10 The diagram shows the energy stores for a mobile (cell) phone and how the energy is transferred between stores.



What describes how the chemical energy is transferred?

- A** electrical work done
 - B** mechanical work done
 - C** electromagnetic waves
 - D** sound waves
- 11 A moving object is brought to rest by a resistive force of 50 N over a distance of 5.0 m.

What is the work done by the force?

- A** 0.10 J
- B** 10 J
- C** 55 J
- D** 250 J

- 12 Which two physical quantities must be used to calculate the power developed by a student running up a flight of steps?
- A force exerted and the vertical height of the steps only
 - B force exerted and the time taken only
 - C work done and the vertical height of the steps only
 - D work done and the time taken only

- 13 A rectangular marble block has dimensions 1 m by 1 m by 5 m and weighs 125 000 N.

The marble block is stored with the long side resting on the ground, as in diagram 1.

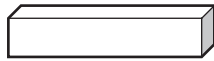


diagram 1



diagram 2

What is the change in the pressure on the ground due to the block when the block is stored as in diagram 2 rather than diagram 1?

- A a decrease of $25\,000\text{ N/m}^2$
 - B an increase of $100\,000\text{ N/m}^2$
 - C an increase of $125\,000\text{ N/m}^2$
 - D no change
- 14 Four students describe the phrase ‘absolute zero’ during a lesson on the particle model.
- Which student is correct?
- A This is the lowest possible temperature.
 - B Particles in a solid start vibrating.
 - C Particles do not have any weight.
 - D Particles have the least gravitational potential energy.

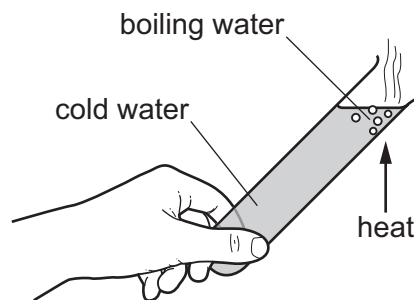
- 15** At the surface of a liquid, the more energetic molecules can escape from the liquid into the atmosphere.

Which name is given to this process?

- A** boiling
- B** condensation
- C** evaporation
- D** melting

- 16** A teacher puts some cold water in a test-tube.

She holds the bottom of the test-tube while heating the top.

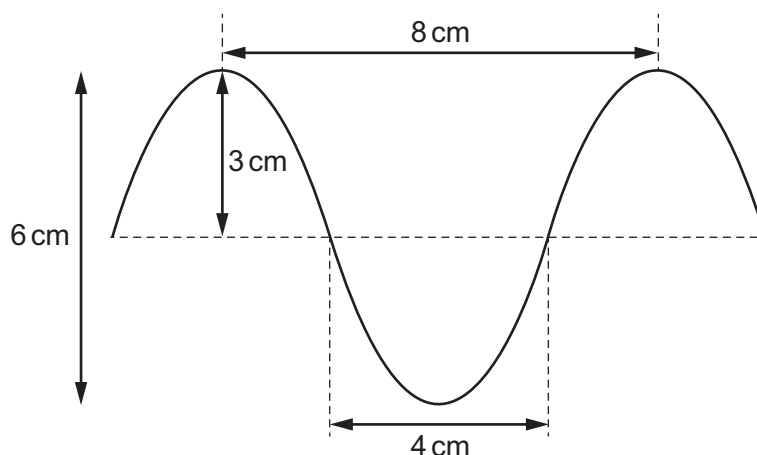


The water at the top boils but she continues to hold the test-tube as the bottom remains cold.

Which conclusion about water is made from this experiment?

- A** Water is a bad conductor.
- B** Water is a bad convector.
- C** Water is a good conductor.
- D** Water is a good convector.

17 The diagram shows a wave.



What are the amplitude and the wavelength of this wave?

	amplitude / cm	wavelength / cm
A	3	4
B	3	8
C	6	4
D	6	8

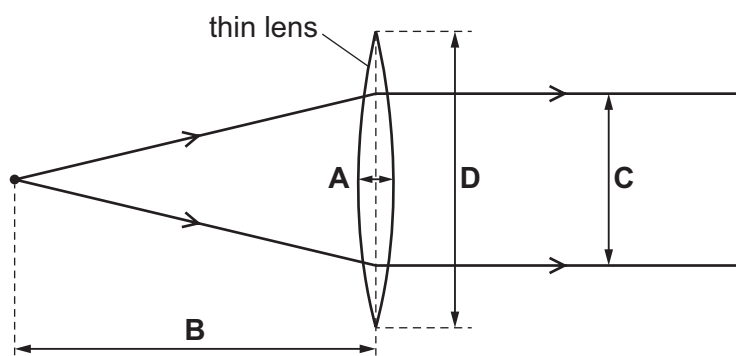
18 A light ray strikes a plane mirror and is reflected.

Which angle is always equal in size to the angle of reflection?

- A** the angle between the incident ray and the mirror
- B** the angle between the incident ray and the normal to the mirror
- C** the angle between the reflected ray and the mirror
- D** the angle between the reflected ray and the incident ray

19 The diagram shows two diverging rays of light passing through a lens and emerging parallel to each other.

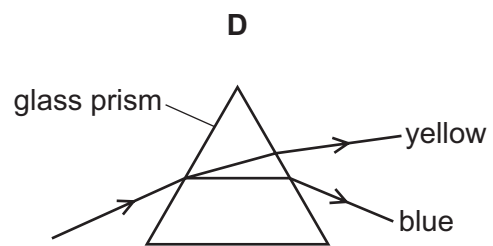
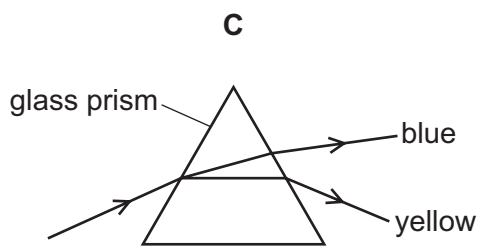
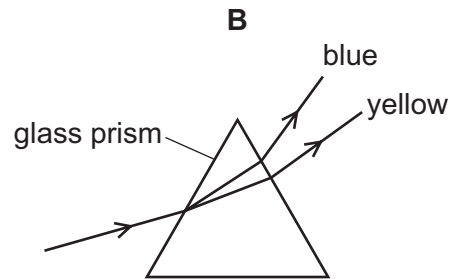
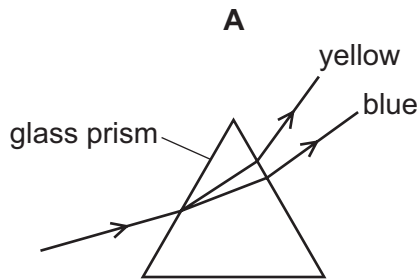
Which labelled distance is the focal length of the lens?



20 A beam of light consists of yellow and blue light.

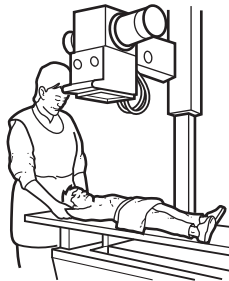
The beam of light is incident on a glass prism.

Which diagram is correct?



21 The two devices shown use different types of electromagnetic waves.

medical scanning



remote controller



Which types of waves are used in these devices?

	medical scanning	remote controller
A	ultraviolet	infrared
B	ultraviolet	microwaves
C	X-rays	infrared
D	X-rays	microwaves

- 22** A boy shouts and hears the echo from a tall building 2.2 s later.

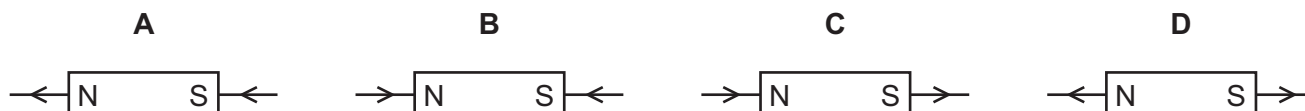
The speed of sound in air is 330 m/s.

How far away from the boy is the building?

- A** 150 m **B** 300 m **C** 360 m **D** 730 m

- 23** The magnetic field of a bar magnet can be represented by magnetic field lines.

Which diagram shows two magnetic field lines correctly?



- 24** A plastic rod is rubbed with a dry cloth. The rod becomes positively charged.

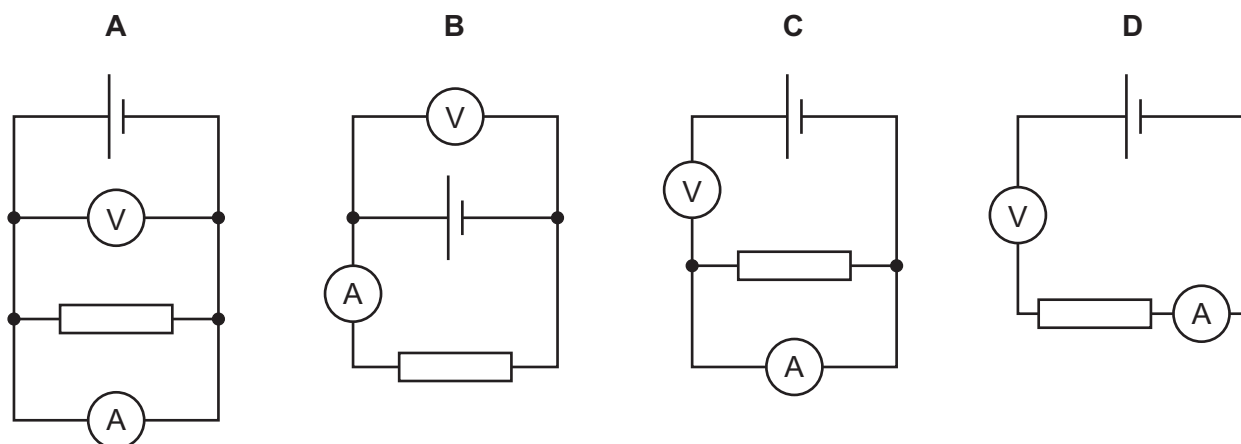
Why has the rod become positively charged?

- A** It has gained electrons.
B It has gained neutrons.
C It has lost electrons.
D It has lost neutrons.

- 25** Which statement about electric current in a conductor is correct?

- A** In a d.c. circuit, the electric current gradually decreases along the conductor.
B In a d.c. circuit, the free electrons flow back and forth.
C In an a.c. circuit, the electric current remains exactly the same all the time.
D In an a.c. circuit, the flow of charge changes direction continually.

- 26** Which circuit can be used to measure the resistance of a resistor?



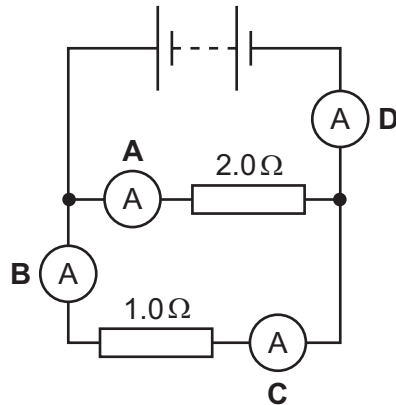
- 27 A lamp rated 12 V, 2.0 A is switched on for 60 s.

How much energy is transferred?

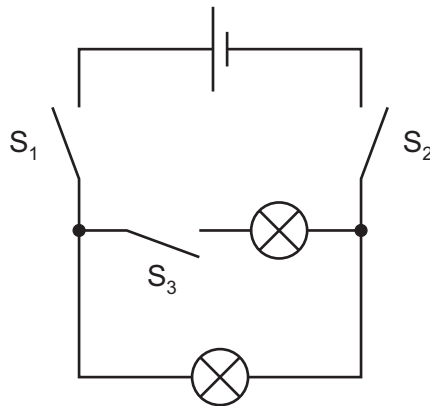
- A 0.40 J B 10 J C 360 J D 1400 J

- 28 The circuit diagram shows a battery connected to two resistors. Four labelled ammeters are connected into the circuit.

Which ammeter shows the largest reading?



- 29 Two lamps are connected in parallel.



Which switches must be closed so that both lamps light?

- A S_1 and S_2 only
 B S_1 and S_3 only
 C S_2 and S_3 only
 D S_1 , S_2 and S_3

30 Which statement about electrical safety is correct?

- A If a device is double insulated, it does not need a fuse.
- B A device that has a normal operating current of 3.0 A must be protected by a 3.0 A fuse.
- C Switches must always be connected into the live supply wire.
- D The metal casing of an electrical device must be connected to the neutral wire.

31 A student investigates the output voltage induced across a coil of wire by a bar magnet.

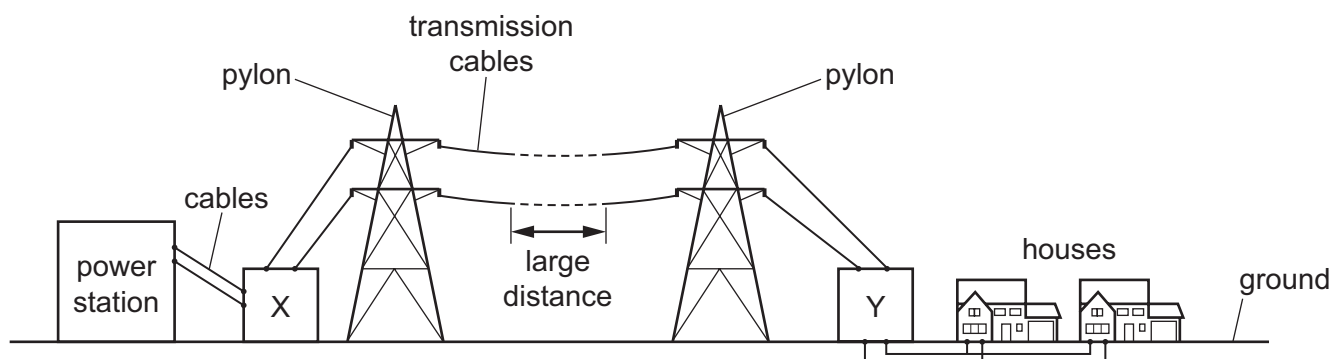
When will the induced voltage have the greatest value?

- A The student slowly moves the bar magnet into the coil of wire.
- B The student leaves the bar magnet stationary in the coil of wire.
- C The student quickly removes the bar magnet from the coil of wire.
- D The student places the bar magnet at rest outside the coil of wire.

32 In which device is the magnetic effect of a current **not** used?

- A electromagnet
- B loudspeaker
- C potential divider
- D relay

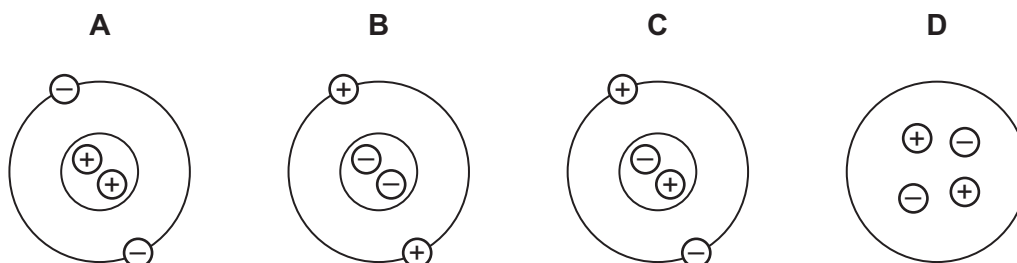
- 33 The diagram represents the transmission of electricity from a power station to homes that are many kilometres away. Two transformers are labelled X and Y.



What type of transformers are X and Y?

	X	Y
A	step-down transformer	step-down transformer
B	step-down transformer	step-up transformer
C	step-up transformer	step-down transformer
D	step-up transformer	step-up transformer

- 34 Which diagram represents the positions of the charged particles of an atom?



- 35 The table shows the composition of three different nuclei.

nucleus	number of protons	number of neutrons
X	3	3
Y	3	4
Z	4	3

Which nuclei are isotopes of the same element?

- A** X, Y and Z **B** X and Y only **C** X and Z only **D** Y and Z only

- 36 A sample of a radioactive isotope emits 9600 α -particles per second.

After 40 hours the rate of emission has fallen to 600 α -particles per second.

What is the half-life of this isotope?

- A** 4.0 hours **B** 8.0 hours **C** 10 hours **D** 20 hours

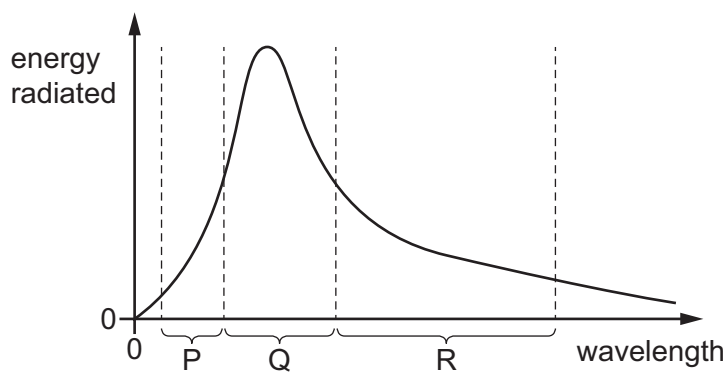
- 37 Which row states a harmful effect and a beneficial effect of ionising radiation on living things?

	harmful effect	beneficial effect
A	kills cancer cells	kills cancer cells
B	kills cancer cells	mutates living cells
C	mutates living cells	kills cancer cells
D	mutates living cells	mutates living cells

- 38 Which statement about the Solar System is correct?

- A** All the planets are rocky.
B Only the Earth has a moon.
C Pluto is a dwarf planet.
D There are many stars in the Solar System.

- 39 The graph shows the energy radiated by the Sun at different wavelengths. Most of the energy is radiated in just three parts of the electromagnetic spectrum, labelled P, Q and R.



Which parts of the electromagnetic spectrum are P and R?

	P	R
A	gamma ray	radio
B	infrared	ultraviolet
C	radio	gamma ray
D	ultraviolet	infrared

40 An astronomer observes redshift in the light from a distant galaxy.

Which statement about redshift is correct?

- A** It is the decrease in the observed wavelength of red light emitted from receding galaxies.
- B** It is evidence that the Universe is contracting and supports the Big Bang Theory.
- C** It is evidence that the Universe is expanding and supports the Big Bang Theory.
- D** Redshift is when light from receding galaxies appears blue.

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Cambridge IGCSE™

PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2023

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 9.8 N (acceleration of free fall = 9.8 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

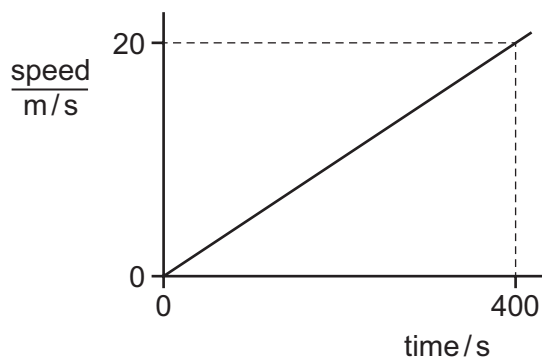
This document has **16** pages.



1 Which unit is a unit of weight?

- A kilogram
- B kilojoule
- C kilometre
- D kilonewton

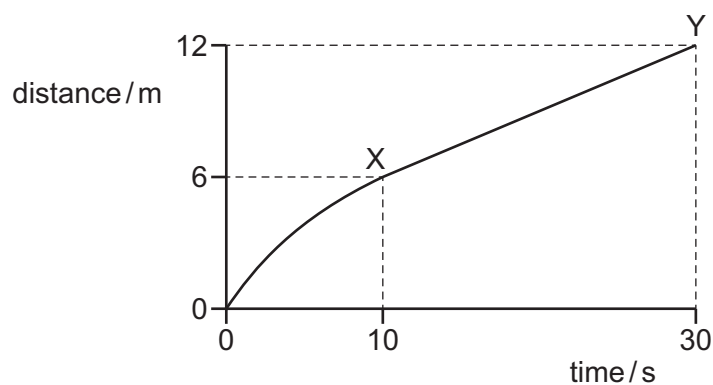
2 The graph represents the motion of a vehicle.



What is the distance travelled by the vehicle in 400 s?

- A 20 m B 400 m C 4000 m D 8000 m

3 The diagram shows a distance–time graph for an object moving in a straight line.



What is the average speed between X and Y?

- A 0.20 m/s B 0.30 m/s C 0.40 m/s D 0.60 m/s

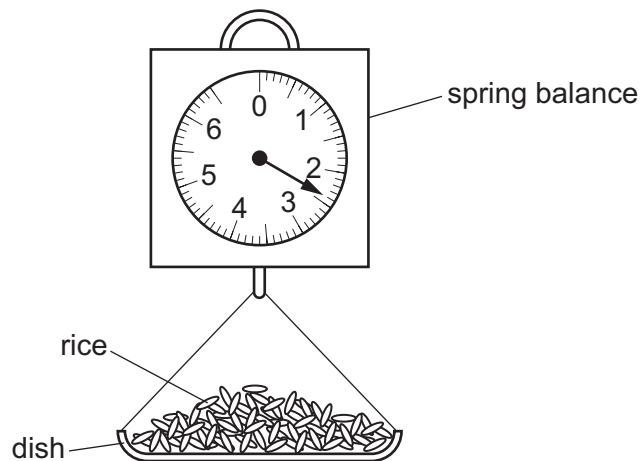
- 4 A space rocket travels to the Moon.

The acceleration of free fall is greater on the Earth than it is on the Moon.

How do the mass and weight of the space rocket on the Moon compare with their values on the Earth?

	mass on the Moon	weight on the Moon
A	less than on the Earth	same as on the Earth
B	less than on the Earth	more than on the Earth
C	same as on the Earth	less than on the Earth
D	same as on the Earth	more than on the Earth

- 5 A shopkeeper pours rice into a dish that hangs from a spring balance. He records the reading.

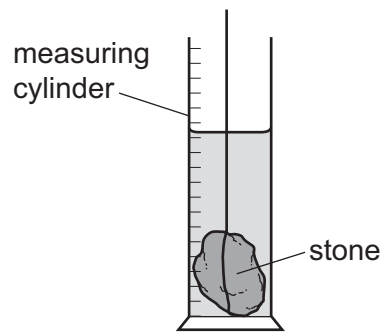


A customer buys some pasta. The shopkeeper notices that the reading on the spring balance, with just pasta in the dish, is the same as it was with just rice in the dish.

Which quantity **must** be the same for the rice and for the pasta?

- A** density
- B** temperature
- C** volume
- D** weight

- 6 A student determines the density of an irregularly shaped stone. The stone is slowly lowered into a measuring cylinder partly filled with water.

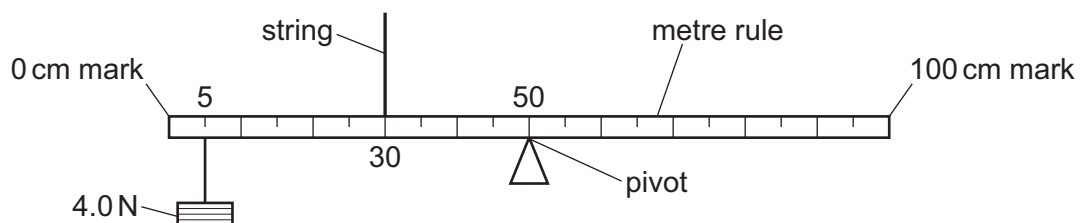


Which other apparatus does the student need to calculate the density of the irregularly shaped stone?

- A a balance
 - B a thermometer
 - C a metre rule
 - D a stop-watch
- 7 Four objects are moving.

Which object has a zero resultant force acting on it?

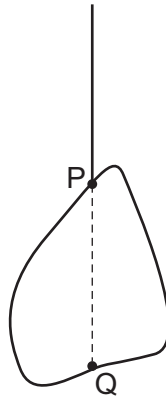
- A the object moving at a decreasing speed
 - B the object moving at an increasing speed
 - C the object moving at a constant speed in a circle
 - D the object moving at a constant speed in a straight line
- 8 The diagram shows a uniform metre rule. The rule is pivoted at its mid-point. A weight of 4.0 N is suspended from the rule at the 5 cm mark. The rule is held by a string at the 30 cm mark. The rule is in equilibrium.



What is the upward force that the string exerts on the rule?

- A 0.67 N
- B 4.0 N
- C 6.0 N
- D 9.0 N

- 9 An irregularly shaped metal plate is freely suspended from a point P and is in equilibrium, as shown. Point Q is vertically below P.



Which statement about the position of the centre of gravity of the metal plate is correct?

- A It is at P.
 - B It is at Q.
 - C It is halfway between P and Q.
 - D Further investigation is needed to determine its position.
- 10 Which row about the change of energy in the energy store must be correct?

	process	energy store	change of energy in store
A	water pumped up to a high-altitude dam	gravitational potential energy of water	increases
B	water pumped up to a high-altitude dam	kinetic energy of water	decreases
C	air passes through a wind turbine	gravitational potential energy of air	increases
D	air passes through a wind turbine	kinetic energy of air	increases

- 11 A rock of weight 50 N falls a vertical distance of 7.0 m from rest.

What is the change in the gravitational potential energy store of the rock?

- A decrease of 7.1 J
- B decrease of 350 J
- C increase of 7.1 J
- D increase of 350 J

- 12** A child runs up a set of stairs four times. The time taken for each run is recorded.

Which time is measured when the child's useful power is greatest?

- A** 10 s **B** 20 s **C** 30 s **D** 40 s

- 13** A woman has a weight of 600 N. She stands on a horizontal floor. The area of her feet in contact with the floor is 0.050 m^2 .

What is the pressure she exerts on the floor?

- A** $1.2 \times 10^3 \text{ N/m}^2$
B $2.4 \times 10^3 \text{ N/m}^2$
C $1.2 \times 10^4 \text{ N/m}^2$
D $2.4 \times 10^4 \text{ N/m}^2$

- 14** Gases, liquids and solids are made up of small particles.

Which row gives the relative separation of the particles?

	gas	liquid	solid
A	far apart	far apart	far apart
B	far apart	far apart	close together
C	far apart	close together	close together
D	close together	close together	far apart

- 15** Which statements about evaporation of water are correct?

- 1 Evaporation causes the remaining liquid to cool.
- 2 During evaporation, the more energetic particles escape from the surface of the liquid.
- 3 Evaporation only happens at 100°C .

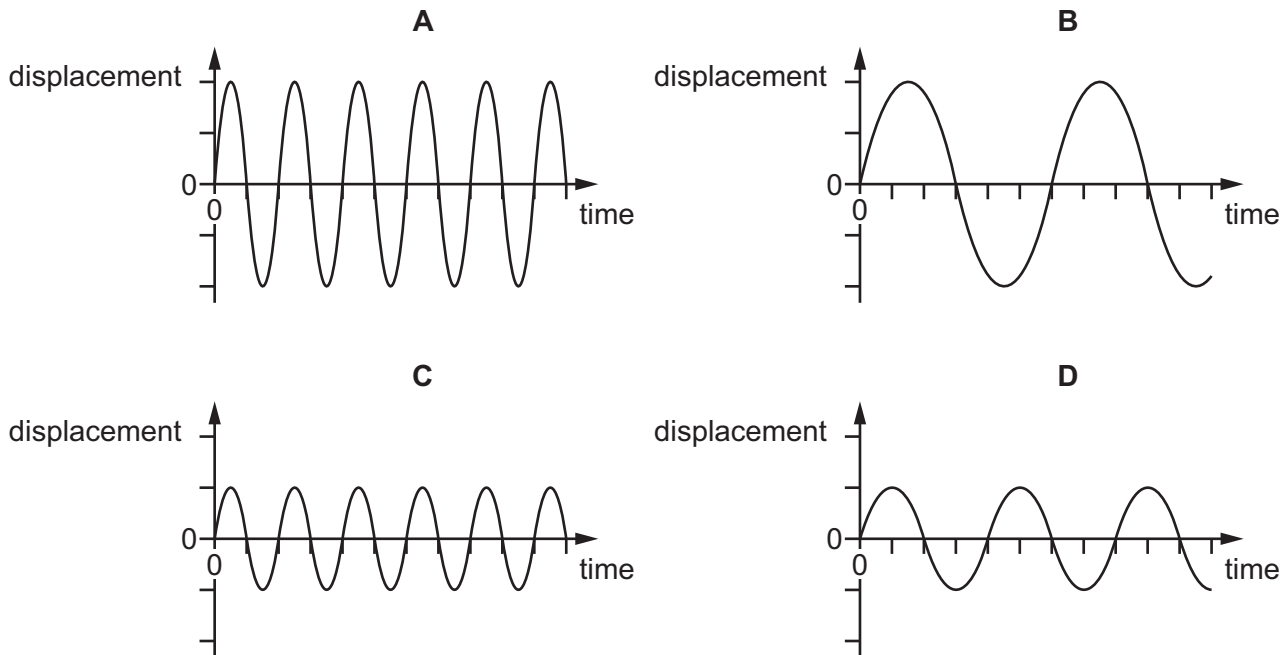
- A** 1, 2 and 3 **B** 1 and 2 only **C** 1 and 3 only **D** 2 and 3 only

- 16** Which surface is the worst absorber of infrared radiation?

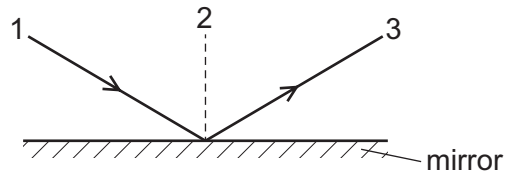
- A** dull black
B dull white
C shiny black
D shiny white

- 17 The diagrams show graphs of displacement against time for four waves. All the graphs are drawn to the same scale.

Which wave has the largest amplitude and the highest frequency?



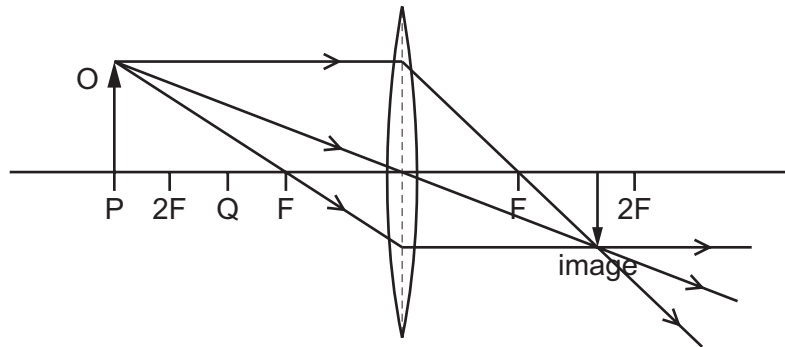
- 18 A student draws a diagram to show the directions of a light ray reflecting off a plane mirror.



What are the correct terms for the lines drawn?

	normal	incident ray	reflected ray
A	1	2	3
B	1	3	2
C	2	1	3
D	2	3	1

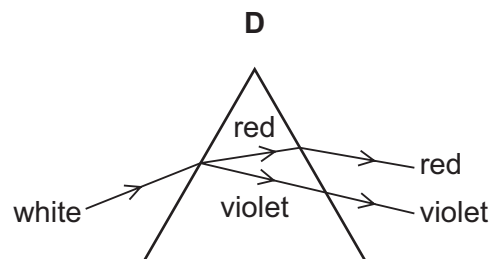
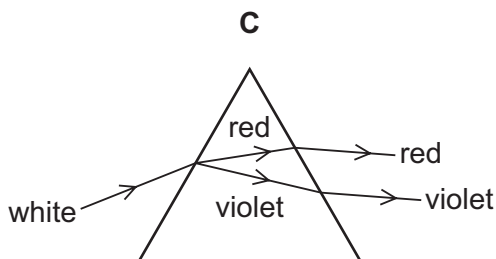
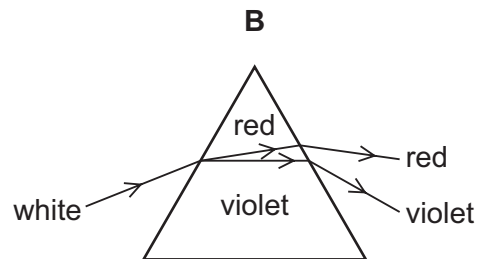
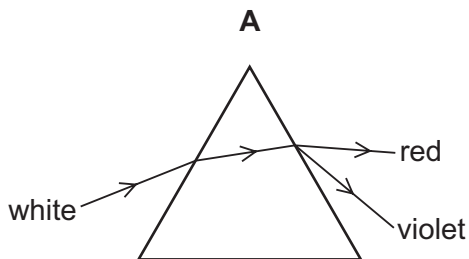
- 19** An object O is placed at point P near to a thin converging lens. The diagram shows three rays from the top of O passing through the lens. Each point F is one focal length from the centre of the lens. Each point $2F$ is two focal lengths from the centre of the lens.



The object O is moved to point Q on the diagram.

Which type of image is produced when the object O is at point Q ?

- A** inverted and the same size as the object
 - B** inverted and enlarged
 - C** upright and the same size as the object
 - D** upright and enlarged
- 20** Which diagram shows the dispersion of white light by a glass prism?



21 Which statement is correct?

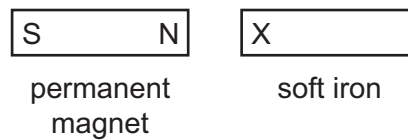
- A A remote controller emits microwave radiation.
- B A remote controller emits infrared radiation.
- C A remote controller emits ultrasound.
- D A remote controller emits ultraviolet radiation.

22 Student X fires a starting pistol which produces smoke and sound. Student Y is standing 100 m away and sees the smoke the instant it is produced. The speed of sound in air is 340 m/s.

What is the time delay between student Y seeing the smoke and hearing the sound?

- A 0.29 s B 0.59 s C 1.7 s D 3.4 s

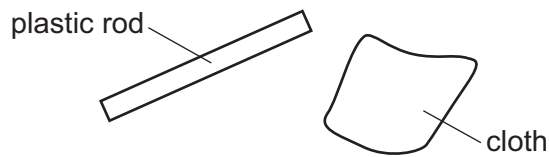
23 An unmagnetised piece of soft iron is placed close to a strong permanent magnet, as shown.



What is the induced polarity of end X of the soft iron and in which direction does the magnetic force act on the soft iron?

	polarity of end X	direction of force on the soft iron
A	N	to the left
B	N	to the right
C	S	to the left
D	S	to the right

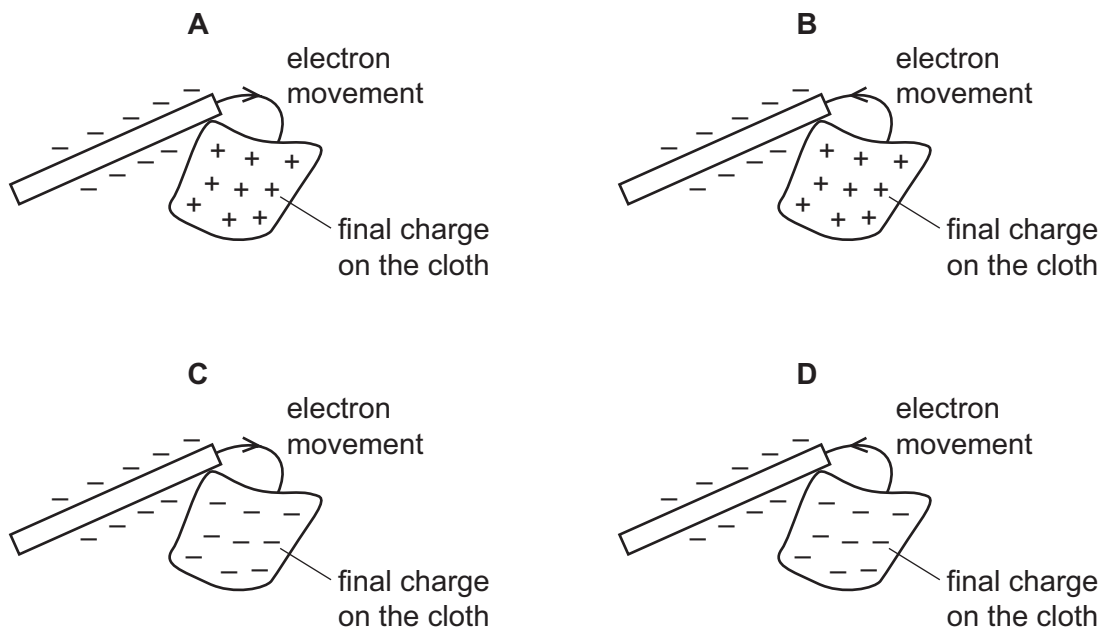
- 24** A plastic rod is rubbed with a cloth.



The rod and the cloth both become charged as electrons move between them.

The rod becomes negatively charged.

Which diagram shows how the rod becomes negatively charged and shows the final charge on the cloth?

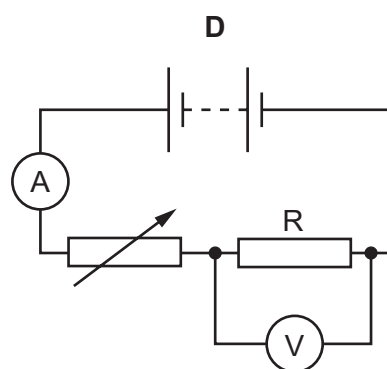
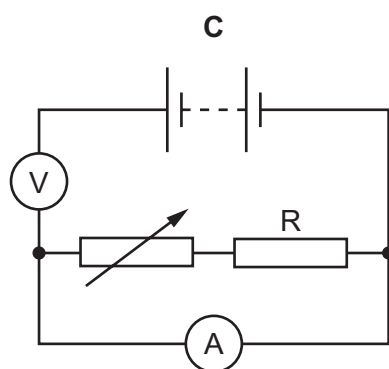
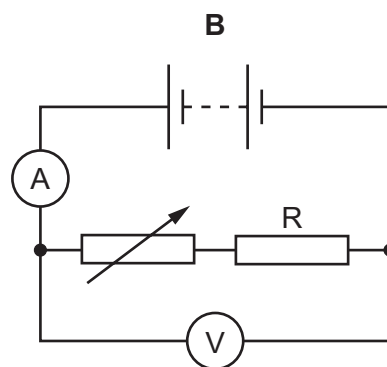
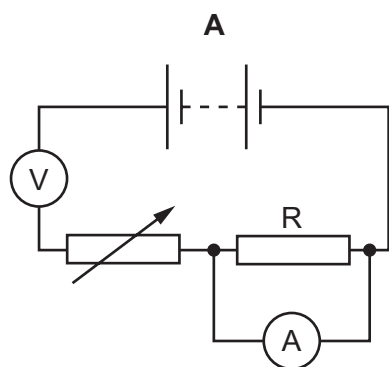


- 25** Which particles move to cause a current in a copper wire?

- A** copper atoms
- B** electrons from the copper atoms
- C** protons from the copper nuclei
- D** neutrons from the copper nuclei

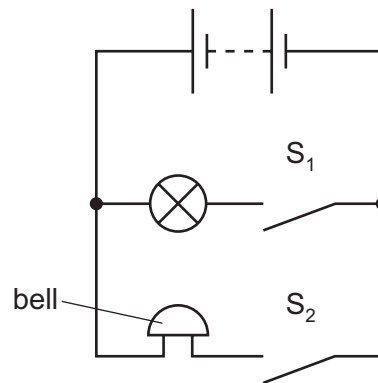
- 26 Four students draw a circuit diagram of the apparatus used to measure the resistance of resistor R .

Which circuit is correct?



27 The battery on a bicycle is connected in parallel to its lamp and bell.

The circuit includes two switches, S_1 and S_2 .



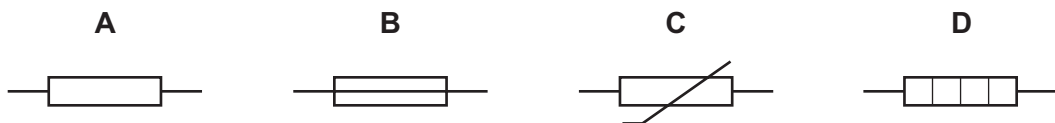
The cyclist closes S_1 to light the lamp.

She then also closes S_2 to sound the bell.

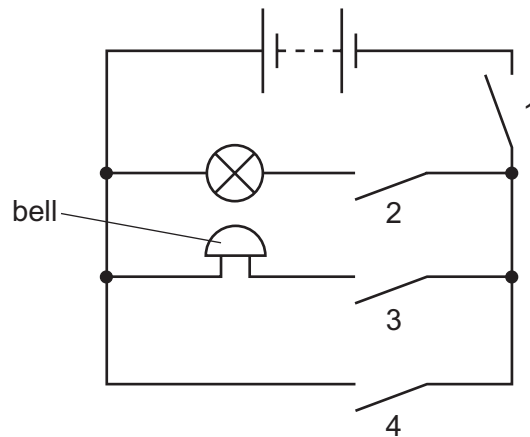
What happens to the current in the battery and the power output from the battery when the cyclist closes S_2 ?

	current in the battery	power output from the battery
A	increases	increases
B	increases	stays the same
C	stays the same	increases
D	stays the same	stays the same

28 Which symbol represents an electric heater?



29 A student connects the circuit shown.



Which switches must be closed for the bell to ring without lighting the lamp?

- A** 1 and 2 only **B** 1 and 3 only **C** 1, 3 and 4 **D** 2, 3 and 4

30 A double-insulated electrical appliance must be connected safely to the electricity supply.

Which statement is correct?

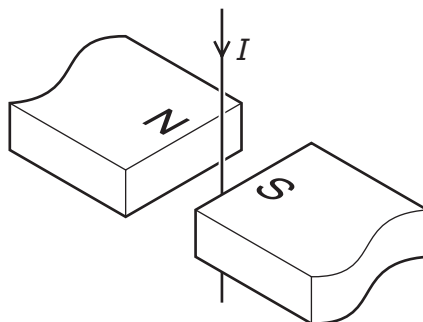
- A** It must be connected with a fuse and an earth wire.
B It can be connected with a fuse only.
C It can be connected with an earth wire only.
D It does not need a fuse or an earth wire.

31 A simple electric generator induces an electromotive force (e.m.f.).

Which modification would increase the induced e.m.f.?

- A** Increase the number of turns in the coil of the generator.
B Increase the distance between the magnetic poles.
C Reduce the strength of the magnetic field around the coil.
D Reverse the direction of the magnetic field.

- 32 The diagram shows a vertical wire carrying a current I placed between the poles of a magnet.



What is the direction of the force on the wire exerted by the magnetic field?

- A from N to S
 B from S to N
 C horizontal and at right angles to the direction from N to S
 D parallel to the wire
- 33 What is an advantage of transmitting electricity at a high voltage?
- A It is faster.
 B It is safer.
 C Less energy is wasted.
 D Less equipment is needed.
- 34 How are positive and negative ions formed from atoms?

	positive ion	negative ion
A	add positive charge to the nucleus	add an electron to the atom
B	add positive charge to the nucleus	remove positive charge from the nucleus
C	remove an electron from the atom	add an electron to the atom
D	remove an electron from the atom	remove positive charge from the nucleus

- 35 A nucleus of element X is represented as ${}^{56}_{26}\text{X}$.

Which is an isotope of element X?

- A ${}^{26}_{56}\text{X}$ B ${}^{54}_{26}\text{X}$ C ${}^{56}_{24}\text{X}$ D ${}^{54}_{28}\text{X}$

36 Which statement about the random decay of the nuclei in a sample of uranium-238 is correct?

- A** The probabilities of an alpha-particle, a beta-particle or a gamma ray being emitted from a nucleus in the sample are equal.
- B** The probability of a nucleus in the sample decaying decreases as time passes.
- C** The probability of a nucleus decaying in any ten minute interval is the same for all the nuclei in the sample.
- D** The probability of a nucleus in the sample decaying increases as time passes.

37 The count rate due to a sample of a radioactive isotope is measured for 80 minutes.

time / minutes	count rate counts/second
0	480
20	380
40	300
60	240
80	190

What is the half-life of the isotope?

- A** 20 minutes **B** 40 minutes **C** 60 minutes **D** 80 minutes

38 Which planet in our Solar System is nearest to the Sun and what is the nature of the planet?

	planet	nature
A	Mercury	rocky
B	Mercury	gaseous
C	Venus	rocky
D	Venus	gaseous

39 The Sun consists mostly of two elements.

What are these two elements?

- A** helium and nitrogen
- B** hydrogen and helium
- C** hydrogen and oxygen
- D** oxygen and nitrogen

40 Which statement about the Milky Way is correct?

- A** It is a galaxy.
- B** It is a group of galaxies.
- C** It is a group of stars outside our own galaxy.
- D** It is a group of stars which are part of our galaxy.

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Question	Answer	Marks
1	C	1
2	B	1
3	B	1
4	C	1
5	B	1
6	A	1
7	A	1
8	A	1
9	A	1
10	D	1
11	C	1
12	C	1
13	C	1
14	A	1
15	A	1
16	D	1
17	C	1
18	A	1
19	D	1
20	C	1
21	C	1
22	B	1
23	B	1
24	C	1
25	C	1
26	A	1
27	D	1
28	A	1

Question	Answer	Marks
29	C	1
30	B	1
31	C	1
32	D	1
33	B	1
34	C	1
35	C	1
36	A	1
37	B	1
38	D	1
39	A	1
40	C	1



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Question	Answer	Marks
1	D	1
2	B	1
3	C	1
4	B	1
5	B	1
6	D	1
7	B	1
8	C	1
9	B	1
10	A	1
11	D	1
12	D	1
13	B	1
14	A	1
15	C	1
16	A	1
17	B	1
18	B	1
19	B	1
20	D	1
21	C	1
22	C	1
23	A	1
24	C	1
25	D	1
26	B	1
27	D	1
28	D	1

Question	Answer	Marks
29	D	1
30	C	1
31	C	1
32	C	1
33	C	1
34	A	1
35	B	1
36	C	1
37	C	1
38	C	1
39	D	1
40	C	1



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Question	Answer	Marks
1	D	1
2	C	1
3	B	1
4	C	1
5	D	1
6	A	1
7	D	1
8	D	1
9	D	1
10	A	1
11	B	1
12	A	1
13	C	1
14	C	1
15	B	1
16	D	1
17	A	1
18	C	1
19	B	1
20	B	1
21	B	1
22	A	1
23	C	1
24	B	1
25	B	1
26	D	1
27	A	1
28	D	1

Question	Answer	Marks
29	B	1
30	B	1
31	A	1
32	C	1
33	C	1
34	C	1
35	B	1
36	C	1
37	C	1
38	A	1
39	B	1
40	A	1



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PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

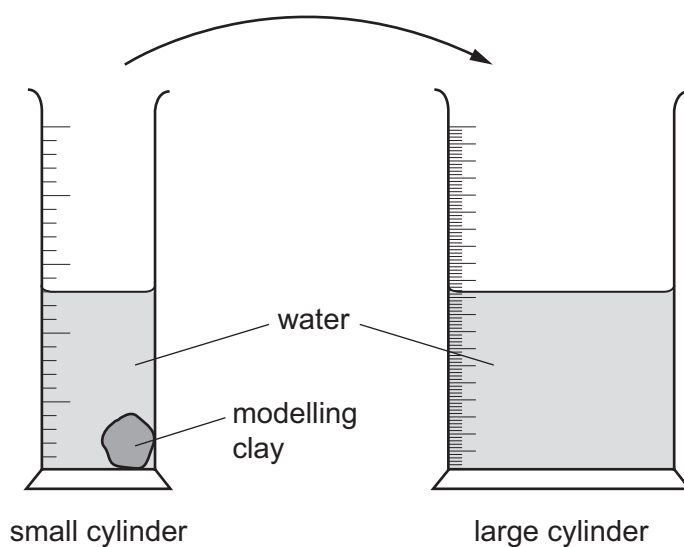
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



- 1 A lump of modelling clay is moved from a small measuring cylinder to a large measuring cylinder that has twice the diameter.

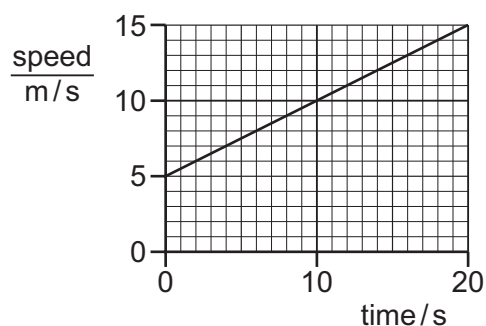


The reading on the small measuring cylinder goes down by 20 cm^3 .

By how much does the reading on the large cylinder go up?

- A** 10 cm^3 **B** 20 cm^3 **C** 40 cm^3 **D** 80 cm^3

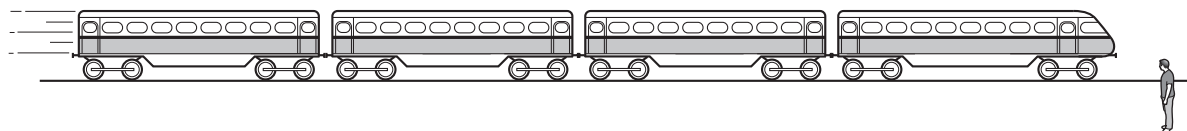
- 2 The graph shows the motion of a car.



Which row correctly describes this motion?

	acceleration	distance travelled / m
A	constant	200
B	constant	300
C	increasing at a constant rate	200
D	increasing at a constant rate	300

- 3 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A** 20 m **B** 38 m **C** 40 m **D** 80 m
- 4 Four students are given two different objects, P and Q.

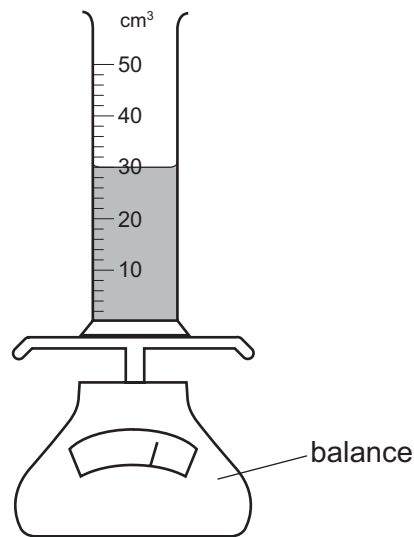
Each student measures the mass of P and the weight of Q.

The results are shown in the table.

Which row gives a possible result?

	mass of object P	weight of object Q
A	10 kg	10 kg
B	10 kg	10 N
C	10 N	10 kg
D	10 N	10 N

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



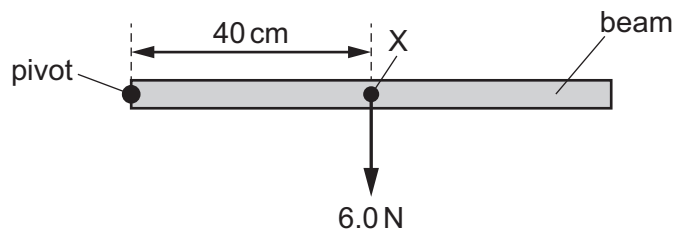
Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

What is the density of the liquid?

- A** 0.60 g/cm^3 **B** 0.67 g/cm^3 **C** 1.5 g/cm^3 **D** 1.7 g/cm^3

- 6 A beam is pivoted at one end, as shown.



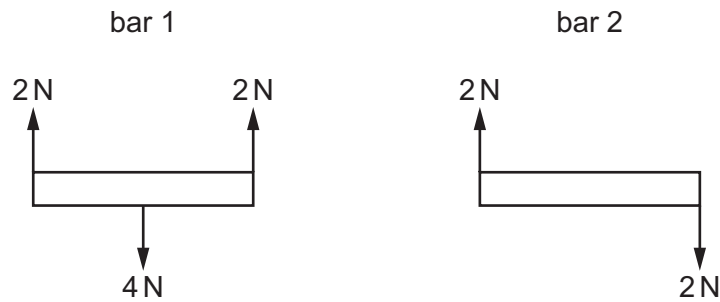
The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A** vertically downwards at 20 cm to the left of X
B vertically downwards at 20 cm to the right of X
C vertically upwards at 20 cm to the left of X
D vertically upwards at 20 cm to the right of X

- 7 The diagram shows two identical bars of negligible weight. All the forces acting on each bar are marked.



Which bars are in equilibrium?

- A** bar 1 and bar 2
B bar 1 only
C bar 2 only
D neither bar 1 nor bar 2
- 8 Four objects have different base areas and their centres of mass are in different positions.

Which object is most stable?

	base area	position of centre of mass
A	large	high
B	large	low
C	small	high
D	small	low

- 9 In which form is energy stored by stretching a spring?

- A** chemical energy
B elastic potential energy
C gravitational potential energy
D thermal energy

- 10 What is meant by the power of an engine?

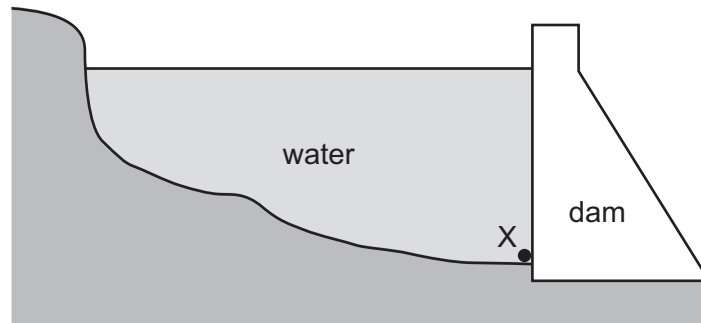
- A** the energy that the engine transfers per unit time
B the maximum force that the engine can exert
C the maximum weight that the engine can lift
D the total energy that the engine transfers

- 11** A force of 14 N is applied to the head of a nail. This causes a pressure of 25 N/mm^2 at the tip of the nail.

What is the cross-sectional area of the tip of the nail?

- A** 0.56 mm^2 **B** 11 mm^2 **C** 39 mm^2 **D** 350 mm^2

- 12** The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A** the depth of the water at X
B the length of the reservoir
C the surface area of the water
D the thickness of the dam wall
- 13** The properties of two states of matter are listed.
- state 1 The molecules move quickly and randomly. There is a large distance between the molecules.
- state 2 The molecules vibrate about fixed positions. The molecules are closely packed together.

What are states 1 and 2?

	state 1	state 2
A	gas	liquid
B	liquid	solid
C	solid	liquid
D	gas	solid

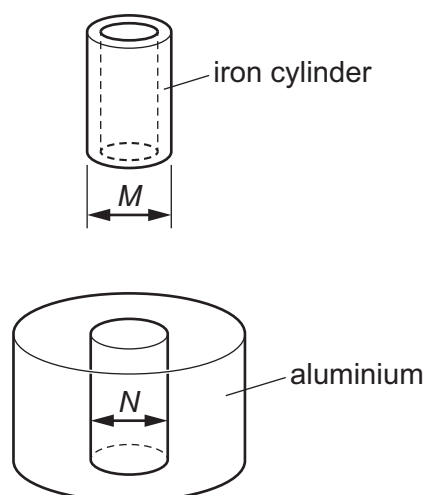
- 14 A gas in a container is cooled but the volume of the gas does not change.

Which row describes the changes in the pressure of the gas and the average kinetic energy of the gas particles?

	pressure of gas	average kinetic energy of gas particles
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 15 The iron cylinder of an engine is to be fitted into a piece of aluminium.

The outside diameter M of the iron cylinder is slightly larger than the diameter N of the hole in the aluminium.

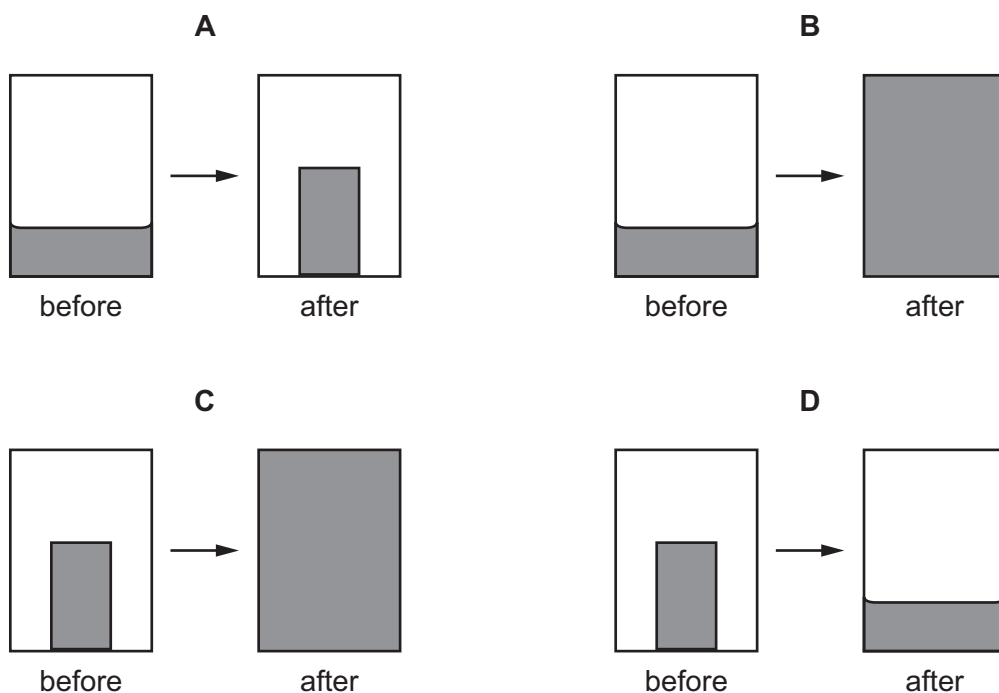


What is the best action to fit the cylinder into the aluminium?

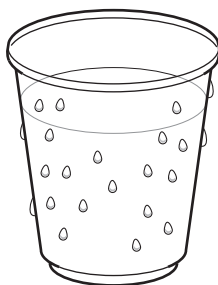
- A** Cool the aluminium and cool the iron.
- B** Cool the aluminium and heat the iron.
- C** Heat the aluminium and cool the iron.
- D** Heat the aluminium and heat the iron.

- 16** A solid in a closed container is heated until it completely melts.

Which diagram shows the shape and the volume of the solid before and after heating.



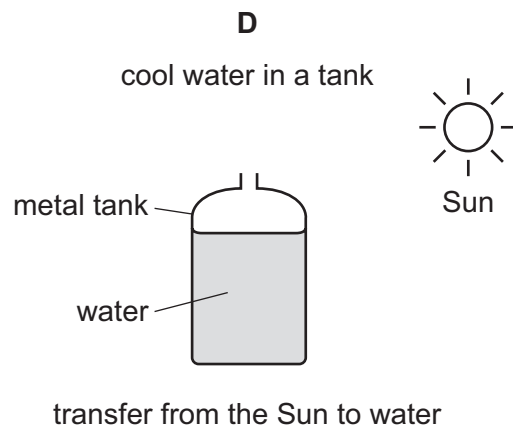
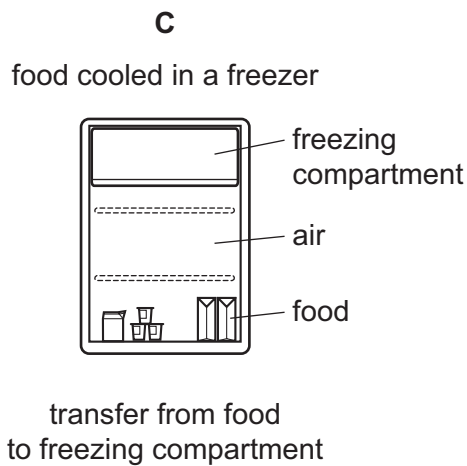
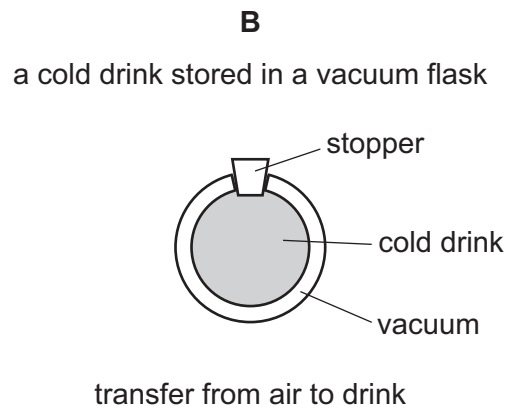
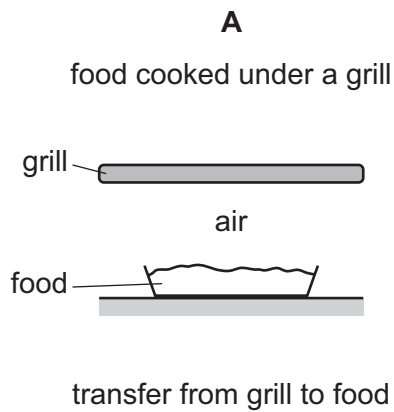
- 17** A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A** condensation
- B** conduction
- C** convection
- D** evaporation

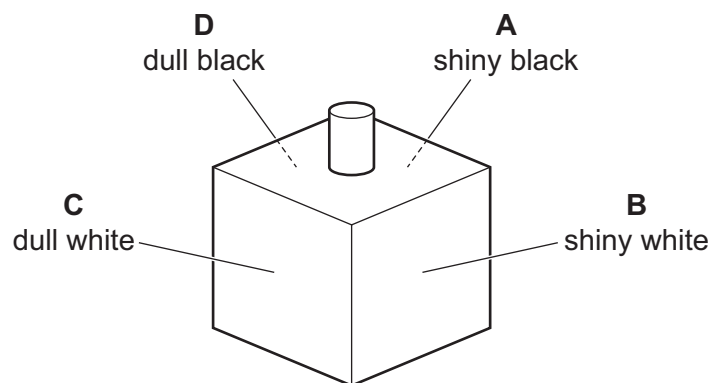
18 In which situation is the main transfer of thermal energy by convection?



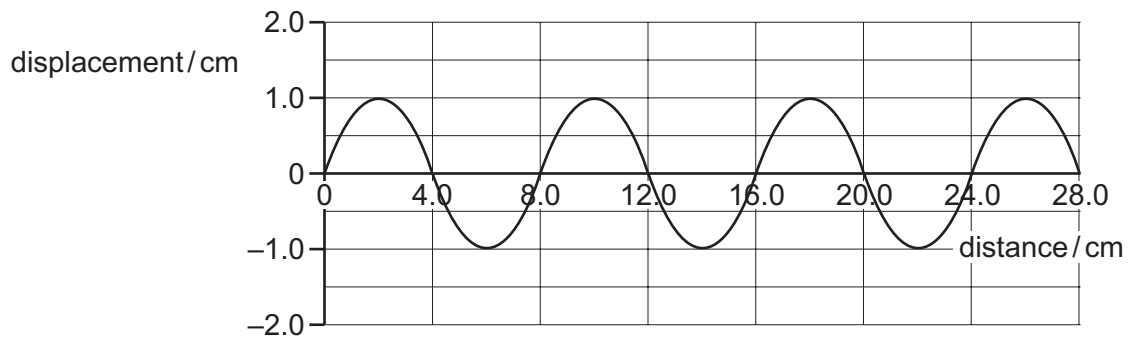
19 A metal container is the shape of a hollow cube.

The four sides of the container have different surface finishes.

Which side is the best emitter of radiation?



20 The diagram shows a wave.



Which row is correct?

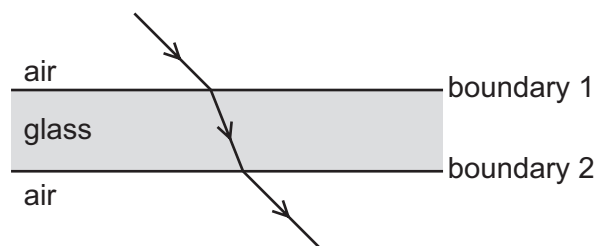
	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

21 A girl is sitting on a rock in the sea looking at passing waves. She notices that five complete wavelengths pass her in 20 s.

What is the frequency of this wave?

- A** 0.25 Hz **B** 4.0 Hz **C** 15 Hz **D** 100 Hz

22 A ray of light passes from air through a sheet of glass and out the other side, as shown.

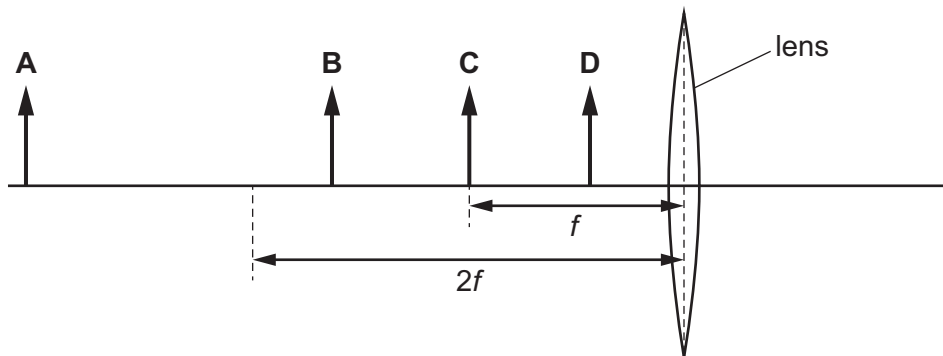


Which two angles are equal to each other?

- A** angle of incidence at boundary 1 and angle of incidence at boundary 2
B angle of incidence at boundary 1 and angle of refraction at boundary 1
C angle of incidence at boundary 1 and angle of refraction at boundary 2
D angle of refraction at boundary 1 and angle of refraction at boundary 2

- 23** An object is placed in front of a converging lens. The lens has a focal length f .

In which labelled position should the object be placed in order to produce a real image that is smaller than the object?

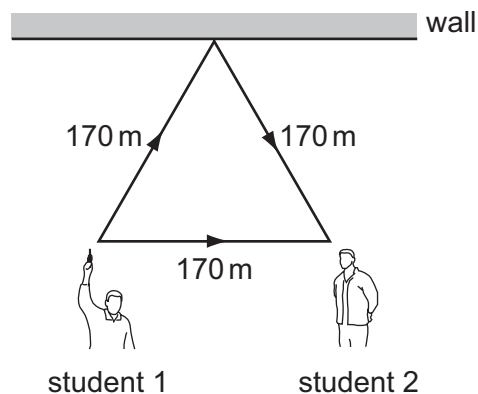


- 24** Which statement correctly compares radio waves and X-rays?

- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
- B** Radio waves have a longer wavelength and the same speed in a vacuum.
- C** Radio waves have a shorter wavelength and a greater speed in a vacuum.
- D** Radio waves have a shorter wavelength and the same speed in a vacuum.

- 25** Student 1 and student 2 stand 170 m apart as shown.

Student 1 fires a starting pistol. Student 2 hears the sound twice, once by the direct route and once from the reflection from the wall.



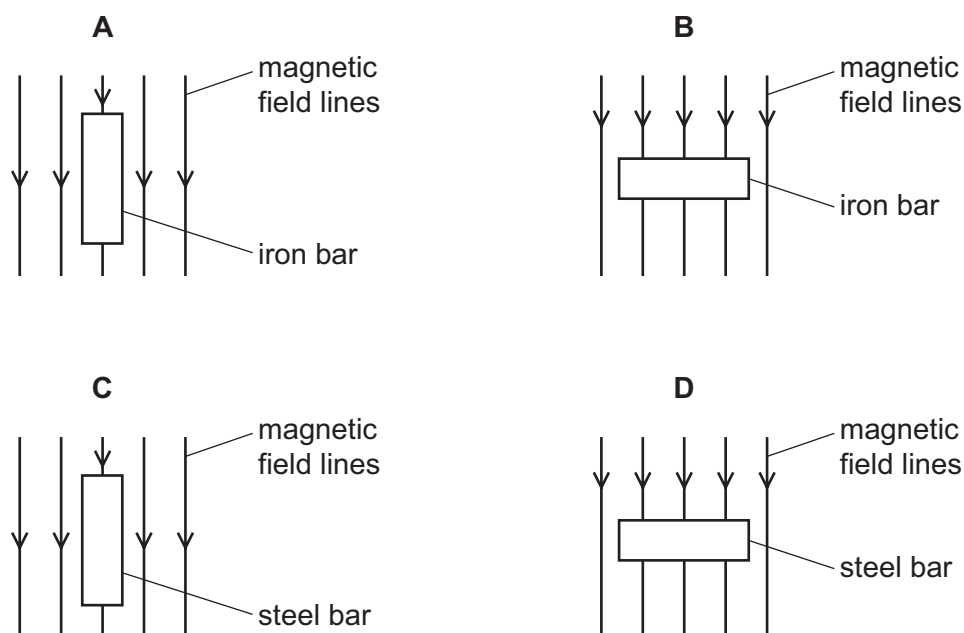
The speed of sound in air is 340 m/s.

What is the interval between hearing the two sounds?

- A** 0.25 s
- B** 0.50 s
- C** 1.0 s
- D** 2.0 s

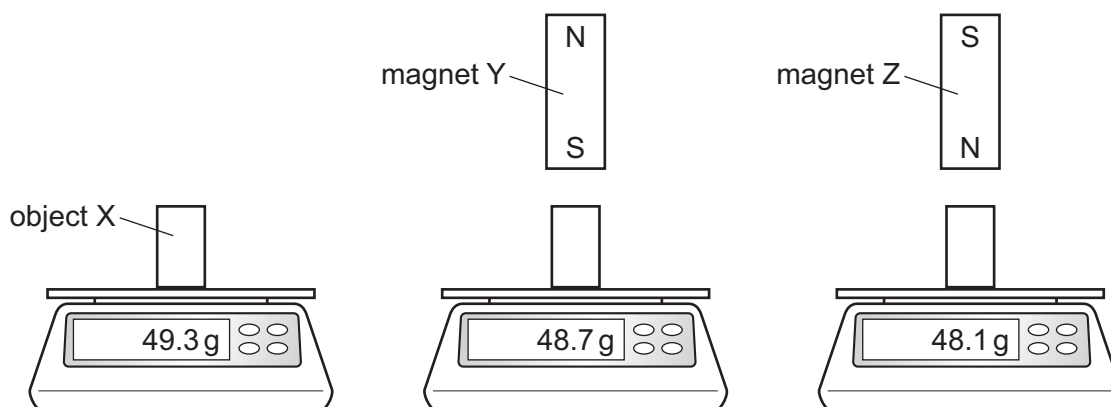
- 26** A student attempts to make a permanent magnet by hammering metal bars of the same size in the same magnetic field.

In which case is the strongest permanent magnet produced?



- 27** A student places object X on a balance. The student first brings magnet Y and then magnet Z close to object X and observes the readings on the balance. The distance between Y and X is the same as the distance between Z and X.

The diagram shows the results of the experiment.



Which statement explains the results?

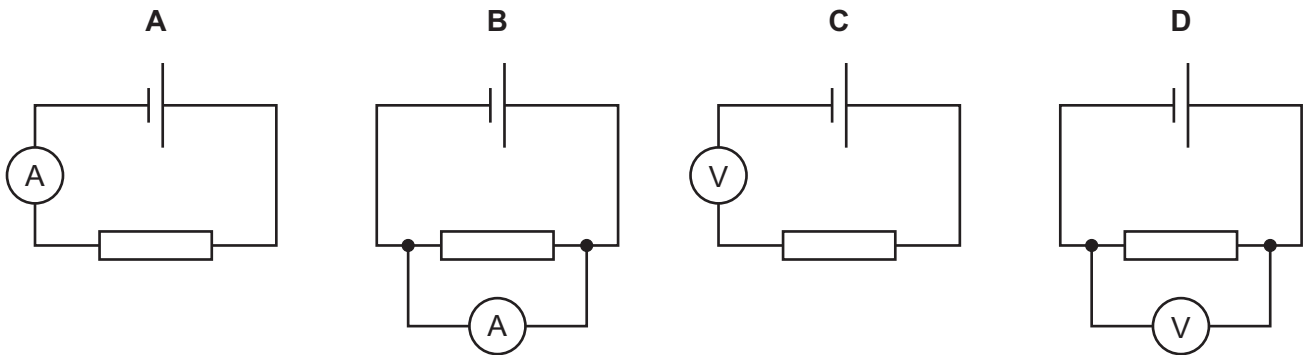
- A** Object X is an iron block and magnet Y is stronger than magnet Z.
- B** Object X is an iron block and magnet Y is weaker than magnet Z.
- C** Object X is a permanent magnet and magnet Y is stronger than magnet Z.
- D** Object X is a permanent magnet and magnet Y is weaker than magnet Z.

28 A girl rubs a plastic rod with a cloth. The plastic rod then repels a positively charged object.

Which row is correct?

	the state of the plastic rod	what happened when the rod was rubbed
A	negatively charged	it gained some electrons
B	negatively charged	it lost some protons
C	positively charged	it lost some electrons
D	positively charged	it gained some protons

29 Which circuit shows a meter being used correctly to measure the current in a resistor?



30 A resistor and a battery are connected in series.

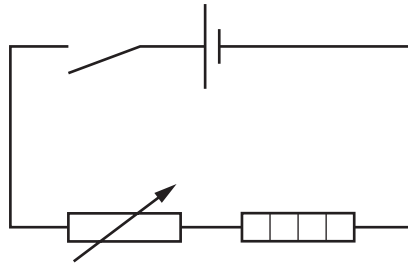
The value of the resistor is $20\ \Omega$.

The potential difference (p.d.) of the battery is 4.0 V .

What is the current in the resistor?

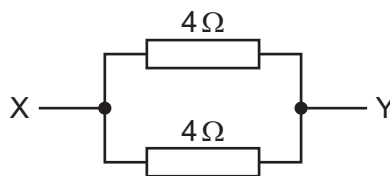
- A** 0.20 A **B** 4.0 A **C** 5.0 A **D** 80 A

- 31 The circuit shown includes a cell.



Which other components does the circuit contain?

- A lamp, voltmeter and switch
 - B resistor, thermistor and bell
 - C switch, variable resistor and heater
 - D switch, heater and fuse
- 32 Two $4\ \Omega$ resistors are connected in parallel.



What is the combined resistance between X and Y?

- A less than $4\ \Omega$
 - B $4\ \Omega$
 - C $8\ \Omega$
 - D more than $8\ \Omega$
- 33 Which statement about identical lamps in a parallel circuit is **not** correct?
- A If one lamp blows, the others remain switched on.
 - B The current in each lamp is different.
 - C The lamps can be switched on and off separately.
 - D The lamps have the same voltage across each of them.

34 Why is a fuse used in an electrical circuit?

- A** so that the current can have only one value
- B** to prevent the current becoming too large
- C** to provide a path to earth if a fault occurs
- D** to save electrical energy

35 The information describes the currents in three different circuits.

Circuit P has a steady current of 0.52 A in one direction.

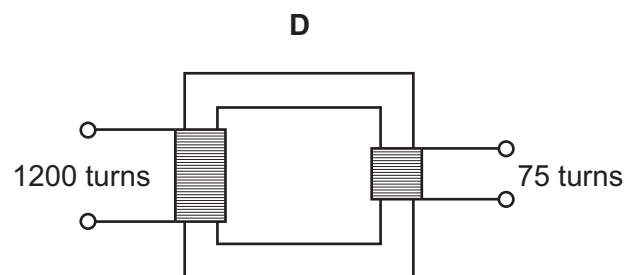
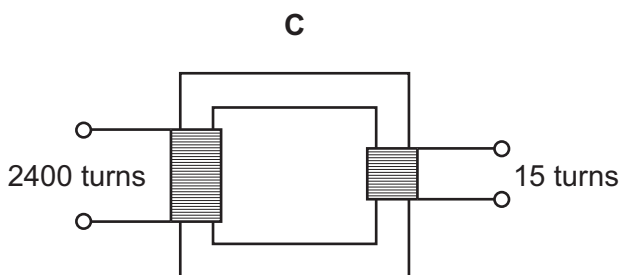
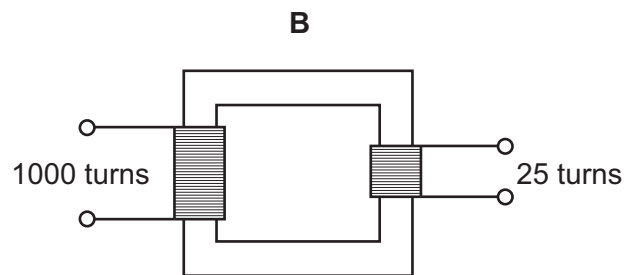
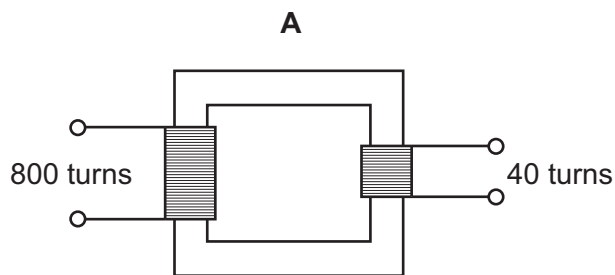
Circuit Q has a current that continually changes between 0.25 A and 0.35 A but is always in the same one direction.

Circuit R has a peak current of 0.52 A that changes direction periodically.

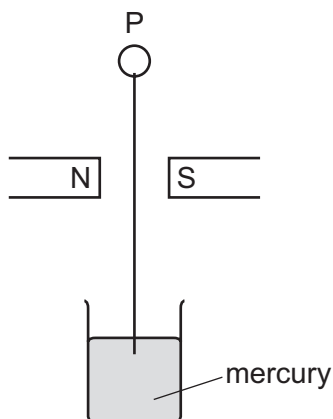
Which circuits contain a direct current?

- A** P only **B** P and Q **C** Q and R **D** R only

36 Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



- 37 The diagram shows a wire hanging from a metal loop P and dipping into a bath of mercury.



The wire is hanging vertically between the N and S poles of a magnet.

The loop P is then connected to the positive terminal of a battery and the mercury is connected to the negative terminal.

The wire swings out of the page.

In which direction does the wire move when P is connected to the negative terminal of the battery and the mercury is connected to the positive terminal?

- A The wire swings into the page.
 - B The wire swings out of the page.
 - C The wire swings to the left.
 - D The wire swings to the right.
- 38 The atoms of an element can exist as different isotopes.

What is the difference between atoms of different isotopes of the same element?

- A They have different numbers of electrons.
- B They have different numbers of protons and different numbers of neutrons.
- C They have different numbers of protons only.
- D They have different numbers of neutrons only.

- 39** A radioactive source has a half-life of 0.5 hours.

A detector near the source shows a reading of 6000 counts per second.

Background radiation can be ignored.

What is the reading on the detector 1.5 hours later?

- A** 750 counts per second
 - B** 1500 counts per second
 - C** 2000 counts per second
 - D** 3000 counts per second
- 40** Which statement explains why radioactive materials need to be handled carefully?
- A** γ -rays are part of the electromagnetic spectrum.
 - B** Radioactive decay is a random process.
 - C** Radioactive materials have a half-life.
 - D** The radiation given out is ionising.

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

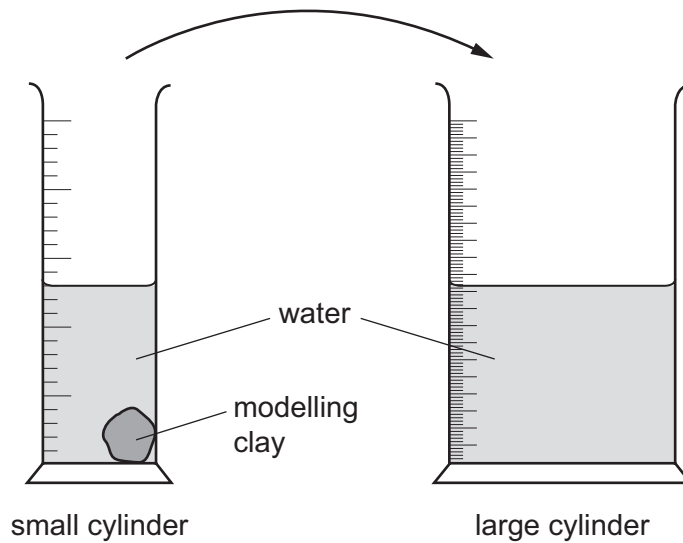
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 A lump of modelling clay is moved from a small measuring cylinder to a large measuring cylinder that has twice the diameter.



The reading on the small measuring cylinder goes down by 20 cm^3 .

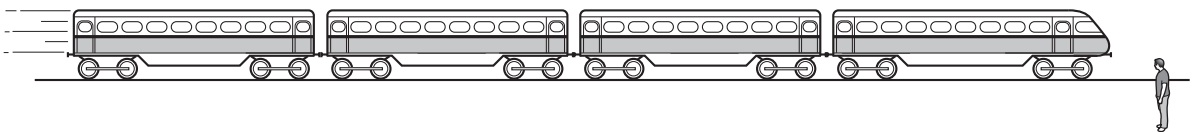
By how much does the reading on the large cylinder go up?

- A** 10 cm^3 **B** 20 cm^3 **C** 40 cm^3 **D** 80 cm^3

- 2 What is used to determine the distance travelled by an object in motion?

- A** the area under a distance–time graph
B the area under a speed–time graph
C the gradient of a distance–time graph
D the gradient of a speed–time graph

- 3 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A** 20 m **B** 38 m **C** 40 m **D** 80 m

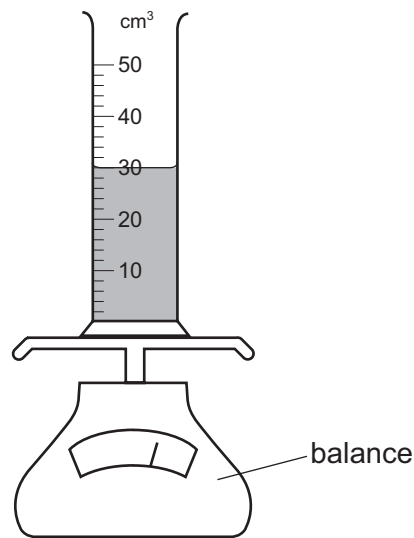
- 4 An object is moved from point X to point Y.

The acceleration of free fall at X is different from that at Y.

Which statement about the object at Y is correct?

- A Both its mass and its weight are different from those at X.
- B Both its mass and its weight are the same as those at X.
- C Its mass is the same as at X but its weight is different.
- D Its weight is the same as at X but its mass is different.

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

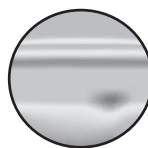
What is the density of the liquid?

- A 0.60 g/cm^3 B 0.67 g/cm^3 C 1.5 g/cm^3 D 1.7 g/cm^3

- 6 A spaceship approaches and passes a planet.



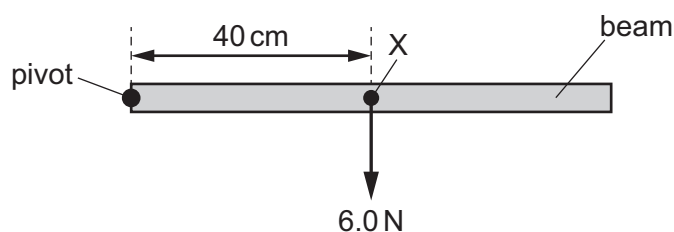
spaceship



planet

What can the force of gravity between the spaceship and planet achieve?

- A** It can change the direction of the spaceship, or slow it down, but not speed it up.
 - B** It can change the direction of the spaceship, or speed it up, but not slow it down.
 - C** It can slow down the spaceship, or speed it up, but not change its direction.
 - D** It can change the direction of the spaceship, slow it down, or speed it up.
- 7 A beam is pivoted at one end, as shown.



The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

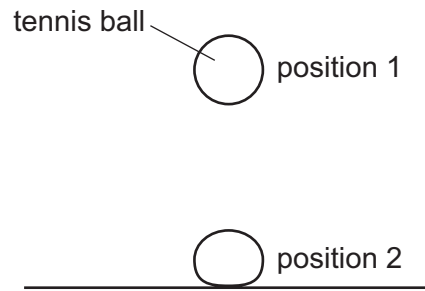
A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A** vertically downwards at 20 cm to the left of X
 - B** vertically downwards at 20 cm to the right of X
 - C** vertically upwards at 20 cm to the left of X
 - D** vertically upwards at 20 cm to the right of X
- 8 What are the conditions for an object to be in equilibrium?

	forces on object	moment on object
A	no resultant force	no resultant moment
B	no resultant force	resultant moment
C	resultant force	no resultant moment
D	resultant force	resultant moment

- 9 A tennis ball is dropped from position 1. It falls vertically onto a hard surface at position 2.



Which energy changes have taken place between position 1 and position 2?

- A** gravitational potential → kinetic → chemical
B gravitational potential → kinetic → elastic (strain)
C kinetic → gravitational potential → chemical
D kinetic → gravitational potential → elastic (strain)
- 10 A force does work moving an object in the direction of the force.

Which change in the force and distance **always** increases the work done?

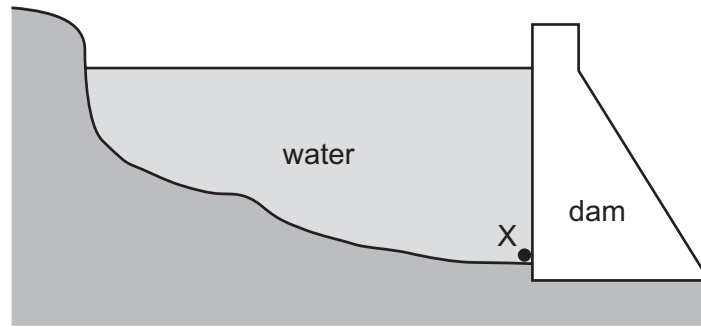
	force	distance
A	greater	same
B	greater	smaller
C	same	smaller
D	smaller	smaller

- 11 The table shows the weights and base areas of four metal blocks.

Which block exerts the greatest pressure on its base?

	weight / N	area of base / m ²
A	3 000	0.20
B	10 000	0.50
C	16 000	2.0
D	20 000	1.5

- 12 The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A the depth of the water at X
 - B the length of the reservoir
 - C the surface area of the water
 - D the thickness of the dam wall
- 13 A particular state of matter consists of molecules that move freely in random directions at high speed. The average speed of the molecules is decreasing.

Which state of matter is being described?

- A a gas cooling
- B a gas being heated
- C a solid cooling
- D a liquid being heated

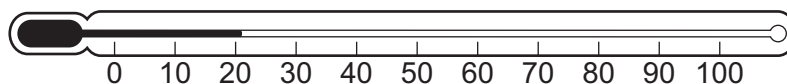
- 14 Small pollen particles are suspended in water.

When viewed with a microscope, the pollen particles can be seen to be moving about irregularly.

What causes this movement?

- A The pollen particles are being bombarded by the heavier particles of the water.
- B The pollen particles are being bombarded by the lighter particles of the water.
- C The temperature of the water is higher than that of the pollen particles.
- D The temperature of the water is lower than that of the pollen particles.

- 15 The diagram shows a liquid-in-glass thermometer.



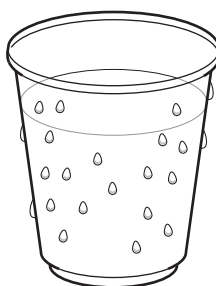
Which physical property of the thermometer is used to measure temperature?

- A** expansion of glass
B expansion of liquid
C mass of glass
D mass of liquid
- 16 An engineer investigates the increase in temperature of the oil in a car engine when it is first switched on.

Which row is correct?

	change in internal energy	explanation
A	increase	The random kinetic energy of the particles increases.
B	increase	The oil evaporates when it is heated.
C	decrease	The potential energy of the particles increases.
D	decrease	The oil changes state to a gas when it is heated.

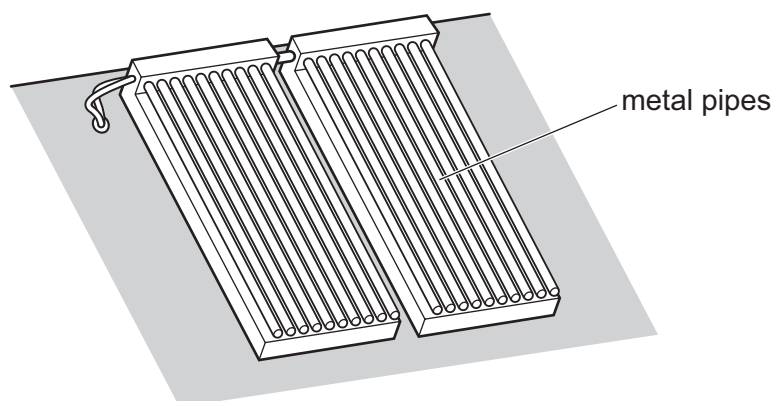
- 17 A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A** condensation
B conduction
C convection
D evaporation

- 18** The diagram shows solar water panels on the roof of a house. The panels absorb energy from the Sun to heat up the water in the metal pipes.



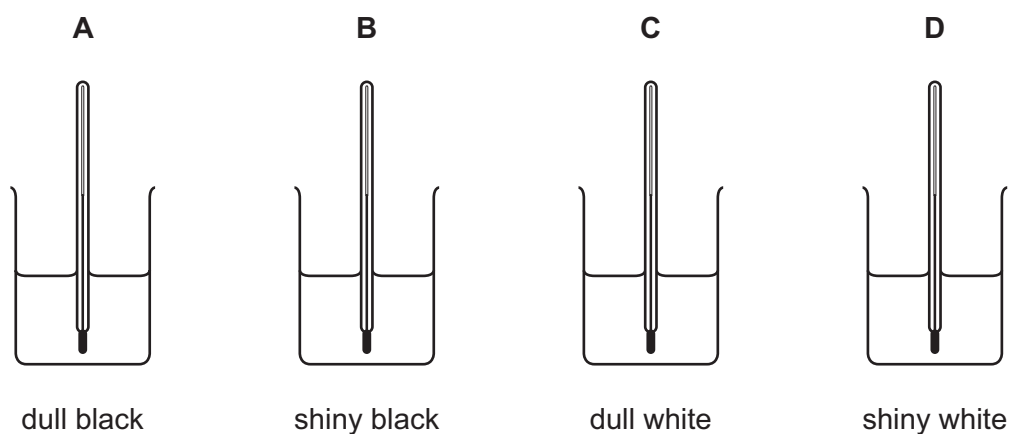
Which row describes how energy is transferred from the Sun and through the metal of the pipes?

	from the Sun	through the metal of the pipes
A	conduction	conduction
B	conduction	convection
C	radiation	conduction
D	radiation	convection

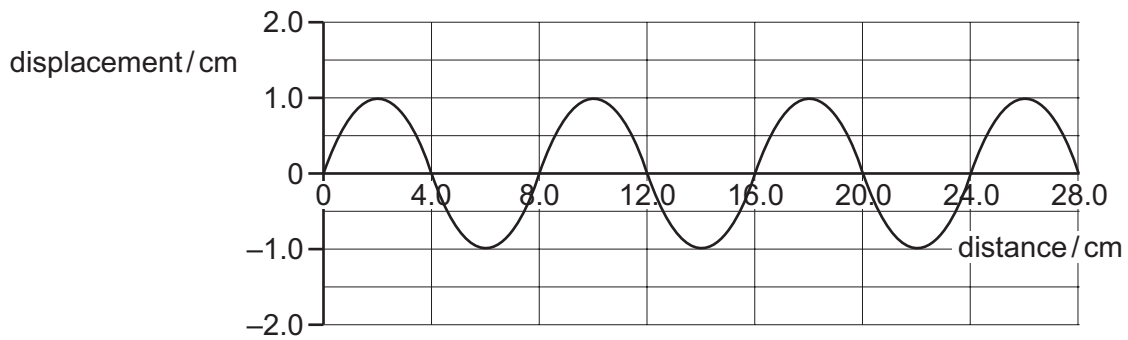
- 19** Four beakers containing equal volumes of water at 10 °C are placed outside in full sunshine on a hot day.

The four beakers are identical except for their surface colour and texture.

Which beaker will heat up the quickest?



20 The diagram shows a wave.



Which row is correct?

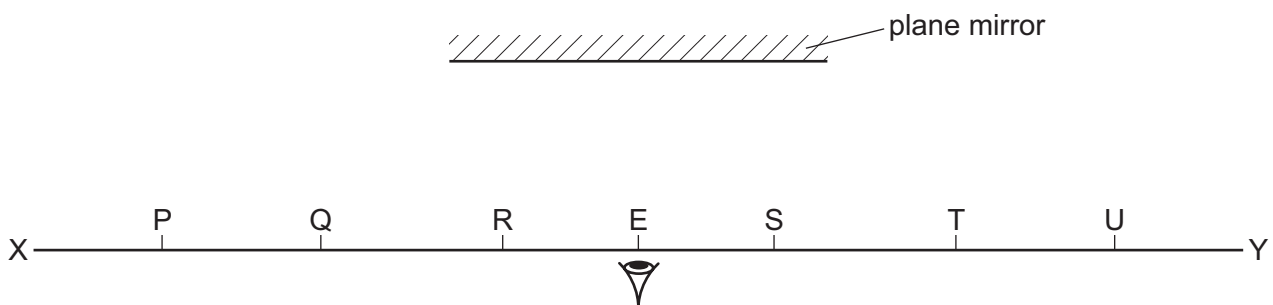
	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

21 A girl is sitting on a rock in the sea looking at passing waves. She notices that five complete wavelengths pass her in 20 s.

What is the frequency of this wave?

- A** 0.25 Hz **B** 4.0 Hz **C** 15 Hz **D** 100 Hz

22 A student uses one eye to look at images in a plane mirror.



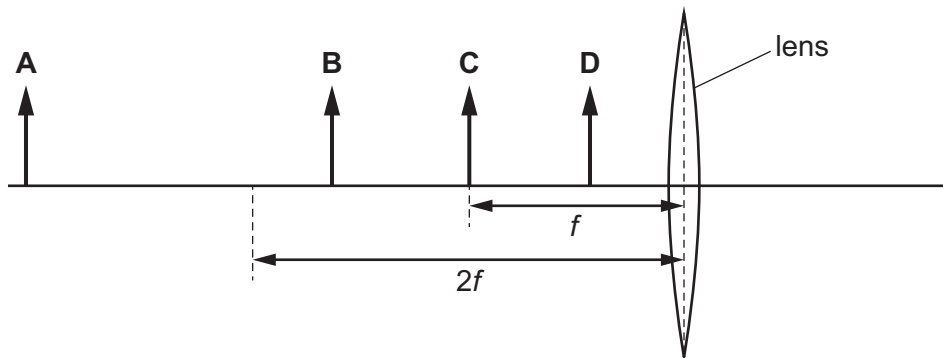
Objects are placed on the line XY.

Which objects give rise to images that can be seen by the eye at E?

- A** P, Q, R, S, T and U
B Q, R, S and T only
C P and U only
D R and S only

- 23** An object is placed in front of a converging lens. The lens has a focal length f .

In which labelled position should the object be placed in order to produce a real image that is smaller than the object?

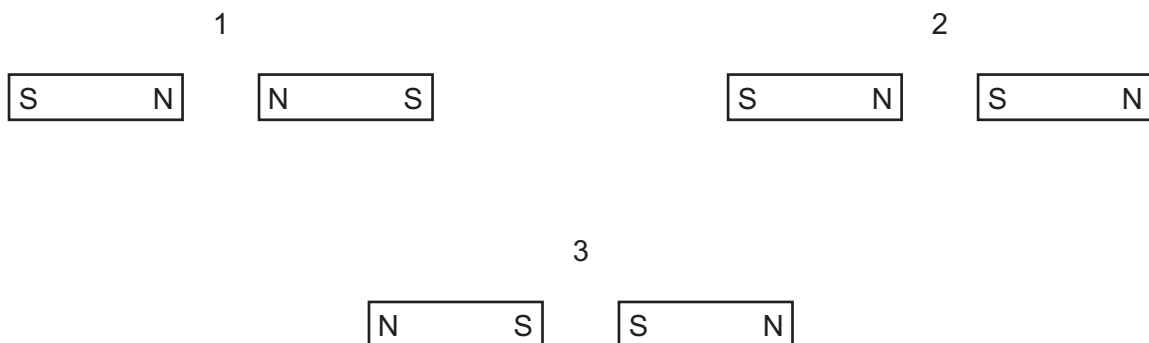


- 24** Microwaves, green light and infrared are three types of electromagnetic radiation.

What is their order when listed by wavelength from the shortest wavelength to the longest?

- A** green light → infrared → microwaves
B green light → microwaves → infrared
C infrared → green light → microwaves
D microwaves → infrared → green light
- 25** Which statement correctly compares radio waves and X-rays?
- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
B Radio waves have a longer wavelength and the same speed in a vacuum.
C Radio waves have a shorter wavelength and a greater speed in a vacuum.
D Radio waves have a shorter wavelength and the same speed in a vacuum.

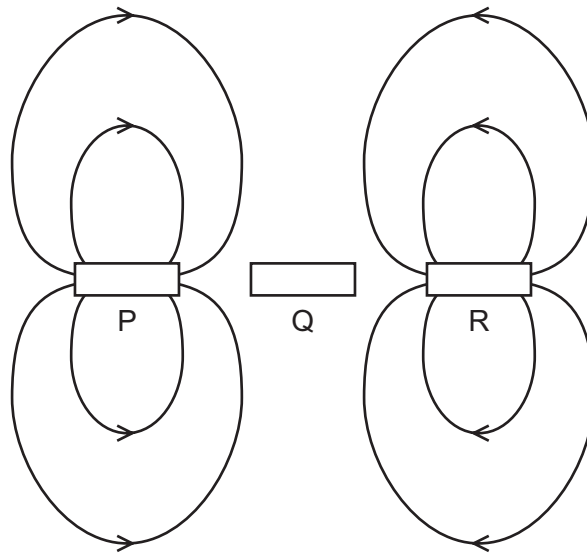
- 26** The three diagrams each show two magnets.



In which diagrams do the two magnets attract each other?

- A** 1 only **B** 1 and 3 **C** 2 only **D** 2 and 3

- 27 The diagram shows the magnetic fields around three objects, P, Q and R, placed close to each other.



Which row shows the nature of each of the objects?

	P	Q	R
A	permanent magnet N S	copper rod	permanent magnet N S
B	permanent magnet N S	copper rod	permanent magnet S N
C	permanent magnet N S	iron rod	permanent magnet N S
D	permanent magnet N S	iron rod	permanent magnet S N

- 28 Iron filings are picked up by an electromagnet.

The current in the electromagnet is switched off and the filings fall on a plastic bench.

A plastic comb is rubbed with a woollen cloth and held just above the iron filings.

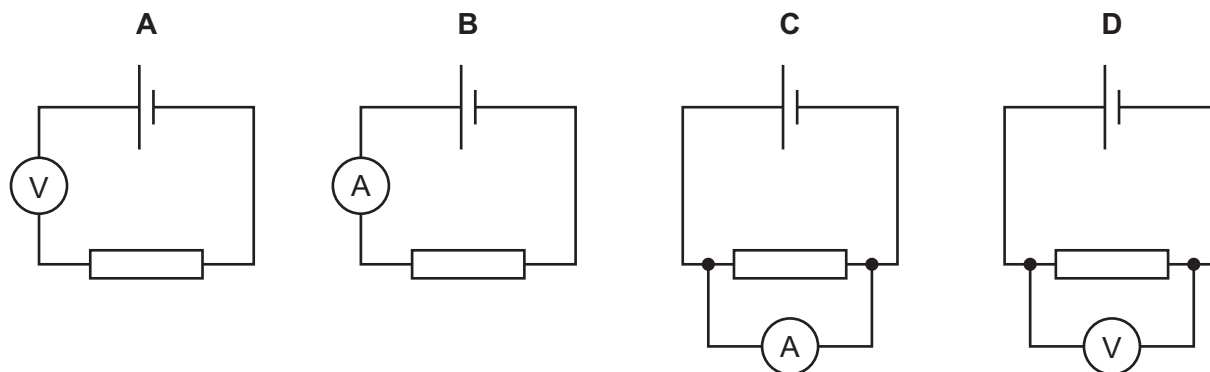
Some of the filings jump and stick to the comb.

Which statement correctly explains the last observation?

- A** The comb has been electrically charged by rubbing with the cloth.
- B** The comb has been magnetised by rubbing with the cloth.
- C** The filings have been electrically charged by the electromagnet.
- D** The filings have been magnetised by the electromagnet.

29 Four circuits are set up.

In which circuit does the meter measure the potential difference (p.d.) across the resistor?

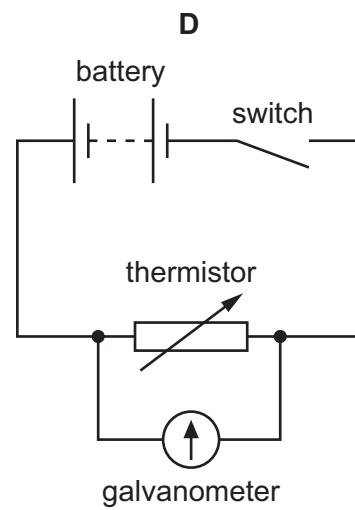
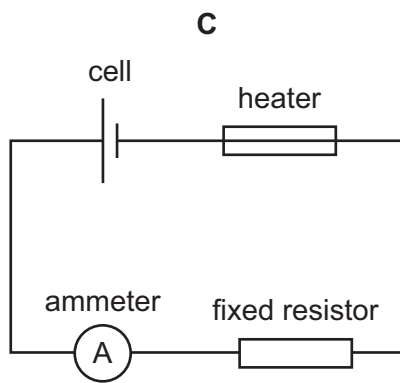
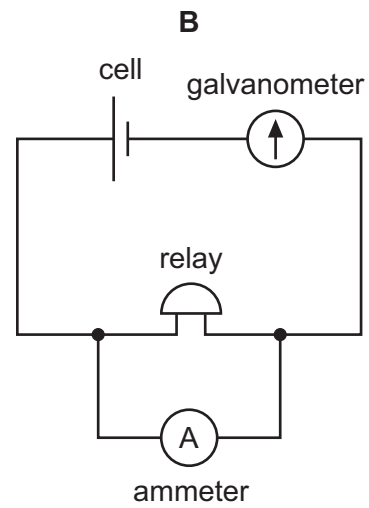
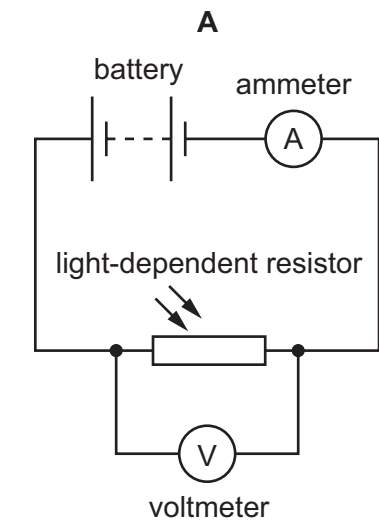


30 A resistor has a potential difference (p.d.) of 12 V across it and a current of 0.60 A in it.

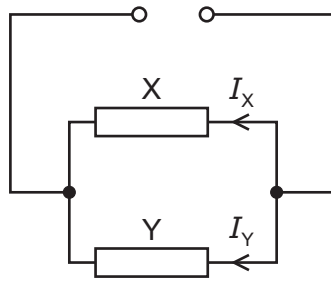
What is the resistance of the resistor?

- A** $0.050\ \Omega$ **B** $2.0\ \Omega$ **C** $7.2\ \Omega$ **D** $20\ \Omega$

31 Which circuit is correctly labelled?



32 The diagram shows an electric circuit.

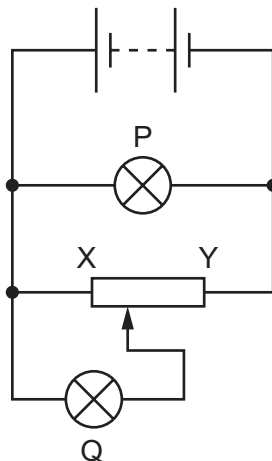


I_X is the current in resistor X. I_Y is the current in resistor Y.

Which statement describes the current from the power supply?

- A greater than I_X and greater than I_Y
- B greater than I_X and less than I_Y
- C less than I_X and greater than I_Y
- D less than I_X and less than I_Y

33 The diagram shows a battery connected to a potential divider and to two lamps, P and Q.



The slider on the potential divider is moved from end X to end Y of the resistor.

Which row shows the effect on the brightness of each lamp?

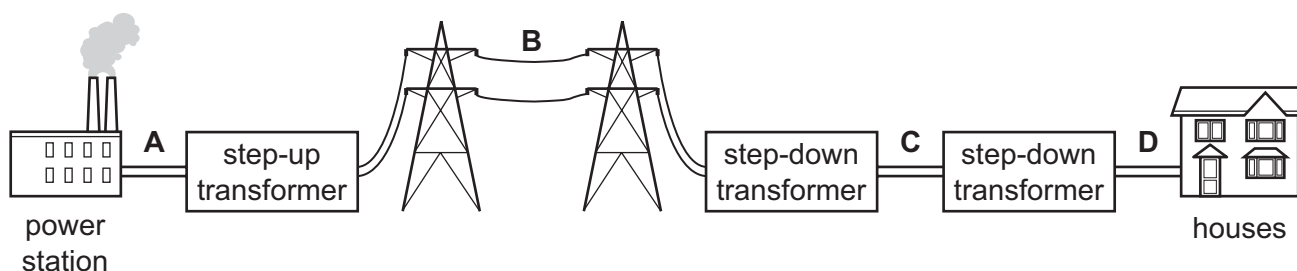
	brightness of P	brightness of Q
A	brighter	brighter
B	brighter	dimmer
C	unchanged	brighter
D	unchanged	dimmer

34 Why is a fuse used in an electrical circuit?

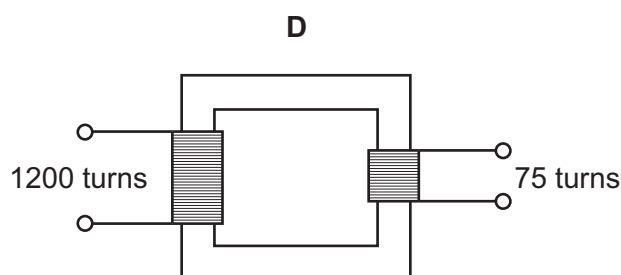
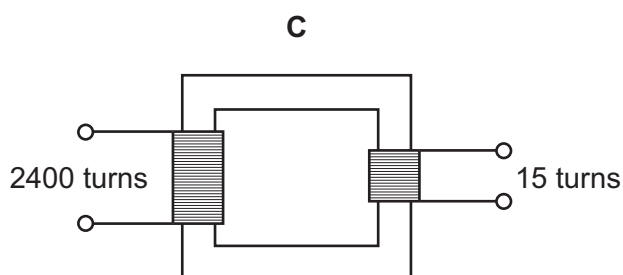
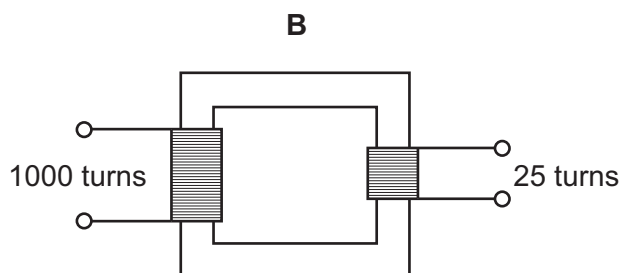
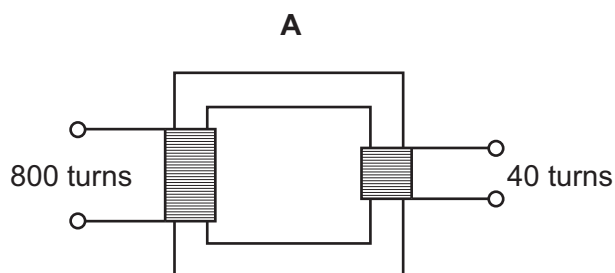
- A so that the current can have only one value
- B to prevent the current becoming too large
- C to provide a path to earth if a fault occurs
- D to save electrical energy

35 Electrical power is transmitted from power stations to homes using the National Grid.

In which part of the National Grid is the voltage highest?



36 Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



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- 37** The coils in two electric motors are identical in size, but motor 1 is observed to spin more quickly than motor 2.

Three suggestions are made to explain this observation.

- 1 The current in the coil of motor 1 is greater than the current in the coil of motor 2.
- 2 The number of turns on the coil of motor 1 is greater than on the coil of motor 2.
- 3 The magnets in motor 1 are stronger than the magnets in motor 2.

Which suggestions give a possible explanation for this observation?

- A** 1 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 38** Which statement describes two atoms of different isotopes of an element?

- A** two atoms with the same nucleon number but different proton number
B two atoms with a different nucleon number but the same proton number
C two atoms with the same nucleon number and the same proton number
D two atoms with a different nucleon number and different proton number

- 39** A radioactive source has a half-life of 0.5 hours.

A detector near the source shows a reading of 6000 counts per second.

Background radiation can be ignored.

What is the reading on the detector 1.5 hours later?

- A** 750 counts per second
B 1500 counts per second
C 2000 counts per second
D 3000 counts per second

- 40** Some nuclei are unstable. They emit radiation and change into nuclei of a different element.

What is this process called?

- A** convection
B electromagnetic induction
C radioactive decay
D the motor effect



Cambridge IGCSE™

PHYSICS

0625/11

Paper 1 Multiple Choice (Core)

May/June 2022

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
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- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

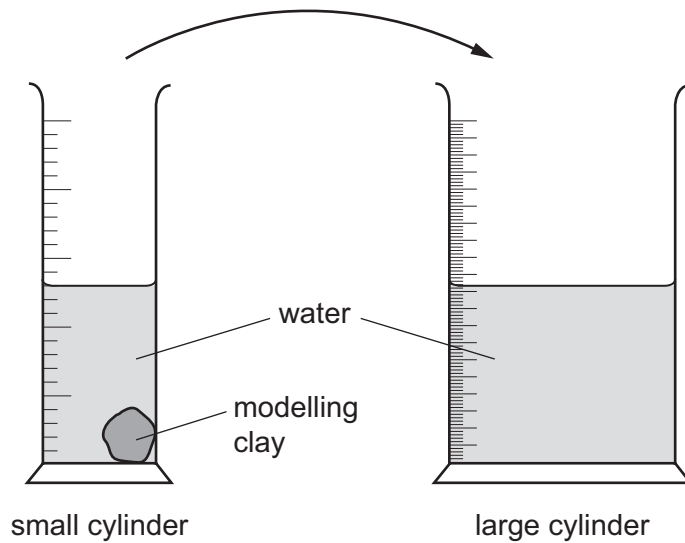
INFORMATION

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- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



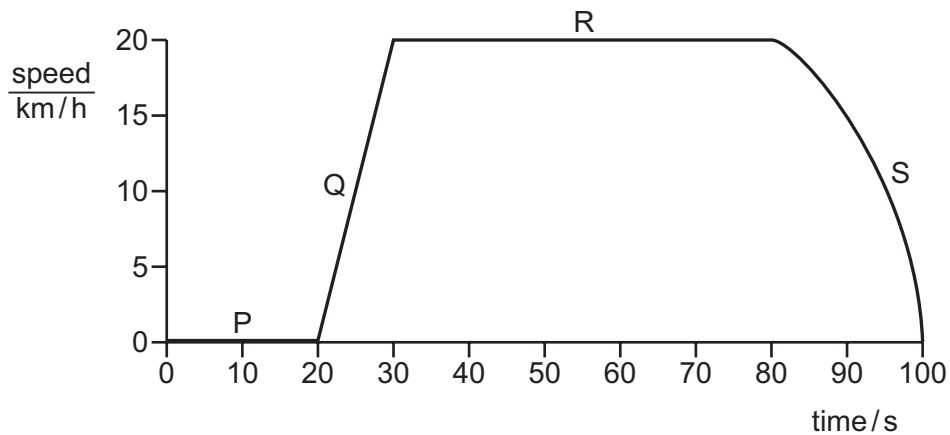
- 1 A lump of modelling clay is moved from a small measuring cylinder to a large measuring cylinder that has twice the diameter.



The reading on the small measuring cylinder goes down by 20 cm^3 .

By how much does the reading on the large cylinder go up?

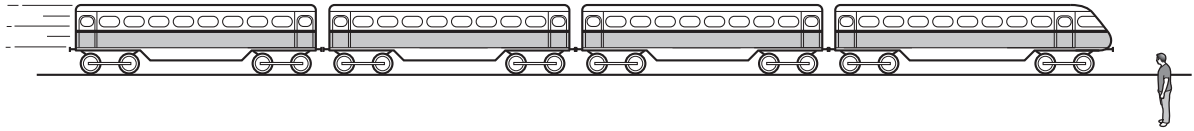
- A** 10 cm^3 **B** 20 cm^3 **C** 40 cm^3 **D** 80 cm^3
- 2 The speed–time graph for a train is shown.



Which regions of the graph show the train moving?

- A** P, Q, R and S
B Q, R and S only
C Q and S only
D R only

- 3 A man stands next to a railway track.



A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A** 20 m **B** 38 m **C** 40 m **D** 80 m
- 4 A spring balance operates by the compression of a spring. It has been calibrated on the Earth in grams.

A beam balance operates by balancing standard masses against the unknown mass to be measured.

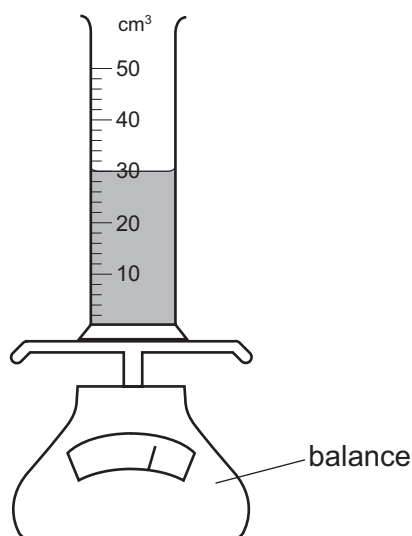
The same unknown mass is measured with each balance on the Earth and on the Moon.

The gravitational field strength on the Earth is greater than that on the Moon.

How would the measurements on the Earth compare with those on the Moon?

	spring balance measurements	beam balance measurements
A	larger on the Earth than on the Moon	larger on the Earth than on the Moon
B	larger on the Earth than on the Moon	same on the Earth as on the Moon
C	same on the Earth as on the Moon	larger on the Earth than on the Moon
D	same on the Earth as on the Moon	same on the Earth as on the Moon

- 5 A measuring cylinder contains 30 cm^3 of a liquid.



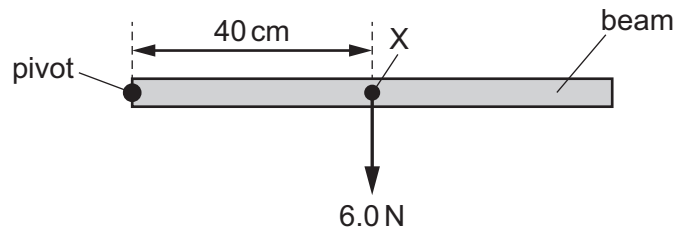
Some more of the liquid is added until the liquid level reaches the 50 cm^3 mark.

The reading on the balance increases by 30 g.

What is the density of the liquid?

- A** 0.60 g/cm^3 **B** 0.67 g/cm^3 **C** 1.5 g/cm^3 **D** 1.7 g/cm^3
- 6 Which pieces of apparatus are the most appropriate for an experiment to plot an extension–load graph of a spring?
- A** ruler, newton meter, clamp and stand
B balance, stop-watch, ruler
C light gate, ruler, newton meter
D stop-watch, balance, measuring cylinder

- 7 A beam is pivoted at one end, as shown.



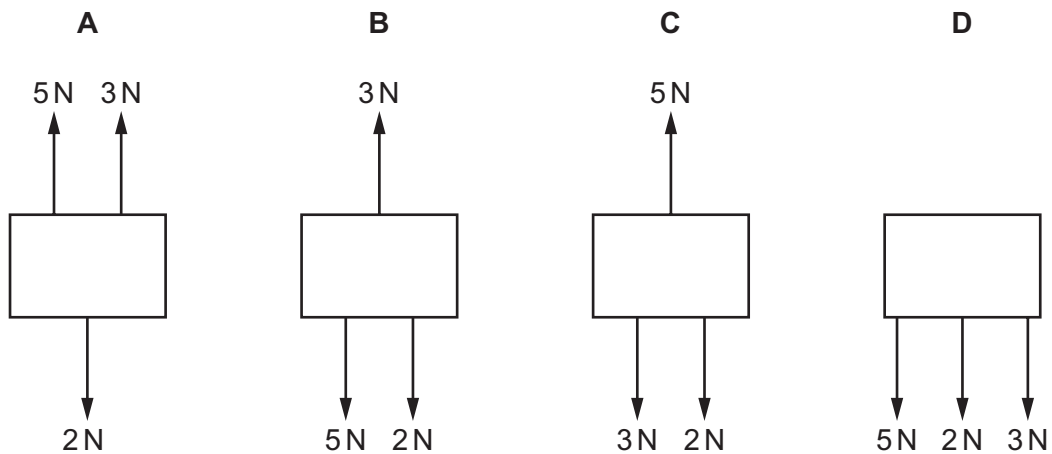
The beam weighs 6.0 N and its weight acts at a point X 40 cm from the pivot.

A force of 4.0 N is applied to the beam causing it to balance horizontally.

In which direction and where is the 4.0 N force applied?

- A vertically downwards at 20 cm to the left of X
 - B vertically downwards at 20 cm to the right of X
 - C vertically upwards at 20 cm to the left of X
 - D vertically upwards at 20 cm to the right of X
- 8 The diagrams show the forces acting on four moving objects.

Which object is moving at a constant speed?



- 9 A mass hangs vertically from a spring.

The mass is raised to a point P and is then released.

The mass oscillates repeatedly between point P and a lower point Q.

Which energies alternately increase and decrease throughout the oscillations?

- A** gravitational potential energy, kinetic energy and elastic energy
B gravitational potential energy and kinetic energy only
C gravitational potential energy, kinetic energy and internal energy
D internal energy and elastic energy
- 10 A force F acts on an object and the object moves a distance d in the direction of the force.

What is the work done on the object?

- A** $\frac{F}{d}$ **B** $\frac{d}{F}$ **C** $F \times d$ **D** $\frac{1}{(F \times d)}$

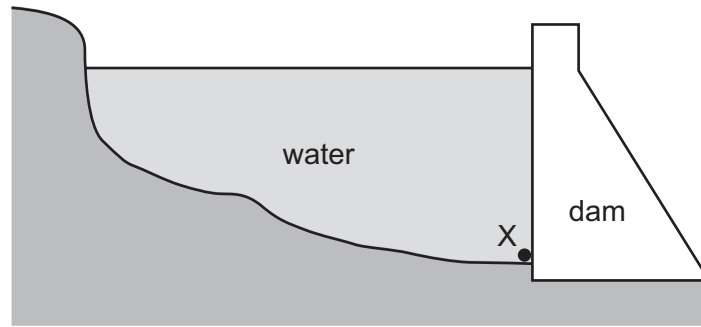
- 11 Liquid of mass 92 kg is contained in a rectangular tank.

The area of the base of the tank is 0.23 m^2 .

What is the pressure exerted by the liquid on the base of the tank?

- A** $2.5 \times 10^{-4} \text{ N/m}^2$
B $2.5 \times 10^{-3} \text{ N/m}^2$
C 400 N/m^2
D 4000 N/m^2

- 12 The diagram shows a deep reservoir formed by a dam.



On what does the pressure at X depend?

- A the depth of the water at X
 - B the length of the reservoir
 - C the surface area of the water
 - D the thickness of the dam wall
- 13 The list gives three properties of different states of matter.

- 1 They cannot be compressed significantly.
- 2 They can flow.
- 3 They always completely fill their container.

Which properties are correct for liquids?

- A 1 and 2 only
 - B 2 and 3 only
 - C 1 and 3 only
 - D 1, 2 and 3
- 14 Two thermometers, P and Q, give the same reading at room temperature.

The bulb of thermometer Q is wrapped in gauze and dipped in a beaker of water at room temperature.

Air at room temperature is blown over the two thermometer bulbs.

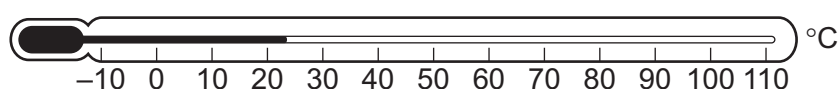
Which statement correctly describes and explains what happens?

- A P reads higher than Q because bulb P absorbs energy from the air.
- B P reads higher than Q because water evaporates from the gauze and cools bulb Q.
- C P reads lower than Q because the wet gauze insulates bulb Q.
- D P reads lower than Q because water evaporates from the gauze and warms bulb Q.

- 15 What happens to the volumes of liquid metal and of solid metal when heated at constant pressure?

	volume of liquid metal	volume of solid metal
A	decreases	increases
B	decreases	no change
C	increases	increases
D	increases	no change

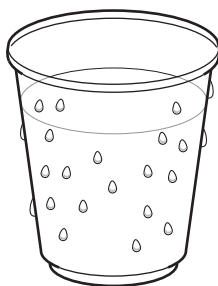
- 16 The diagram shows a liquid-in-glass thermometer.



A student wishes to check the marking of the upper fixed point on this thermometer.

What should she do?

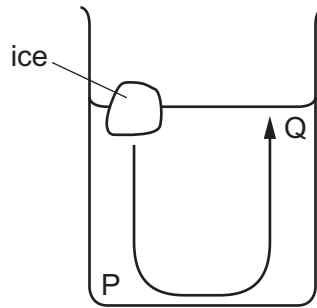
- A** Put the bulb in a beaker of boiling sea water.
 - B** Put the bulb in a beaker of boiling pure water.
 - C** Put the bulb in a beaker of ice and salt.
 - D** Put the bulb in a beaker of pure melting ice.
- 17 A glass contains an iced drink on a warm and humid day. Water starts to form on the outside of the glass.



What is the name of the effect by which the water forms?

- A** condensation
- B** conduction
- C** convection
- D** evaporation

- 18 The diagram shows a convection current caused by a piece of ice placed in a beaker of water at room temperature.



Which row correctly compares the temperatures and densities at water points P and Q?

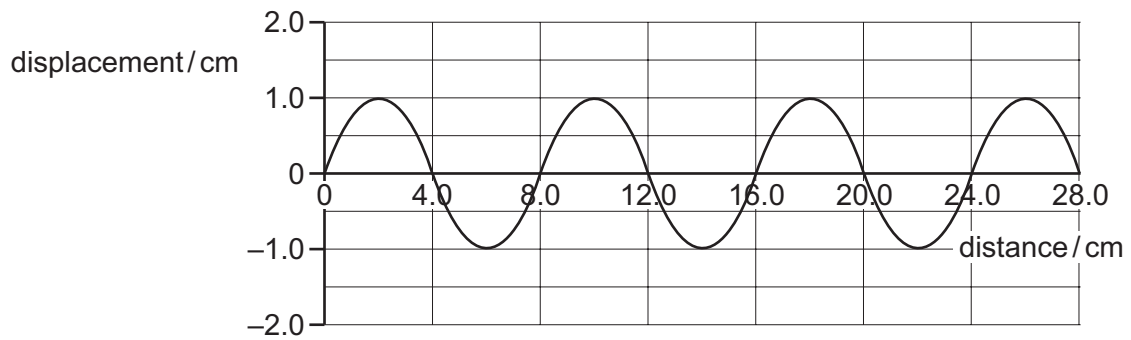
	temperature at P	density at P
A	higher than at Q	higher than at Q
B	higher than at Q	lower than at Q
C	lower than at Q	higher than at Q
D	lower than at Q	lower than at Q

- 19 The outside of one of two identical shiny metal containers is painted dull black. The containers are filled with equal masses of hot water at the same temperature.

Why does the dull black container cool more quickly?

- A** Black surfaces are better conductors than shiny surfaces.
- B** Black surfaces are better emitters of radiation than shiny surfaces.
- C** Black surfaces are better reflectors of radiation than shiny surfaces.
- D** Black surfaces are worse absorbers of radiation than shiny surfaces.

20 The diagram shows a wave.



Which row is correct?

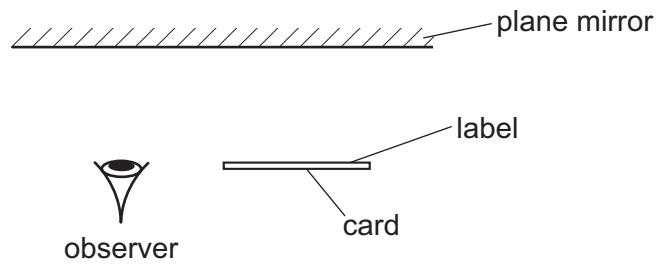
	amplitude of the wave / cm	wavelength of the wave / cm
A	1.0	4.0
B	1.0	8.0
C	2.0	4.0
D	2.0	8.0

21 A girl is sitting on a rock in the sea looking at passing waves. She notices that five complete wavelengths pass her in 20 s.

What is the frequency of this wave?

- A** 0.25 Hz **B** 4.0 Hz **C** 15 Hz **D** 100 Hz

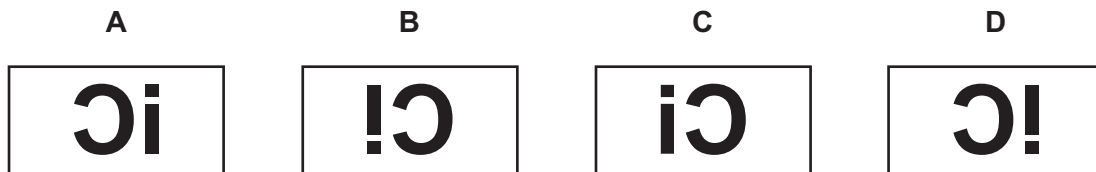
- 22 A card is placed in front of a plane mirror so that its label is facing the mirror, as shown.



The label is shown.

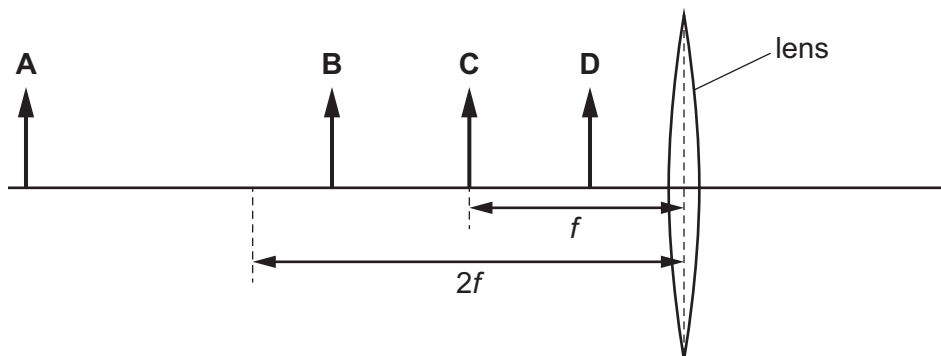


How does the image of the label formed by the mirror appear to the observer?



- 23 An object is placed in front of a converging lens. The lens has a focal length f .

In which labelled position should the object be placed in order to produce a real image that is smaller than the object?



- 24 Which part of the electromagnetic spectrum is used by a remote controller for a television?

- A infrared waves
- B microwaves
- C radio waves
- D visible light

25 Which statement correctly compares radio waves and X-rays?

- A** Radio waves have a longer wavelength and a greater speed in a vacuum.
- B** Radio waves have a longer wavelength and the same speed in a vacuum.
- C** Radio waves have a shorter wavelength and a greater speed in a vacuum.
- D** Radio waves have a shorter wavelength and the same speed in a vacuum.

26 Point X is near to a bar magnet, as shown.

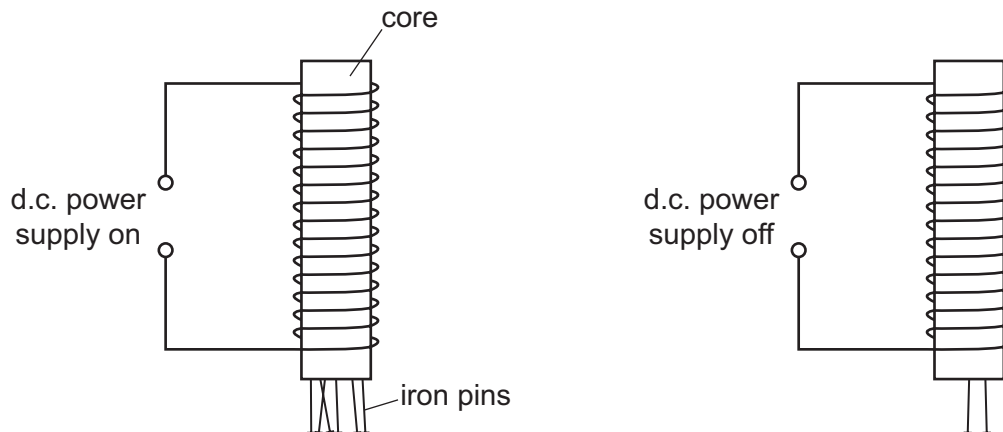
• X



Which arrow indicates the direction of the magnetic field of the bar magnet at point X?



- 27** A student counts how many iron pins an electromagnet picks up when its power supply is switched on. Then, she counts how many pins are picked up when the power supply is switched off.

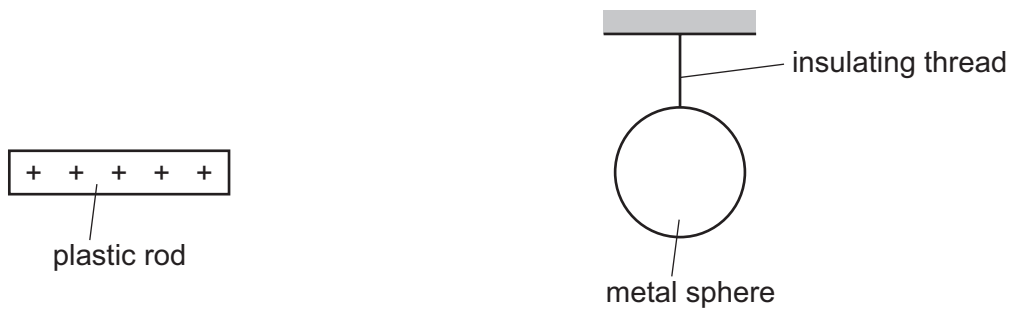


She repeats the experiment using cores made of different materials. The results are shown.

Which core is made out of soft iron?

	pins picked up with the power supply on	pins picked up with the power supply off
A	0	0
B	2	7
C	8	5
D	12	0

- 28 The diagram shows a charged plastic rod and an uncharged metal sphere. The metal sphere is suspended by an insulating thread.



The plastic rod is then moved close to the metal sphere.

Which row is correct?

	observation	the overall state of the metal sphere
A	The rod attracts the sphere.	charged
B	The rod attracts the sphere.	uncharged
C	The rod repels the sphere.	charged
D	The rod repels the sphere.	uncharged

- 29 A large battery is labelled with various items of information about the battery.

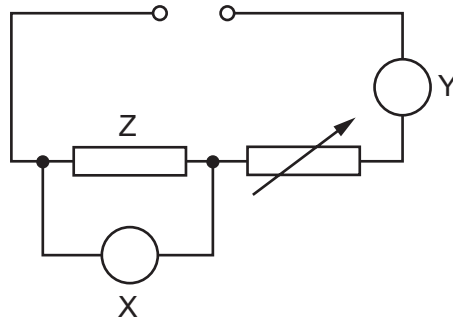
12 V	30 kg	216 kJ	680 A
------	-------	--------	-------

One of these items of information is the electromotive force (e.m.f.) of the battery.

What is the e.m.f. of the battery?

- A** 12 V **B** 30 kg **C** 216 kJ **D** 680 A

- 30 The circuit shown is being used to measure the resistance of resistor Z.



What is the correct combination of meters to determine the resistance of resistor Z?

	meter X	meter Y
A	ammeter	ammeter
B	ammeter	voltmeter
C	voltmeter	ammeter
D	voltmeter	voltmeter

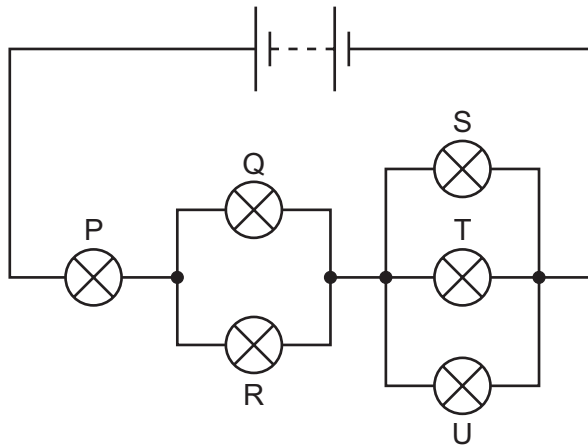
- 31 The diagram shows the circuit diagram symbol of an electrical component.



Which component does the symbol represent?

- A** light-dependent resistor
 - B** relay coil
 - C** thermistor
 - D** variable resistor
- 32 What is the advantage of connecting lamps in parallel in a lighting circuit?
- A** A smaller fuse is needed to protect the lamps.
 - B** If the filament of one lamp breaks, the remaining lamps stay lit.
 - C** The current taken from the supply is less.
 - D** The lamps use less power.

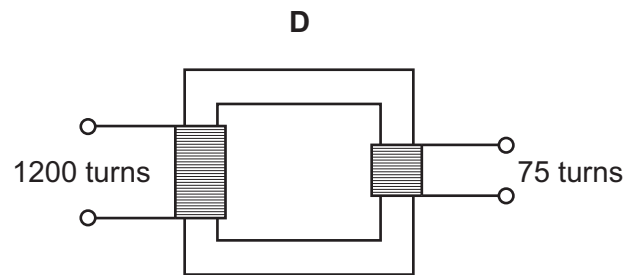
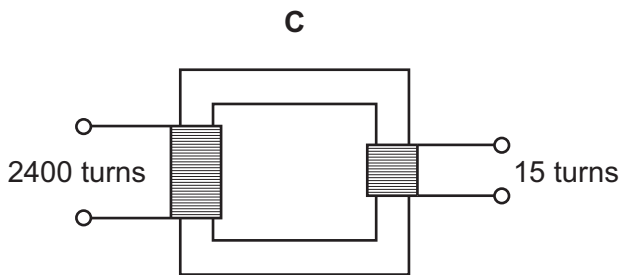
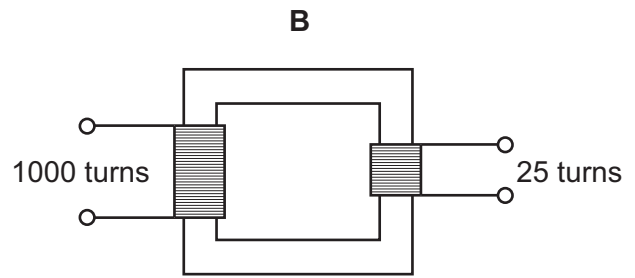
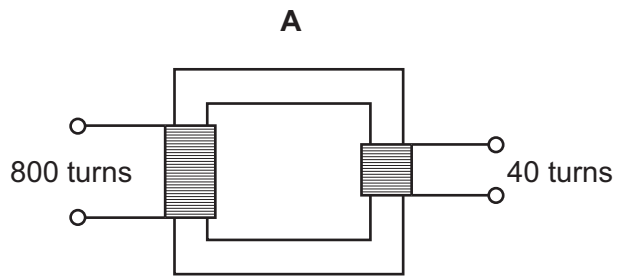
33 The diagram shows a circuit of six identical lamps connected to a battery.



Which lamps are brightest?

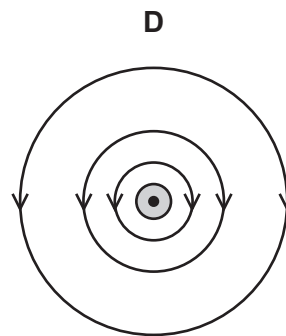
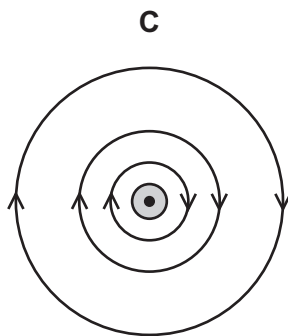
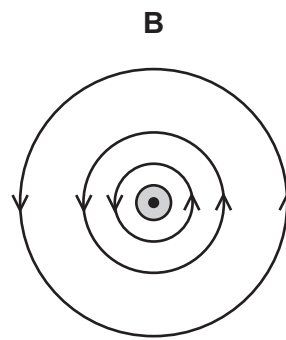
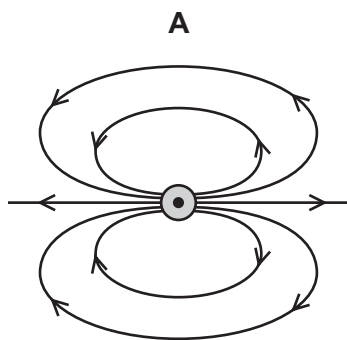
- A P only
 - B Q and R only
 - C S, T and U only
 - D P, Q, R, S, T and U are equally bright
- 34 Why is a fuse used in an electrical circuit?
- A so that the current can have only one value
 - B to prevent the current becoming too large
 - C to provide a path to earth if a fault occurs
 - D to save electrical energy

35 Which transformer can change a 240 V a.c. input into a 15 V a.c. output?



36 The diagrams show patterns around a straight wire carrying a current perpendicularly out of the page.

Which pattern represents the magnetic field due to the current in the wire?



- 37** A current-carrying coil in a magnetic field experiences a turning effect.

Three suggestions for increasing the turning effect are given.

- 1 Increase the number of turns on the coil.
- 2 Increase the current in the coil.
- 3 Increase the strength of the magnetic field.

Which suggestions are correct?

- A** 1 and 2 only **B** 1 and 3 only **C** 2 and 3 only **D** 1, 2 and 3

- 38** A model of an atom consists of small particles orbiting a central nucleus.

Where is the positive charge in an atom?

- A** on the orbiting particles
B in the nucleus
C in the space between the nucleus and the orbiting particles
D spread throughout the atom

- 39** A radioactive source has a half-life of 0.5 hours.

A detector near the source shows a reading of 6000 counts per second.

Background radiation can be ignored.

What is the reading on the detector 1.5 hours later?

- A** 750 counts per second
B 1500 counts per second
C 2000 counts per second
D 3000 counts per second

- 40** Which statement about the radioactive decay of a substance is correct?

- A** It cannot be predicted when a particular nucleus will decay.
B Placing a radioactive substance inside a lead-lined box prevents it from decaying.
C The decay always produces poisonous gases.
D The rate of decay increases if the substance is dissolved in water.

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May/June 2022

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Question	Answer	Marks
1	B	1
2	A	1
3	D	1
4	B	1
5	C	1
6	D	1
7	B	1
8	B	1
9	B	1
10	A	1
11	A	1
12	A	1
13	D	1
14	A	1
15	C	1
16	D	1
17	A	1
18	C	1
19	D	1
20	B	1
21	A	1
22	C	1
23	A	1
24	B	1
25	B	1
26	C	1
27	B	1
28	C	1

Question	Answer	Marks
29	A	1
30	A	1
31	C	1
32	A	1
33	B	1
34	B	1
35	B	1
36	D	1
37	A	1
38	D	1
39	A	1
40	D	1



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Question	Answer	Marks
1	B	1
2	B	1
3	D	1
4	C	1
5	C	1
6	D	1
7	D	1
8	A	1
9	B	1
10	A	1
11	B	1
12	A	1
13	A	1
14	B	1
15	B	1
16	A	1
17	A	1
18	C	1
19	A	1
20	B	1
21	A	1
22	B	1
23	A	1
24	A	1
25	B	1
26	C	1
27	B	1
28	A	1

Question	Answer	Marks
29	D	1
30	D	1
31	A	1
32	A	1
33	C	1
34	B	1
35	B	1
36	D	1
37	D	1
38	B	1
39	A	1
40	C	1



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Question	Answer	Marks
1	B	1
2	B	1
3	D	1
4	B	1
5	C	1
6	A	1
7	D	1
8	C	1
9	A	1
10	C	1
11	D	1
12	A	1
13	A	1
14	B	1
15	C	1
16	B	1
17	A	1
18	C	1
19	B	1
20	B	1
21	A	1
22	C	1
23	A	1
24	A	1
25	B	1
26	D	1
27	D	1
28	B	1

Question	Answer	Marks
29	A	1
30	C	1
31	C	1
32	B	1
33	A	1
34	B	1
35	D	1
36	B	1
37	D	1
38	B	1
39	A	1
40	A	1



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PHYSICS

0625/13

Paper 1 Multiple Choice (Core)

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

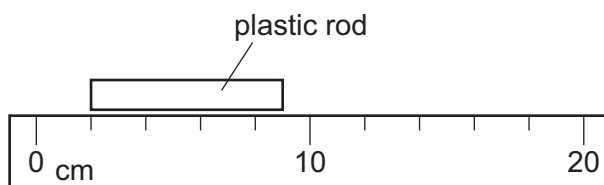
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages. Any blank pages are indicated.



- 1 The diagram shows a plastic rod alongside a ruler.



What is the length of the rod?

- A** 2.5 cm **B** 3.5 cm **C** 7.0 cm **D** 9.0 cm
- 2 Two stones of different weights fall at the same time from a table. Air resistance may be ignored.

What will happen and why?

	what will happen	why
A	Both stones hit the floor at the same time.	Acceleration of free fall is constant.
B	Both stones hit the floor at the same time.	They fall at constant speed.
C	The heavier stone hits the floor first.	Acceleration increases with weight.
D	The heavier stone hits the floor first.	Speed increases with weight.

- 3 A cyclist rides 300 m up a slope in 50 s.

She then rides down the slope in 25 s.

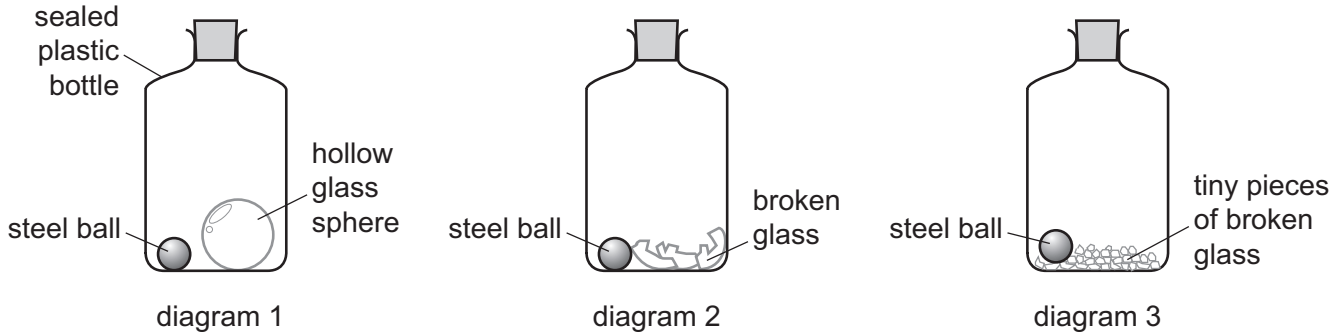
What is her average speed for the whole journey?

- A** 4.0 m/s **B** 8.0 m/s **C** 9.0 m/s **D** 16 m/s

- 4 Diagram 1 shows a sealed plastic bottle containing a hollow glass sphere and a steel ball.

Diagram 2 shows the same bottle after it has been shaken.

Diagram 3 shows the same bottle after it has been shaken again until the broken glass is in tiny pieces.



The mass of the bottle and contents in diagram 1 is m_1 .

The mass of the bottle and contents in diagram 2 is m_2 .

The mass of the bottle and contents in diagram 3 is m_3 .

Which statement gives the correct relation between m_1 , m_2 and m_3 ?

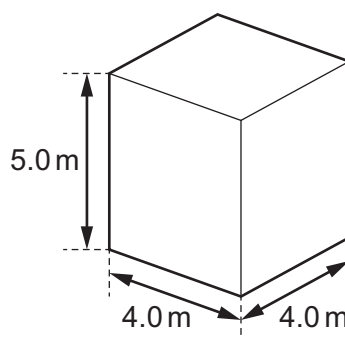
- A** m_1 is equal to m_2 and m_2 is equal to m_3 .
- B** m_1 is greater than m_2 and m_2 is greater than m_3 .
- C** m_1 is less than m_2 and m_2 is greater than m_3 .
- D** m_1 is less than m_2 and m_2 is less than m_3 .
- 5 The table shows the weight of a 15.0 kg mass placed on different planets.

Which planet has a gravitational field strength of 11.1 N/kg?

	planet	weight of 15.0 kg mass/N
A	Jupiter	389
B	Saturn	167
C	Uranus	160
D	Neptune	211

- 6 The tank shown has the dimensions $5.0\text{ m} \times 4.0\text{ m} \times 4.0\text{ m}$.

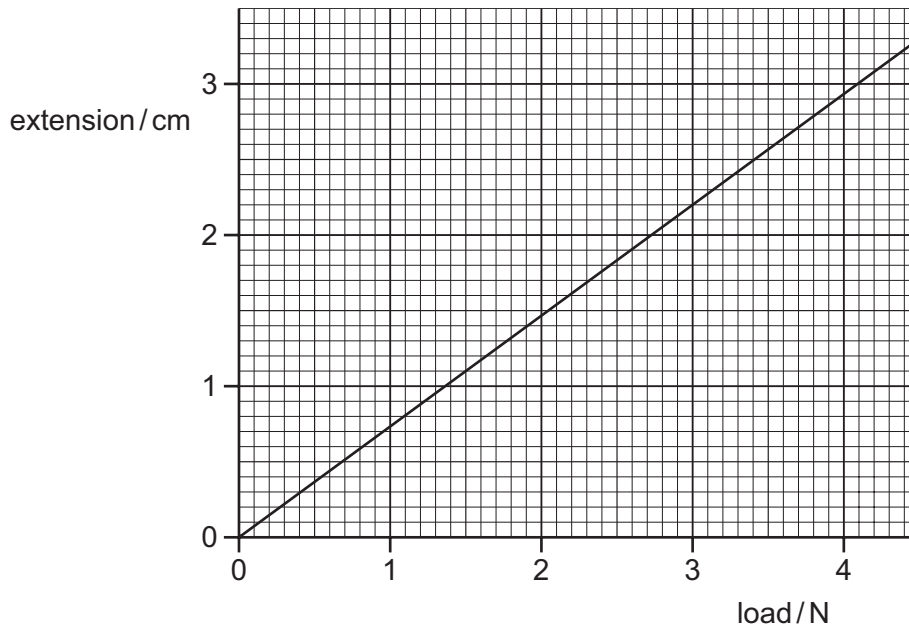
It is completely filled with water of density 1000 kg/m^3 .



What is the mass of water in the tank?

- A** 12.5 kg **B** 62.5 kg **C** 16 000 kg **D** 80 000 kg
- 7 A train travels at a constant speed along a straight track.
- Which statement about the resultant force on the train is correct?
- A** It is bigger than the thrust of the engine.
B It is constantly increasing.
C It is equal to the thrust of the engine.
D It is zero.

- 8 The extension–load graph for a spring is shown. The unstretched length of the spring is 17.0 cm.



When an object is suspended from the spring, the length of the spring is 19.2 cm.

What is the weight of the object?

- A** 1.4 N **B** 1.6 N **C** 2.6 N **D** 3.0 N

- 9 A stream flows out of a lake and down the side of a hill.

What best describes the change in energy stores?

- A** kinetic energy → chemical energy + internal energy
B gravitational potential energy → kinetic energy + internal energy
C gravitational potential energy + kinetic energy → internal energy
D kinetic energy → internal energy + gravitational potential energy

- 10 Energy resources are often used to turn a generator to produce electrical energy.

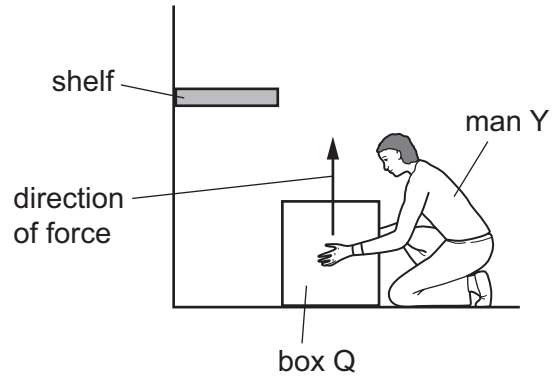
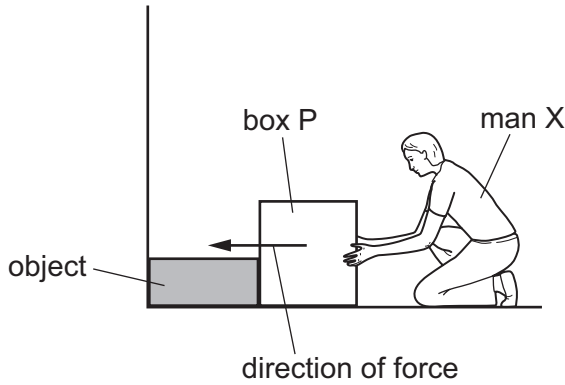
From which resource can electrical energy be produced **without** turning a generator?

- A** radiation from the Sun
B uranium by nuclear fission
C water behind a dam
D wind

- 11** Two men, X and Y, try to move identical heavy boxes, P and Q.

Man X tries to push box P along the floor. The box does not move because an object is in the way.

Man Y lifts box Q from the floor onto a shelf.



Which man does the most work on their box, and which box gains the most energy?

	man doing most work	box gaining most energy
A	X	P
B	X	Q
C	Y	P
D	Y	Q

- 12** A book has a mass of 400 g.

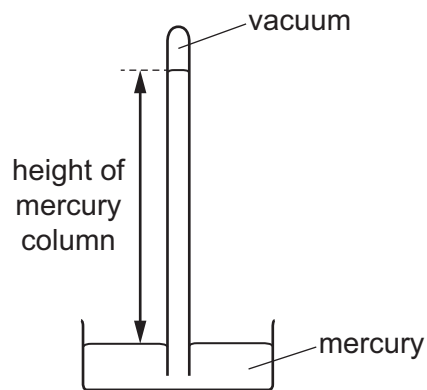
The surface of the book in contact with a table has dimensions $0.10 \text{ m} \times 0.20 \text{ m}$.

The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the table due to the book?

- A** 0.08 N/m^2 **B** 8.0 N/m^2 **C** 20 N/m^2 **D** 200 N/m^2

- 13 The diagram shows a piece of equipment which is used to measure atmospheric pressure.



What is this piece of equipment?

- A barometer
 - B galvanometer
 - C newton meter
 - D thermometer
- 14 A car tyre contains air.

The temperature of the air in the tyre rises. The volume of the air in the tyre remains constant.

Students are asked to suggest how this temperature rise will affect the movement of the air molecules.

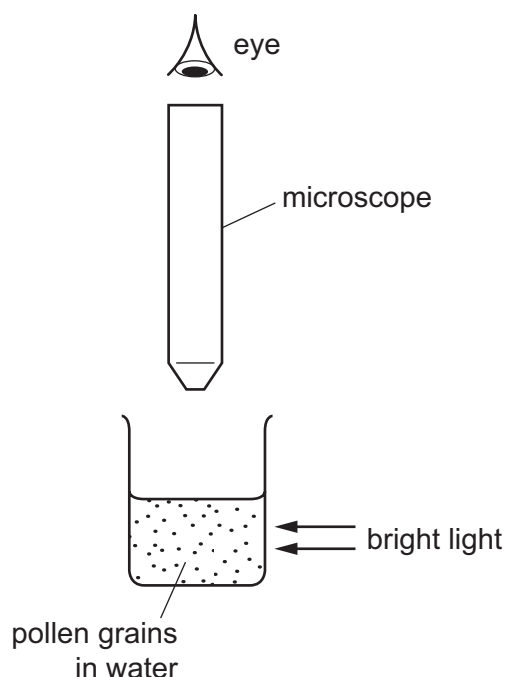
- 1 The air molecules will move faster.
- 2 The air molecules will hit the tyre walls more often.
- 3 The air molecules will hit the tyre walls harder.

Which suggestions are correct?

- A 1, 2 and 3 B 1 and 2 only C 1 and 3 only D 2 and 3 only

- 15** Very small pollen grains are suspended in water. A bright light shines from the side.

When looked at through a microscope, small specks of light are seen to be moving in a random, jerky manner.



What are the moving specks of light?

- A** pollen grains being hit by other pollen grains
 - B** pollen grains being hit by water molecules
 - C** water molecules being hit by other water molecules
 - D** water molecules being hit by pollen grains
- 16** The liquid level in a thermometer rises when the thermometer is placed in hot water.

What causes this?

- A** The liquid contracts.
 - B** The liquid evaporates.
 - C** The liquid expands.
 - D** The liquid freezes.
- 17** Which description of condensation is correct?
- A** a gas changing into a liquid as its molecules lose energy
 - B** a gas changing into a liquid as its molecules gain energy
 - C** a liquid changing into a gas as its molecules lose energy
 - D** a liquid changing into a gas as its molecules gain energy

- 18** The thermal capacity of object Y is greater than that of object Z.

What is a consequence of this?

- A** Object Y needs less thermal energy to melt it than object Z.
- B** Object Y needs less thermal energy to raise its temperature by 1 °C than object Z.
- C** Object Y needs more thermal energy to melt it than object Z.
- D** Object Y needs more thermal energy to raise its temperature by 1 °C than object Z.

- 19** In which states of matter is convection an important method of thermal transfer?

- A** liquids only
- B** solids only
- C** liquids and gases
- D** solids and gases

- 20** Solar heating panels consist of pipes carrying water that absorb radiation from the Sun.

Which texture and colour are the surface of the pipes so that the temperature of the water rises at the quickest rate?

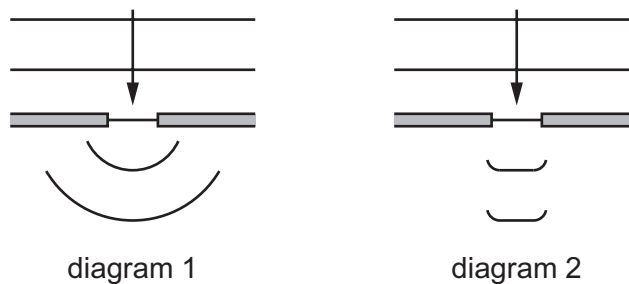
- A** dull black
- B** dull white
- C** shiny black
- D** shiny white

- 21** A child bounces up and down on a trampoline. The instructor asks her to bounce with a higher frequency.

What does the instructor mean?

- A** Bounce higher on the trampoline.
- B** Bounce more times in each minute.
- C** Take more time to bounce.
- D** Bounce with a slower speed.

22 Light waves pass through a gap. The diagrams show two possibilities for what occurs.

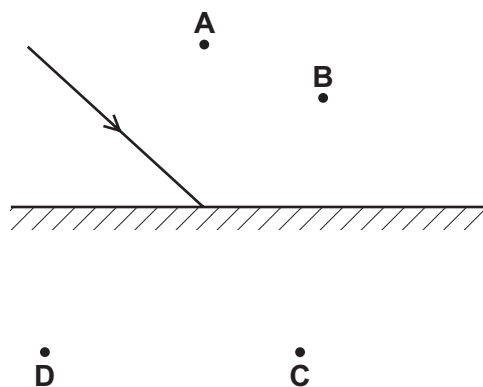


Which row is correct?

	the effect on the wave as it passes through the gap	diagram after it has passed through the gap
A	speed changes	diagram 1 is correct
B	speed changes	diagram 2 is correct
C	speed remains constant	diagram 1 is correct
D	speed remains constant	diagram 2 is correct

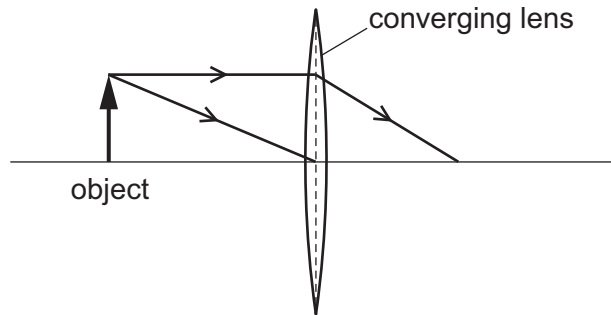
23 The diagram shows a ray of light striking a plane mirror.

Through which point does the reflected ray pass?



- 24** An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

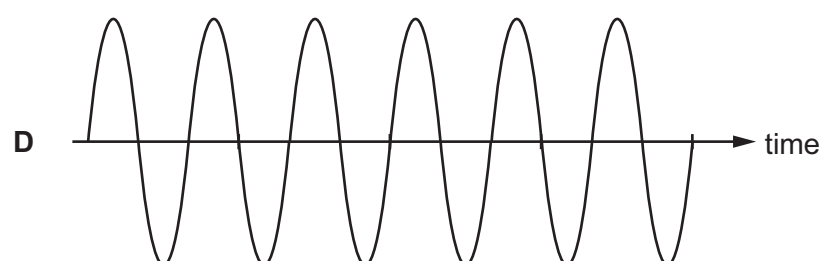
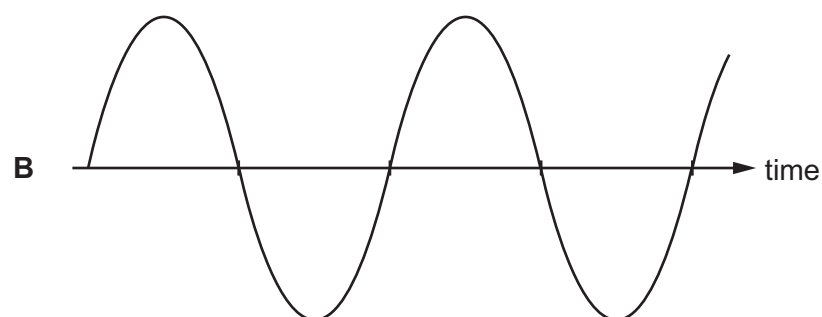
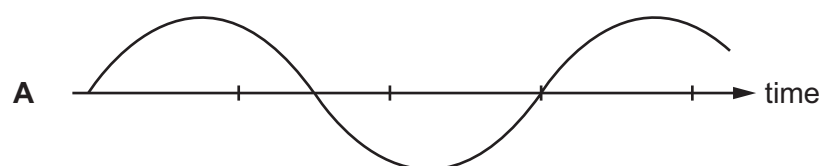
- A** It is larger and inverted.
 - B** It is larger and the same way up.
 - C** It is smaller and inverted.
 - D** It is smaller and the same way up.
- 25** The diagram shows typical wavelengths (in metres) of radio waves, microwaves and X-rays in the electromagnetic spectrum.

In which region are the waves used in TV remote controllers found?



- 26** The diagrams represent the waves produced by four sources of sound. The scales are the same for all the diagrams.

Which sound has the highest frequency?

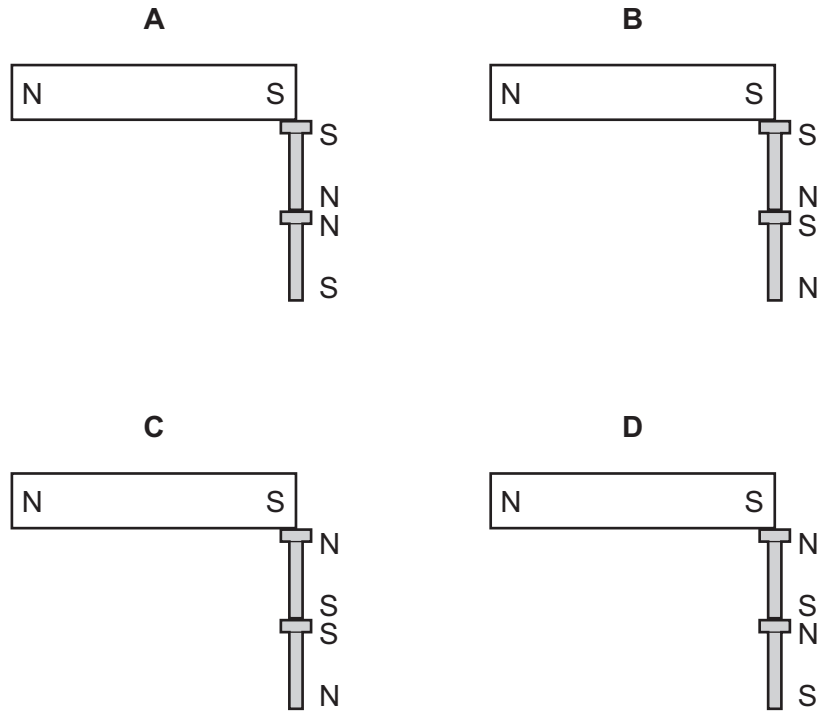


- 27** Which description of ultrasound is correct?

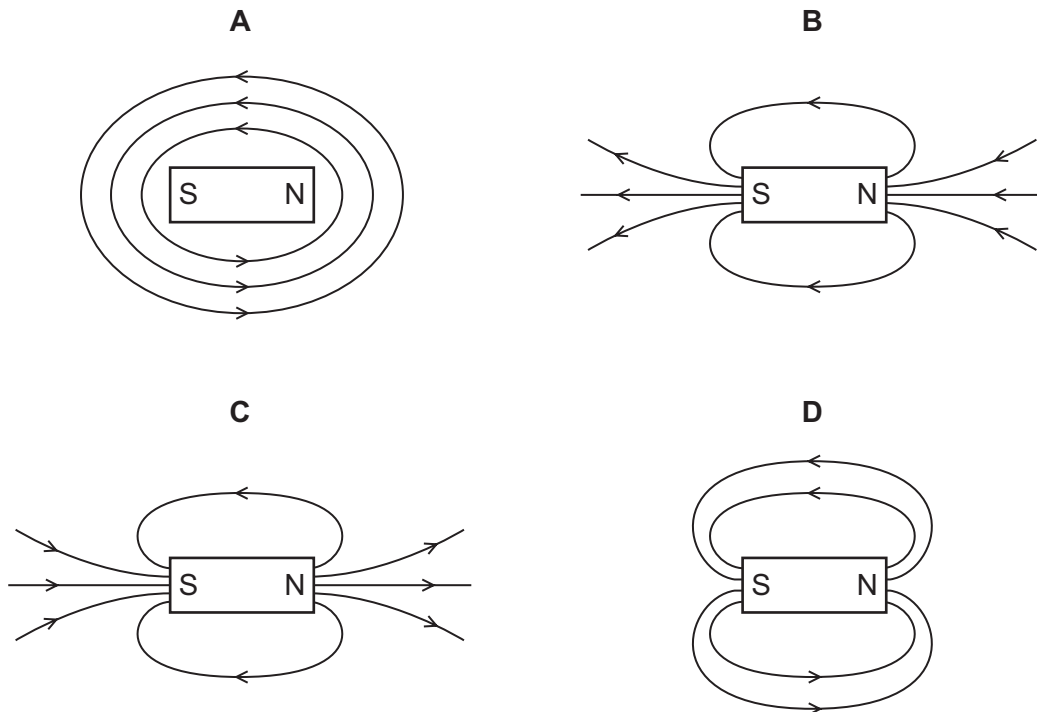
- A** longitudinal waves with a frequency greater than 20 000 Hz
- B** longitudinal waves with a frequency less than 20 Hz
- C** transverse waves with a frequency greater than 20 000 Hz
- D** transverse waves with a frequency less than 20 Hz

28 A bar magnet picks up two steel bolts.

Which diagram shows the magnetic poles induced in the bolts?



29 Which diagram shows the magnetic field around a bar magnet?

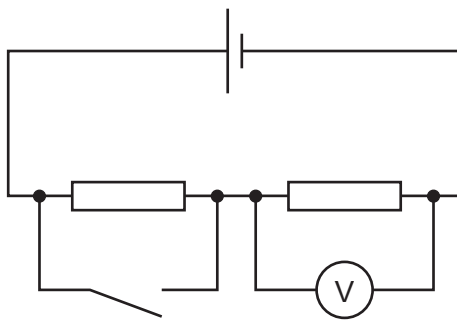


- 30** A student rubs a plastic rod with a cloth.

The rod becomes positively charged.

What has happened to the rod?

- A** It has gained electrons.
 - B** It has gained protons.
 - C** It has lost electrons.
 - D** It has lost protons.
- 31** The diagram shows a circuit containing a cell, two resistors, a switch and a voltmeter.



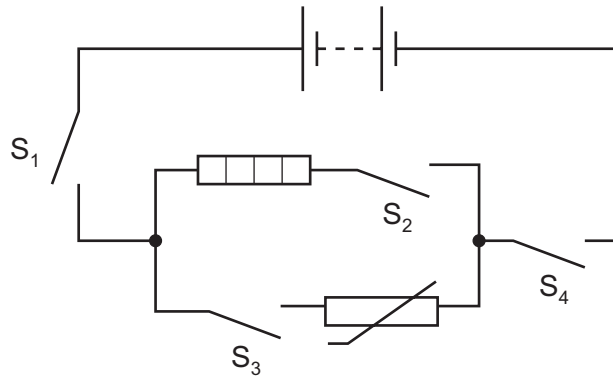
When the switch is open the voltmeter reads 1.5 V.

When the switch is closed the voltmeter reads 2.0 V.

What is the electromotive force (e.m.f.) of the cell?

- A** 0.5 V **B** 1.5 V **C** 2.0 V **D** 3.5 V

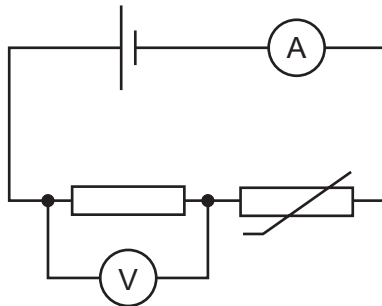
32 The circuit in the diagram contains four switches, S_1 , S_2 , S_3 and S_4 .



Which three switches must be closed for the heater to work?

- A** S_1 , S_2 and S_3
- B** S_1 , S_2 and S_4
- C** S_1 , S_3 and S_4
- D** S_2 , S_3 and S_4

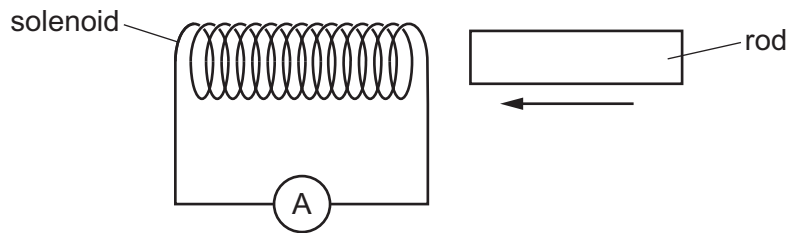
33 The diagram shows a circuit.



What happens to the readings on the voltmeter and on the ammeter when the temperature of the thermistor increases?

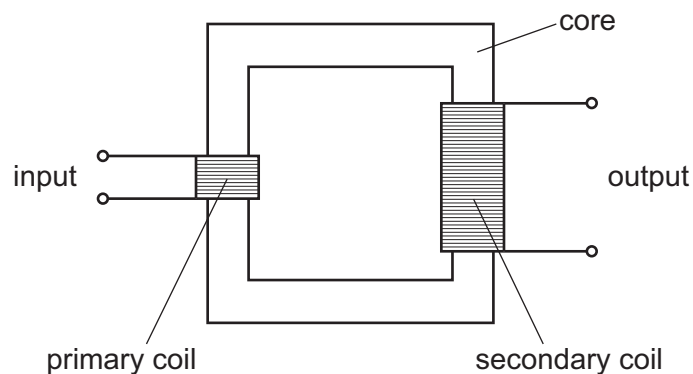
	voltmeter reading	ammeter reading
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 34** A solenoid is connected to a very sensitive ammeter. A rod is inserted into one end of the solenoid. The ammeter shows that there is a small electric current in the solenoid while the rod is moving.



Which rod is being inserted?

- A** a heated copper rod
 - B** a magnetised steel rod
 - C** an uncharged nylon rod
 - D** a radioactive uranium rod
- 35** The diagram shows a transformer with more turns on the secondary coil than on the primary coil.



Which row is correct?

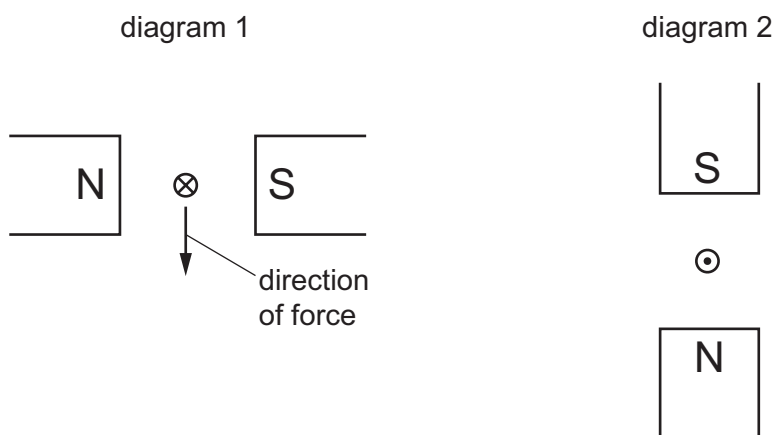
	material of core	material of coils	type of transformer
A	copper	iron	step-up
B	copper	iron	step-down
C	iron	copper	step-up
D	iron	copper	step-down

- 36 Diagram 1 shows a wire carrying an electric current into the page.

The wire is between the poles of a magnet.

A force is produced on the wire acting down towards the bottom of the page.

Diagram 2 shows the situation after the current is reversed and the magnet is turned through 90° .



In which direction does the force act after these changes?

- A towards the top of the page
- B towards the bottom of the page
- C towards the left-hand side of the page
- D towards the right-hand side of the page

- 37 The charge on a proton is e .

What is the charge on an electron and what is the charge on a neutron?

	electron	neutron
A	e	e
B	e	0
C	$-e$	$-e$
D	$-e$	0

- 38 The nuclide notation for sodium-23 is ${}_{11}^{23}\text{Na}$.

How many protons are in a nucleus of sodium-23?

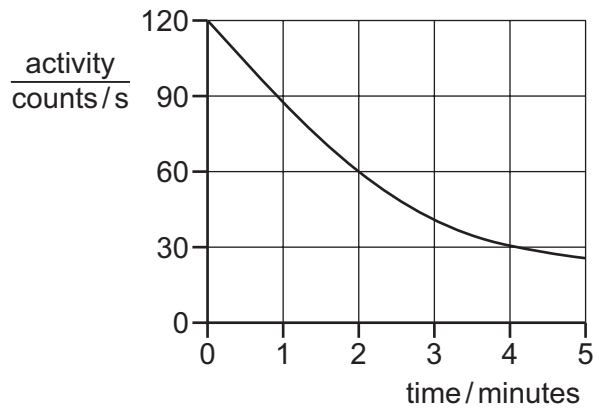
- A 11
- B 12
- C 23
- D 34

- 39 A student is investigating the count rate of a radioactive substance.

How must he adjust his reading for the background count?

- A Add the background count to his reading.
- B Ignore the background count as it will not affect his reading.
- C Subtract the background count from his reading.
- D Take repeat readings to eliminate the background count.

- 40 The graph shows the activity of a radioactive source over a period of time.



What is the half-life of the source?

- A 1.0 minute
- B 2.0 minutes
- C 2.5 minutes
- D 4.0 minutes

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/12

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **20** pages.

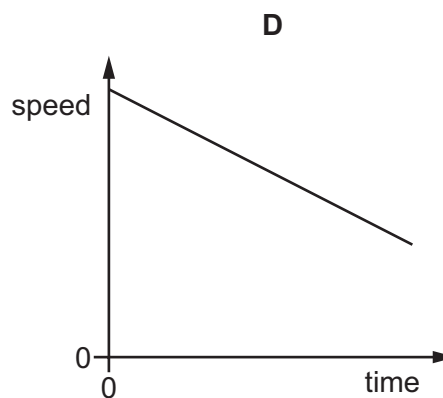
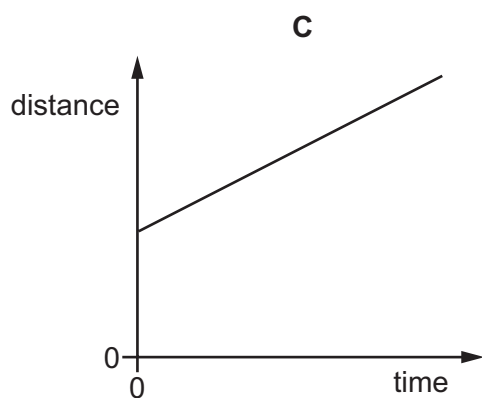
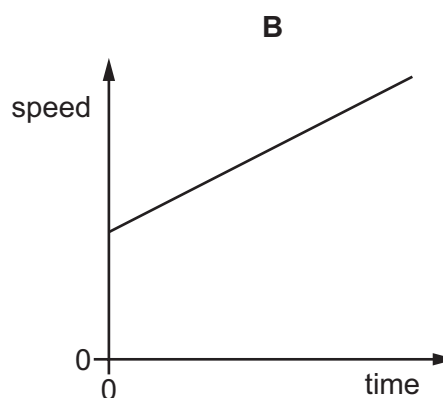
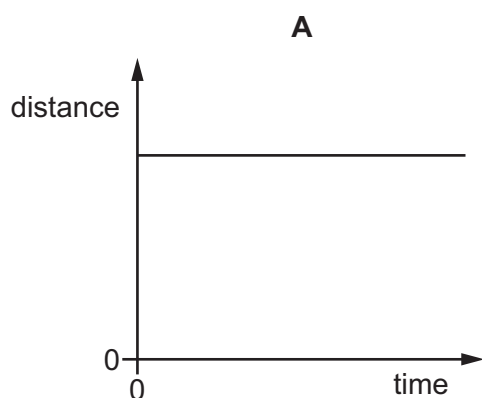


- 1 Which piece of apparatus is the most suitable for measuring the mass of a pencil sharpener?
- A digital balance
 - B measuring cylinder
 - C newton meter
 - D ruler

- 2 A cyclist records his speed and the distance travelled during a journey.

He then plots the data against time for different sections of his journey.

Which graph shows a section when he is moving with constant speed?



- 3 Two stones of different weights fall at the same time from a table. Air resistance may be ignored.

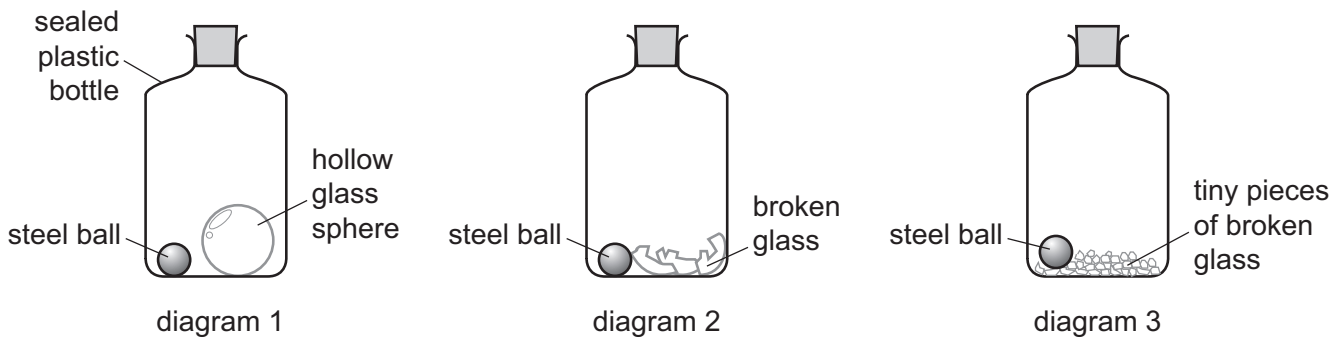
What will happen and why?

	what will happen	why
A	Both stones hit the floor at the same time.	Acceleration of free fall is constant.
B	Both stones hit the floor at the same time.	They fall at constant speed.
C	The heavier stone hits the floor first.	Acceleration increases with weight.
D	The heavier stone hits the floor first.	Speed increases with weight.

- 4 Diagram 1 shows a sealed plastic bottle containing a hollow glass sphere and a steel ball.

Diagram 2 shows the same bottle after it has been shaken.

Diagram 3 shows the same bottle after it has been shaken again until the broken glass is in tiny pieces.



The mass of the bottle and contents in diagram 1 is m_1 .

The mass of the bottle and contents in diagram 2 is m_2 .

The mass of the bottle and contents in diagram 3 is m_3 .

Which statement gives the correct relation between m_1 , m_2 and m_3 ?

- A** m_1 is equal to m_2 and m_2 is equal to m_3 .
- B** m_1 is greater than m_2 and m_2 is greater than m_3 .
- C** m_1 is less than m_2 and m_2 is greater than m_3 .
- D** m_1 is less than m_2 and m_2 is less than m_3 .

- 5 An astronaut wants to know how much she would weigh on different moons.

She measures her mass on Earth to be 82 kg. She researches the values of g on different moons and uses these values to calculate her weight.

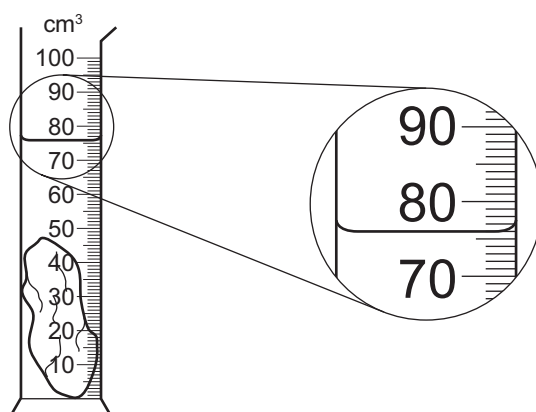
The results are shown.

Which weight is calculated correctly?

	name of moon	$\frac{g}{\text{N/kg}}$	calculated weight / N
A	Callisto	1.2	68
B	Charon	0.28	23
C	Dione	0.23	2.3
D	Umbriel	0.23	0.0028

- 6 A measuring cylinder contains 40 cm^3 of water.

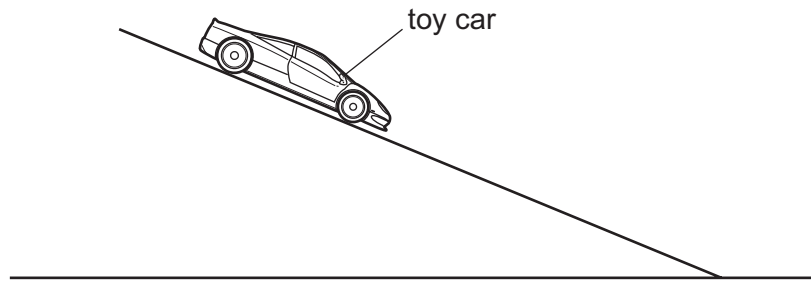
A stone of mass 94 g is lowered into the water so that it is fully submerged as shown.



What is the density of the stone?

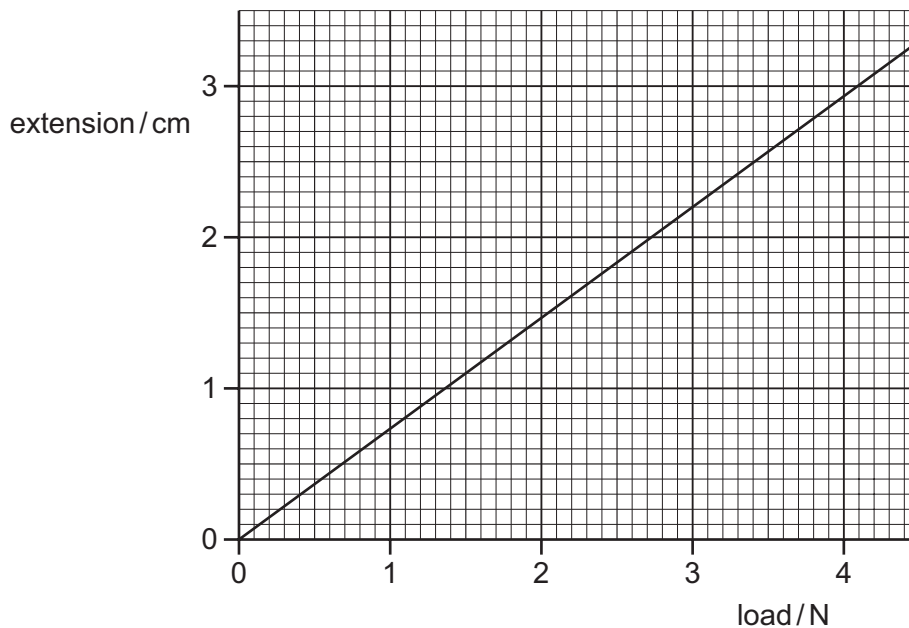
- A** 1.1 g/cm^3 **B** 1.2 g/cm^3 **C** 2.1 g/cm^3 **D** 2.6 g/cm^3

- 7 A toy car travels down a sloping ramp at constant speed.



Which statement about the forces acting on the car is correct?

- A** There are no forces acting on the car vertically.
 - B** There is no resultant force acting on the car.
 - C** There is no gravitational force acting on the car.
 - D** There is no frictional force acting on the car.
- 8 The extension–load graph for a spring is shown. The unstretched length of the spring is 17.0 cm.



When an object is suspended from the spring, the length of the spring is 19.2 cm.

What is the weight of the object?

- A** 1.4 N
- B** 1.6 N
- C** 2.6 N
- D** 3.0 N

- 9 An object is falling through a vacuum.

Which energy transfer is taking place?

- A gravitational potential to kinetic
- B gravitational potential to thermal
- C kinetic to gravitational potential
- D kinetic to thermal

- 10 This question is about four methods used to produce electrical energy.

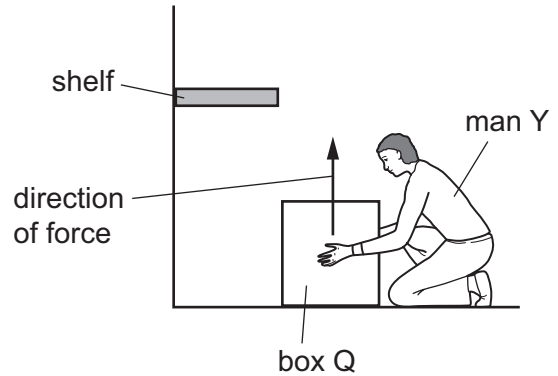
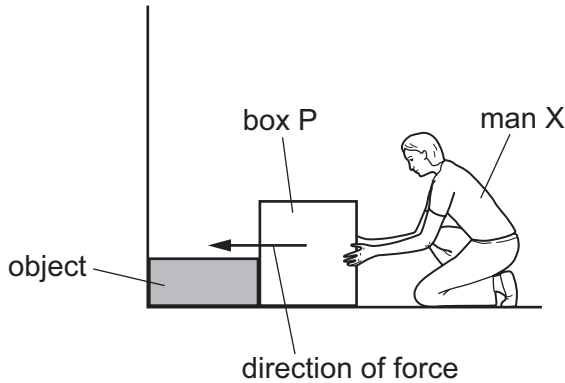
Which method has a correct description?

	method	energy source is renewable	emits carbon dioxide
A	a hydroelectric power station	yes	no
B	a coal-fired power station	no	no
C	a wind turbine	no	yes
D	a nuclear power station	yes	yes

- 11** Two men, X and Y, try to move identical heavy boxes, P and Q.

Man X tries to push box P along the floor. The box does not move because an object is in the way.

Man Y lifts box Q from the floor onto a shelf.



Which man does the most work on their box, and which box gains the most energy?

	man doing most work	box gaining most energy
A	X	P
B	X	Q
C	Y	P
D	Y	Q

- 12** A book has a mass of 400 g.

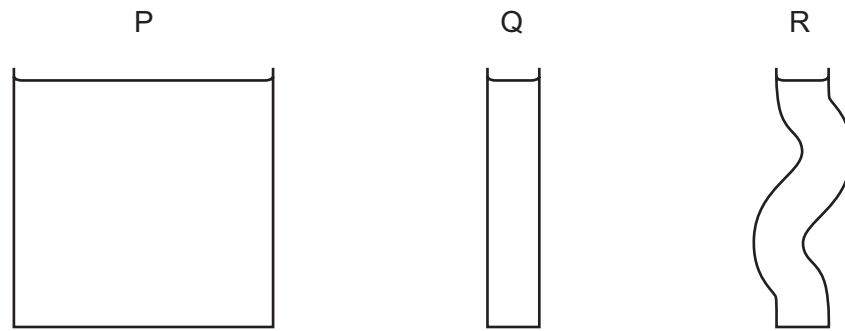
The surface of the book in contact with a table has dimensions $0.10 \text{ m} \times 0.20 \text{ m}$.

The gravitational field strength g is 10 N/kg .

What is the pressure exerted on the table due to the book?

- A** 0.08 N/m^2 **B** 8.0 N/m^2 **C** 20 N/m^2 **D** 200 N/m^2

- 13 The diagrams show three containers, P, Q and R.



Each container contains the same liquid.

The depth of the liquid is the same in each container.

Which statement about the pressure of the liquid at the bottom of the container is correct?

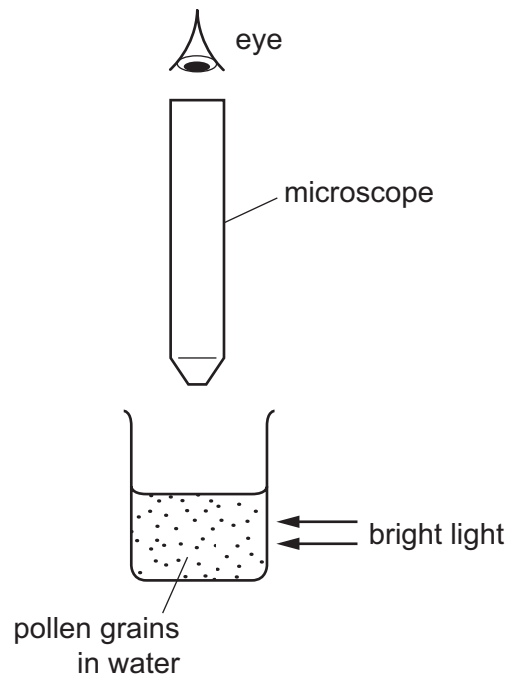
- A The pressure of the liquid at the base of P is greatest.
 - B The pressure of the liquid at the base of Q is greatest.
 - C The pressure of the liquid at the base of R is greatest.
 - D The pressures of the liquid at the bases of P, Q and R are the same.
- 14 A person taking a shower notices that water appears on the inside of the bathroom window, even though the window is some distance from the shower.

Which statement explains this observation?

- A Droplets of water from the hot shower move through the air and onto the window.
- B The colder window causes the steam in the atmosphere to solidify.
- C Water vapour from the atmosphere condenses onto the colder window.
- D The hot water boils and turns into steam.

- 15** Very small pollen grains are suspended in water. A bright light shines from the side.

When looked at through a microscope, small specks of light are seen to be moving in a random, jerky manner.



What are the moving specks of light?

- A** pollen grains being hit by other pollen grains
 - B** pollen grains being hit by water molecules
 - C** water molecules being hit by other water molecules
 - D** water molecules being hit by pollen grains
- 16** A hole is drilled in a metal plate.

What happens to the length of the plate and to the diameter of the hole when the plate is cooled?

	length of plate	diameter of hole
A	decreases	decreases
B	decreases	increases
C	increases	decreases
D	increases	increases

- 17** A liquid-in-glass thermometer uses alcohol as its liquid.

Which property of alcohol is used to measure the temperature?

- A** boiling point
- B** colour
- C** thermal capacity
- D** volume

- 18** The thermal capacity of object Y is greater than that of object Z.

What is a consequence of this?

- A** Object Y needs less thermal energy to melt it than object Z.
- B** Object Y needs less thermal energy to raise its temperature by 1°C than object Z.
- C** Object Y needs more thermal energy to melt it than object Z.
- D** Object Y needs more thermal energy to raise its temperature by 1°C than object Z.

- 19** The table gives some examples of convection and an explanation of why the convection occurs.

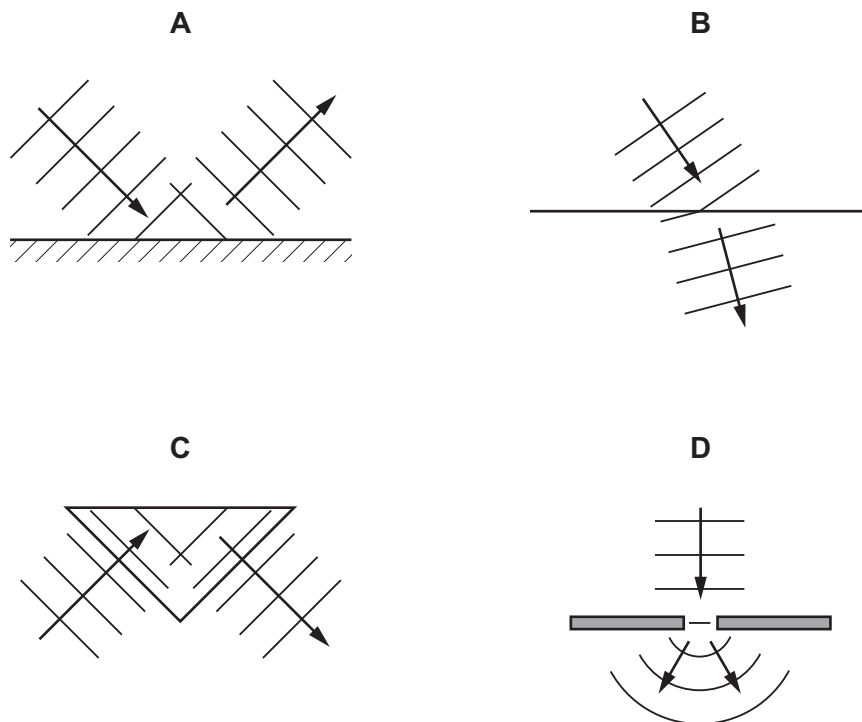
Which row is correct?

	example	explanation
A	air conditioner unit	The unit is placed in a high position to circulate the cold air rising from the floor level and so keeping the room cold.
B	convection oven	The air in the oven becomes more dense when heated so it falls to the bottom heating the food faster.
C	hot-air balloon	Air inside the balloon becomes less dense which causes the balloon to rise.
D	land and sea breezes	A breeze forms due to the warm air above the land moving down towards the sea.

- 20** In which situation is radiation the main method by which energy is transferred?

- A** heating a pan of water using a gas camping stove
- B** energy reaching the Earth from the Sun
- C** heating the air in a room with a radiator
- D** giving gravitational potential energy to a glider when it is lifted by thermal currents

21 Which diagram shows waves diffracting?

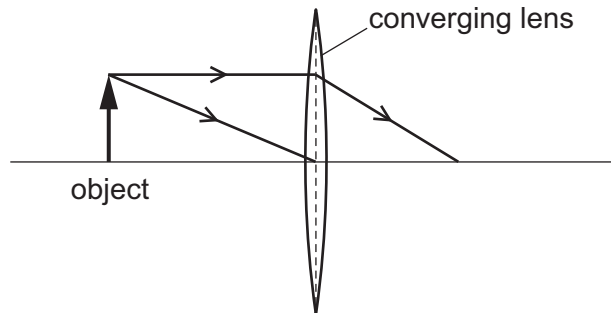


22 Which statement about the image of an object formed in a plane mirror is correct?

- A** It is smaller than the object.
- B** It is the same size as the object.
- C** It is larger than the object.
- D** It is inverted.

- 23** An object is placed in front of a thin converging lens.

The diagram shows the paths of two rays from the top of the object.



An image of the object is formed on a screen to the right of the lens.

How does this image compare with the object?

- A** It is larger and inverted.
 - B** It is larger and the same way up.
 - C** It is smaller and inverted.
 - D** It is smaller and the same way up.
- 24** The table shows information about different colours of light.

colour of light	frequency / Hz
violet	7.2×10^{14}
blue	6.3×10^{14}
yellow	5.2×10^{14}
red	4.5×10^{14}

Using the data, what is the frequency of orange light?

- A** 4.0×10^{14} Hz
- B** 5.0×10^{14} Hz
- C** 6.0×10^{14} Hz
- D** 7.0×10^{14} Hz

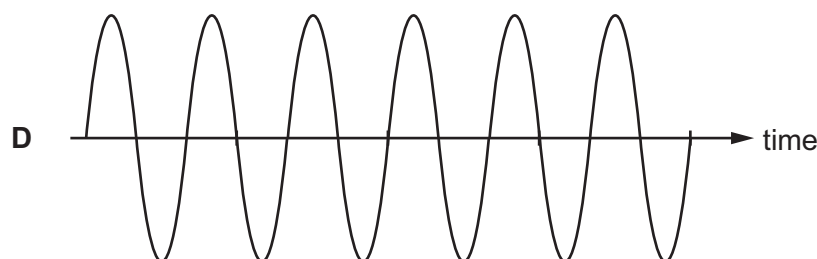
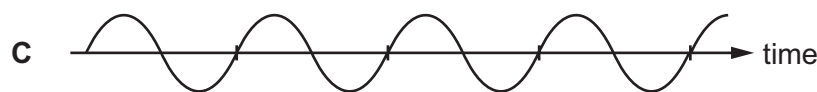
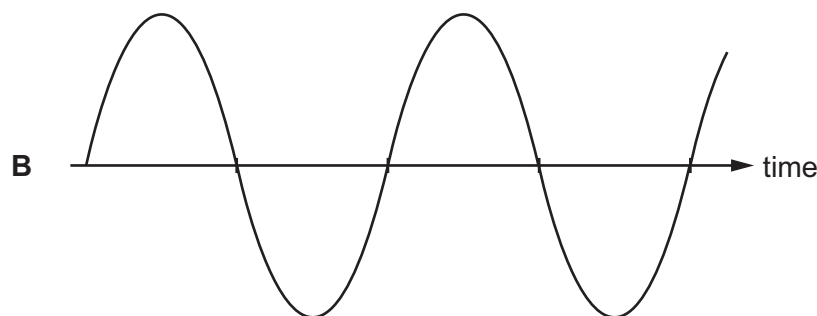
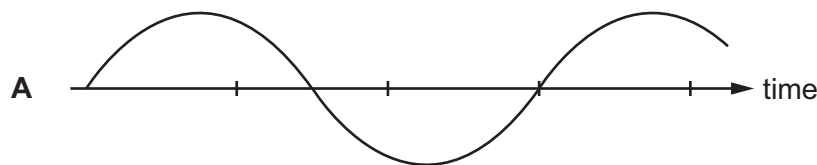
25 Sound is a transfer of energy from an oscillating source.

Which statement describes how sound energy is transferred?

- A** a longitudinal wave with the oscillation parallel to the direction in which energy is transferred
- B** a longitudinal wave with the oscillation perpendicular to the direction in which energy is transferred
- C** a transverse wave with the oscillation parallel to the direction in which energy is transferred
- D** a transverse wave with the oscillation perpendicular to the direction in which energy is transferred

26 The diagrams represent the waves produced by four sources of sound. The scales are the same for all the diagrams.

Which sound has the highest frequency?



- 27 Two metal rods each have a painted end. The painted ends are placed next to the N pole and S pole of a bar magnet in turn.



The rods are made from iron, aluminium or magnetised steel.

rod	placed next to	observation
1	N pole of bar magnet	attracts
1	S pole of bar magnet	attracts
2	N pole of bar magnet	nothing happens
2	S pole of bar magnet	nothing happens

What are rod 1 and rod 2 made from?

	rod 1	rod 2
A	magnetised steel	aluminium
B	iron	magnetised steel
C	aluminium	iron
D	iron	aluminium

28 Three piles of small nails, P, Q and R, are placed on a bench below three electromagnets.

One set of nails is made of copper, one of soft iron and one of steel.

Diagram 1 shows the situation when the electromagnets are switched on.

Diagram 2 shows the situation when the electromagnets are then switched off.

diagram 1
the switches are **closed**

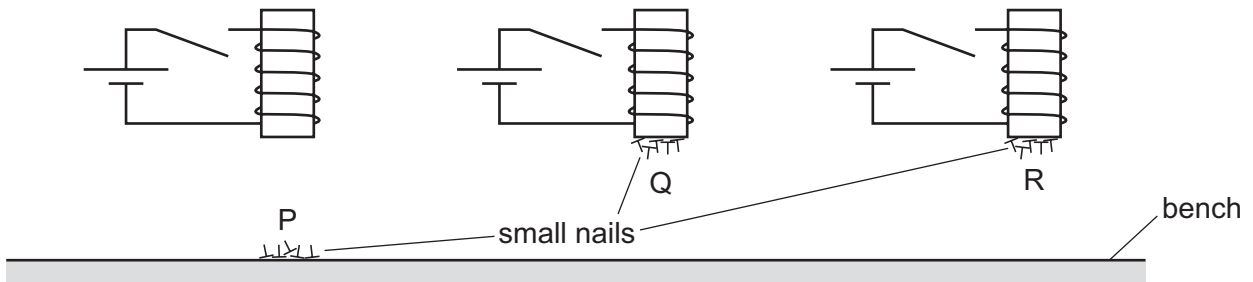
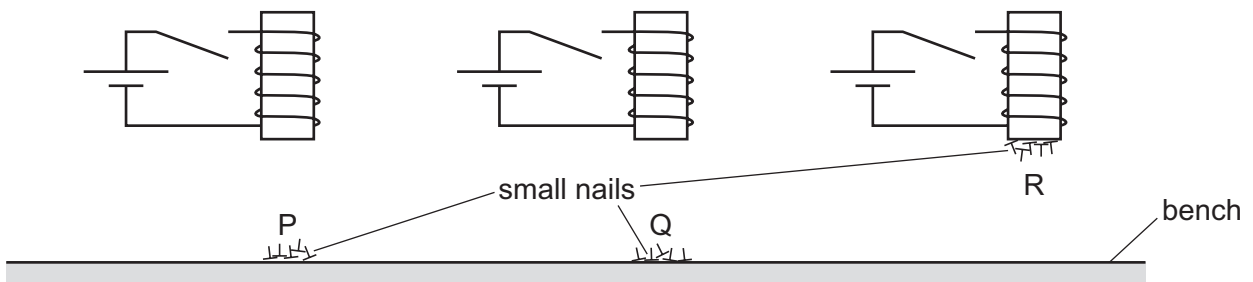


diagram 2
the switches are **open**



Which row correctly identifies the materials from which the nails are made?

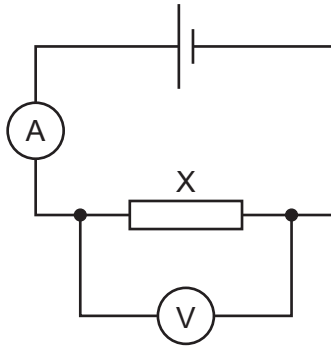
	copper	soft iron	steel
A	P	Q	R
B	P	R	Q
C	Q	P	R
D	Q	R	P

- 29** A student rubs a plastic rod with a cloth.

The rod becomes positively charged.

What has happened to the rod?

- A** It has gained electrons.
 - B** It has gained protons.
 - C** It has lost electrons.
 - D** It has lost protons.
- 30** A student connects a circuit with a resistor X. The reading on the ammeter is 2.0 A.
- The reading on the voltmeter is 6.0 V.

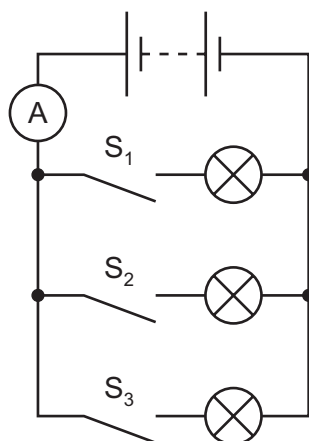


She needs to produce a circuit with a total resistance of $10\ \Omega$.

Which resistor should she add in series to the circuit?

- A** $2\ \Omega$
- B** $3\ \Omega$
- C** $7\ \Omega$
- D** $10\ \Omega$

- 31 The diagram shows a circuit containing a battery, an ammeter, three switches, S_1 , S_2 and S_3 , and three identical lamps.



With only switch S_1 closed, the reading on the ammeter is 0.04 A.

Which row states the **incorrect** ammeter reading for the switch conditions given?

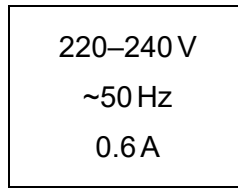
	switch S_1	switch S_2	switch S_3	ammeter reading / A
A	open	open	open	0.00
B	open	closed	open	0.04
C	open	open	closed	0.08
D	closed	closed	closed	0.12

- 32 Two $10\ \Omega$ resistors are connected in series and then in parallel.

What is the combined resistance in each case?

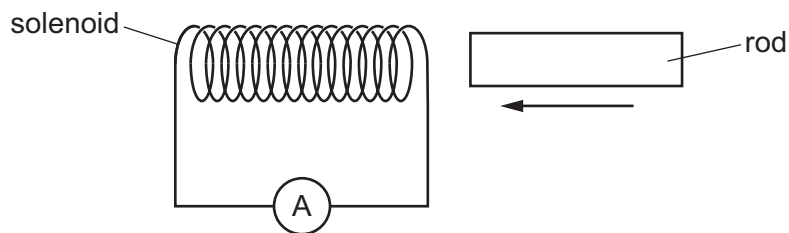
	resistance in series / Ω	resistance in parallel / Ω
A	10	5
B	10	10
C	20	5
D	20	10

- 33 The information on the back of a television is shown.



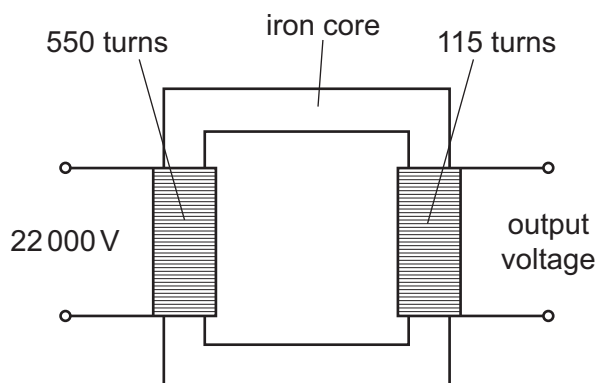
Which fuse provides the best protection for the circuit?

- A** 0.5 A **B** 1 A **C** 5 A **D** 13 A
- 34 A solenoid is connected to a very sensitive ammeter. A rod is inserted into one end of the solenoid. The ammeter shows that there is a small electric current in the solenoid while the rod is moving.



Which rod is being inserted?

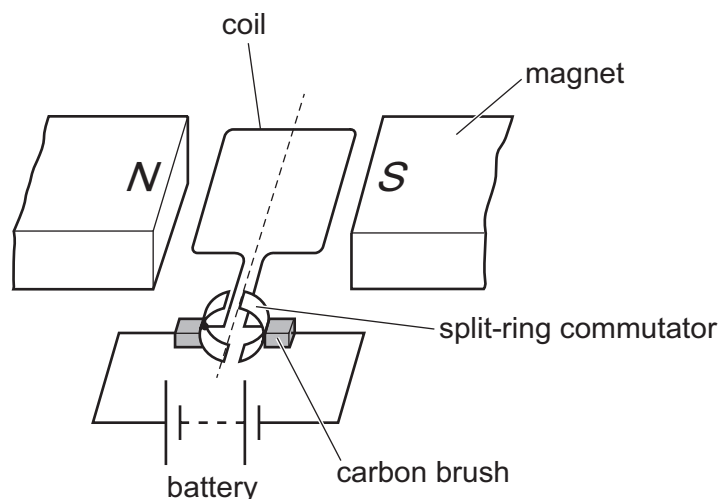
- A** a heated copper rod
B a magnetised steel rod
C an uncharged nylon rod
D a radioactive uranium rod
- 35 The diagram shows a transformer.



What is the output voltage?

- A** 0.35 V **B** 2.9 V **C** 4600 V **D** 105 000 V

36 The diagram shows a d.c. motor.



Which two changes together will always make the coil spin more slowly?

- A Decrease the current in the coil and use a magnet of less strength.
- B Decrease the current in the coil and increase the number of turns on the coil.
- C Increase the current in the coil and use a magnet of less strength.
- D Increase the current in the coil and decrease the number of turns on the coil.

37 The charge on a proton is e .

What is the charge on an electron and what is the charge on a neutron?

	electron	neutron
A	e	e
B	e	0
C	$-e$	$-e$
D	$-e$	0

38 The proton numbers and nucleon numbers of four nuclides are shown.



Which statement is correct?

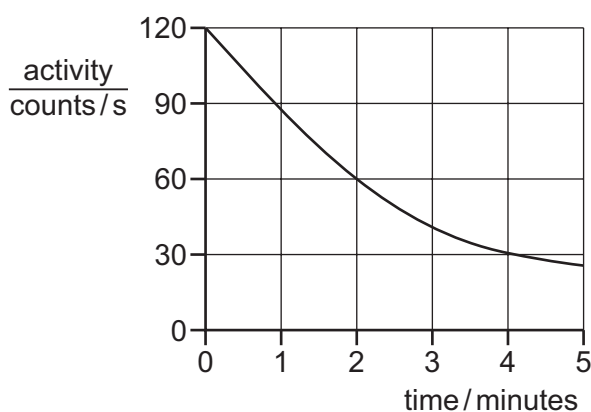
- A Plutonium (Pu) contains one more proton than uranium (U).
- B Thorium (Th) contains one more neutron than radium (Ra).
- C Thorium (Th) contains one more proton than radium (Ra).
- D Uranium (U) contains one more neutron than plutonium (Pu).

39 A radioactive atom decays by emission of a β -particle.

Which row is correct?

	what decays	what happens to the atom
A	the nucleus of the atom	it becomes a different element
B	the nucleus of the atom	it becomes a lighter version of the same element
C	the outer layers of the atom	it becomes a different element
D	the outer layers of the atom	it becomes a lighter version of the same element

40 The graph shows the activity of a radioactive source over a period of time.



What is the half-life of the source?

- A** 1.0 minute **B** 2.0 minutes **C** 2.5 minutes **D** 4.0 minutes

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Cambridge IGCSE™

PHYSICS

Paper 1 Multiple Choice (Core)

0625/11

May/June 2021

45 minutes

You must answer on the multiple choice answer sheet.

You will need: Multiple choice answer sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

INSTRUCTIONS

- There are **forty** questions on this paper. Answer **all** questions.
- For each question there are four possible answers **A**, **B**, **C** and **D**. Choose the **one** you consider correct and record your choice in soft pencil on the multiple choice answer sheet.
- Follow the instructions on the multiple choice answer sheet.
- Write in soft pencil.
- Write your name, centre number and candidate number on the multiple choice answer sheet in the spaces provided unless this has been done for you.
- Do **not** use correction fluid.
- Do **not** write on any bar codes.
- You may use a calculator.
- Take the weight of 1.0 kg to be 10 N (acceleration of free fall = 10 m/s^2).

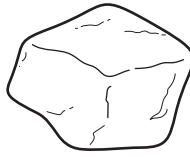
INFORMATION

- The total mark for this paper is 40.
- Each correct answer will score one mark.
- Any rough working should be done on this question paper.

This document has **16** pages.



- 1 The diagram shows a stone of irregular shape.

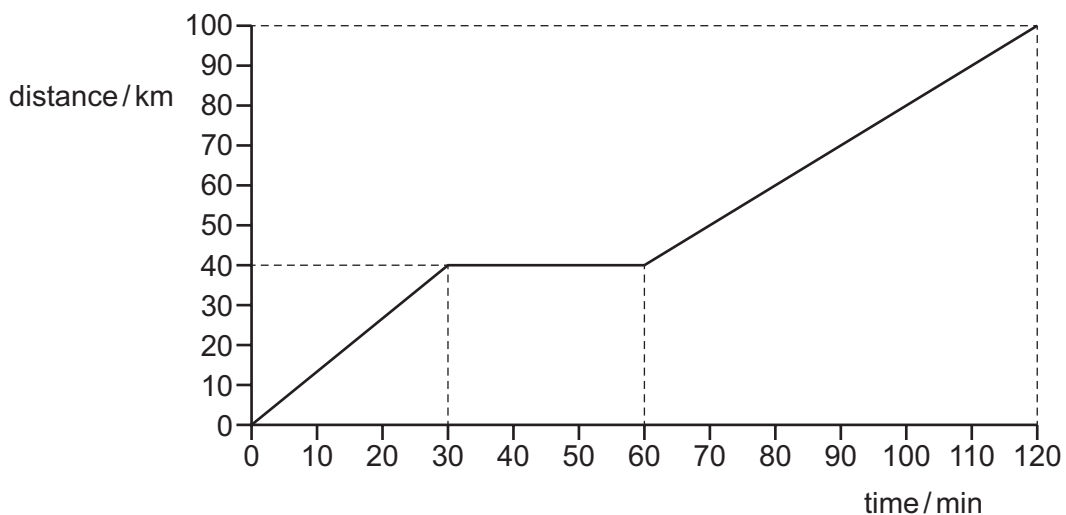


Which property of the stone can be found by lowering it into a measuring cylinder half-filled with water?

- A** length
B mass
C volume
D weight
- 2 Two stones of different weights fall at the same time from a table. Air resistance may be ignored.
 What will happen and why?

	what will happen	why
A	Both stones hit the floor at the same time.	Acceleration of free fall is constant.
B	Both stones hit the floor at the same time.	They fall at constant speed.
C	The heavier stone hits the floor first.	Acceleration increases with weight.
D	The heavier stone hits the floor first.	Speed increases with weight.

- 3 The distance–time graph for a motorway journey is shown.



What is the average speed for the journey?

- A** 50 km/h **B** 67 km/h **C** 70 km/h **D** 83 km/h